Does the VAT Reach Informal Prices?

Evidence from Supply Chains in Peru*

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October 2026

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Abstract

Recent incidence analyses find the VAT progressive because low-income people buy more informally, assuming no pass-through to informal prices. We show that pass-through to an informal price is the formal input share of that price times the pass-through of its formal input. Peru's 2022 chicken exemption fell on informal vendors' formal input in some regions only, because supply chains differ. Where it did, informal prices fell about as predicted; elsewhere, they showed no sustained response. Assuming no pass-through to informal prices would make the VAT on purchased food steeply progressive relative to expenditure; with our model, it is mildly regressive.

JEL codes: H22, H25, H26, O17.
Keywords: value-added tax, informality, pass-through, tax incidence, supply chains, Peru.

*We are deeply grateful to our advisors James Hines Jr., Joel Slemrod, and Damian Vergara. This paper has greatly benefited from conversations with Hoyt Bleakley, Giacomo Brusco, Dario Tortarolo, Mazhar Waseem, Mel Stephens, Paula Lopez, Joaquin Endara, Rosario Aldunate, and Luciana de la Flor as well as many participants in the Public Finance Seminar of the University of Michigan, NTA 2024 Conference, IIPF 2024 Conference, LACEA 2024 Conference, Alumni Workshop UNLP 2024, and M-TAXI 2025. We are also grateful to staff at the Ministry of Agricultural Development and Irrigation in Peru, which provided part of the data for this study and valuable information on how markets operate in the country, as well as to Rackham Graduate School and the Economics Department of the University of Michigan for their financial support in this project. The views expressed here are those of the authors and do not necessarily represent those of the Federal Reserve Bank of Minneapolis or the Federal Reserve System.

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1 Introduction

Indirect taxes such as the value-added tax (VAT) are the primary source of government revenue in many developing countries (Jensen, 2022; Brockmeyer et al., 2024). A growing literature has re-evaluated their role in redistribution. Bachas, Gadenne, and Jensen (2024) use store type as a proxy for tax compliance, assuming that modern stores are fully formal and traditional stores are fully informal, and, in their baseline, that the tax passes through fully to consumer prices in formal stores and not at all in informal ones. Because low-income households spend a larger share of their budget in traditional (informal) stores, this method leads to the conclusion that uniform consumption taxes can be progressive and reduce inequality in developing countries.

How progressive the tax is depends on how much of it passes through to informal prices relative to formal ones, and the assumption that this is zero is unlikely to hold. We show that relative pass-through is not zero whenever a formal firm sits upstream of the informal store, and that this matters: in Peru, the VAT on purchased food is mildly regressive relative to expenditure, where the zero-pass-through accounting finds it steeply progressive.

The reason lies in the input-tax channel created by the VAT's credit mechanism. A formal seller that buys from another formal seller credits the VAT paid on its purchases against the VAT it charges on its sales, so along a fully formal chain the tax becomes a cost only at the final sale. An informal vendor charges no VAT on its sales, but it also cannot claim a credit for the VAT charged by its formal supplier, so that tax becomes a non-recoverable cost (Keen, 2008) that can be passed on to consumers. For the informal consumer, then, the VAT becomes a cost one step earlier: on the informal vendor's formal inputs. How much of the VAT can pass through to an informal price is therefore set by a share we can measure from prices: the formal input share of the informal price, which is zero only when no formal seller lies upstream.

To measure this pass-through, we use a VAT change in Peru that fell on some informal vendors and not others, depending on the nature of their supply chains. Announced in early April and published on April 14, 2022, Law 31452 temporarily exempted five key products of the basic consumption basket (eviscerated chicken, eggs, bread, sugar, and pasta) from the VAT, originally at 18 percent, from May 1 to July 31, 2022.[1] The law exempted the sale of these goods but not of their inputs and, critically, allowed sellers of the exempted goods to claim a fiscal credit for the VAT paid on a specific set of main inputs listed in the legislation. We focus on the case of chicken because of its vertical supply chain and how it interacted with the exemption: the law exempted eviscerated chicken but not its main input, live birds. Chicken reaches informal vendors in traditional markets, which we call *stalls*, through supply chains that cross the formal–informal boundary at different points. The key difference that we exploit is that the nature of the stall's input differs across different geographies. In Lima, the capital, the stalls' last formal supplier is the wholesaler at the live-poultry market, which sells live birds; the birds are slaughtered afterward, outside the formal chain, so the stall's formal input is the live bird, which was not exempted. In the interior of the country, the wholesaler sells eviscerated chicken to the stalls, so the stall's formal input is the exempted

[1] The goal of the policy was to limit the inflationary impact of the international economic crisis.

product.[2]

We formalize this logic in a model of the VAT along a supply chain in which each seller can be formal or informal. The model does two things: it tells us what a tax change should do to informal prices in each chain (pass-through), which is what we test, and it gives us a sufficient statistic that we can measure directly from prices, so that the results can be taken beyond chicken. That statistic is the formal input share of the informal price, δ: the ratio of the price paid to the last formal seller in the chain (which includes VAT that cannot be credited) to the informal price. It equals one only if the informal vendor resells what it buys from formal sellers at the price it paid, and is below one when the vendor has a value added of its own.

The model yields two predictions. First, a change in the general VAT rate falls on every formal input, and the pass-through to the informal price is δ times the pass-through of its formal input.[3] Second, a change in the VAT on a single good, such as an exemption, passes through to the informal price only if it falls on the formal input; when it does, the informal price again passes through δ times what its formal input passes through, as in the first prediction. A change earlier in the chain is offset by the credits of the formal sellers that follow, and a change on a good that the informal vendor produces itself removes a tax that nobody was paying.

We test these predictions using high-frequency price data collected by the Peruvian Ministry of Agriculture, which records prices several times a week at every stage we need, including wholesale markets, stalls, and supermarkets. Throughout the paper, we treat stalls as informal based on their *de facto* behavior. While some stalls are registered with the tax authority, most of their transactions are not reported. We document this with original fieldwork. We surveyed 420 stalls in 42 of the markets in our price data. Our enumerators observed customers' purchases: in 81 percent of markets, none of the observed customers received a receipt. At 318 of the stalls, they also made a test purchase of a randomly drawn amount between 5 and 10 soles, above the value at which a receipt is legally required, and a receipt was issued in only 5 of them (1.6 percent).

Our results are consistent with the model's predictions for both chains. Using a synthetic difference-in-differences design, we compare each chicken price series with a synthetic control built from food products that the exemption did not cover. Supermarkets, at the end of an all-formal chain, passed through 52 percent [43, 60] of the tax over the exemption.[4] In the interior, where a wholesaler we treat as formal sells the exempted eviscerated bird to the stalls, the wholesaler passed through 39 percent [14, 63] and the stalls 38 percent [9, 59]. Both are significantly different from zero: the exemption reached informal consumers because it fell on the stalls' formal input.

[2] We do not observe whether that sale is invoiced; we proxy it by whether an authorized slaughter plant is near the wholesale market. Section 5.2.2 shows that the interior response comes from the ten cities with one.

[3] If formal sellers pass through similar shares of the tax at every stage, so that the formal input moves like a formal retail price, relative pass-through (the pass-through to informal prices relative to formal retail prices) equals δ.

[4] Intervals in brackets are 95 percent confidence intervals from a bootstrap that resamples interior departments and Lima markets with all their price series, chicken and control products alike, so that each department's wholesale and stall series are drawn together, and re-estimates the live-bird adjustment in every draw. Table 4 reports the same intervals for each stage. All estimates except the live bird's own net out the price of live birds.

The model also gives a formula for how much should reach them: the stall's pass-through should be δ times the wholesaler's, where δ, the ratio of the wholesale to the stall price before the exemption, is 0.89. That gives $0.89 \times 39 = 35$ percent [12, 56], close to the 38 percent we estimate directly.

In Lima, the stall's formal input, the live bird, was not exempted, so the model predicts that the stall price does not move. For this exemption, the chain behaves as if it were informal from end to end. In the model, the tax on the live bird still reaches Lima's stall price, but the exemption left it in place. The live bird's price shows no statistically significant response. Lima's stall prices dipped for two weeks around implementation but showed no sustained response over the exemption. Overall, the interior stalls passed through 31 percentage points more than Lima's [3, 54].

We consider three alternative explanations for the fall in informal prices. First, seasonality: when we apply the same design to the same weeks of 2021, when there was no exemption, we find no decline in chicken prices at any stage; if anything, chicken prices rose relative to the control products. Second, if informal prices fell because consumers switched to supermarkets, the formal share of chicken purchases should have increased. In household expenditure data, however, this share did not change significantly during the exemption. Even the upper end of its 95 percent confidence interval, 1.5 percentage points, is well below the 3.9 points that switching would need to produce the fall in informal prices we measure, if every purchase that leaves the stalls goes to supermarkets.[5] Third, if stalls cut prices to compete with supermarkets, stalls located near a supermarket should have cut their prices by more. In Lima, however, we find no statistically significant difference in pass-through between stalls near a supermarket and stalls farther away.[6]

We then ask who gained from the exemption. The model shows that, for this, we need only two things: the pass-through in each chain, formal and informal, and the share of each household's budget spent on the good in each chain. With the pass-through in each chain,[7] the exemption's relief was 0.066 percent of expenditure in the bottom decile and 0.058 in the top, and about 0.11 percent in the second to sixth deciles. The bottom half of the distribution received 31 percent of the relief [27, 35], more than its 26 percent share of total expenditure. Under the conventional accounting of zero pass-through to informal prices (Bachas, Gadenne, and Jensen, 2024), which we call the sales-tax view, only households that buy chicken in supermarkets gain: the bottom deciles, which buy almost none there, gain nothing, and the bottom half receives 4 percent of the relief. We therefore find that the VAT on chicken weighed more, relative to expenditure, on the bottom half of the distribution than on the top half (i.e., a mildly regressive VAT), whereas the sales-tax view puts almost all of it on the top half (i.e., a progressive VAT).

An exemption written on the final good, as Peru's was, misses chains whose last formal sale

[5] Informal supply is held fixed, which gives the smallest shift for a given demand response. The 3.9 points assume a unit elasticity of informal demand; with less elastic demand the required shift is smaller, and switching is rejected as the full explanation for elasticities of about one half or more (Appendix F.1).

[6] The test can be run only in Lima, where supermarkets are dense enough to compare markets by distance.

[7] Outside Lima, provinces with an authorized slaughter plant take the response of the interior cities with one; provinces without one get zero, the model's prediction, which the estimate for those cities does not reject (Sections 5.2.2 and 6.1).

is of an earlier good, such as the live bird in Lima; there, only an exemption on the input reaches informal consumers. A tax cut meant to help poorer households should therefore fall on the last formal sale in the chains that supply the informal stores where they shop, which need not be the good they buy.

Finally, we use the model for a counterfactual: removing the VAT on all purchased food. A change in the general rate would fall on every formal input, so it would reach every informal price with a formal seller upstream, by δ times the pass-through of its formal input. If formal prices passed the tax through fully, as Bachas, Gadenne, and Jensen (2024) assume, each household would gain in proportion to the tax its food purchases carry today. With informal prices carrying the tax on their formal inputs, that tax is mildly regressive relative to expenditure: it falls from 1.6 percent of expenditure in the bottom decile to 1.2 in the top, because food is a larger share of poorer households' budgets. The sales-tax view finds it steeply progressive: it rises from 0.08 percent of expenditure in the bottom decile to 0.60 in the top. For the VAT on all consumption we do not know each good's formal input share, but the share of the tax that an informal purchase carries must lie between zero, as the sales-tax view assumes, and one. Within those bounds the tax remains mildly progressive: the top decile's rate is 1.4 times the bottom decile's in our central case, against 2.1 under the sales-tax view, and close to proportional when every informal purchase carries the full tax. In every case, then, the VAT is less progressive than the sales-tax view implies, because that view is the one case in which informal purchases carry none of the tax.

Related literature. Our paper contributes to four strands of the literature. First, our work speaks to the literature on how informality and tax evasion change the conclusions drawn from conventional incidence analysis (Kopczuk et al., 2016; Harju, Kosonen, and Skans, 2018; Slemrod and Velayudhan, 2022; Bachas, Gadenne, and Jensen, 2024; Brockmeyer et al., 2024; Brusco and Velayudhan, 2025). Bachas, Gadenne, and Jensen (2024) show that, if the tax passes through fully to formal prices and not at all to informal ones, consumption taxes are progressive in developing countries, because poorer households buy a larger share of what they consume in informal stores.[8] We build on their approach in four ways. We show which tax changes reach the informal price; we measure the formal input share of the informal price from posted prices; we estimate pass-through at both ends of the same chain, which lets us test the relationship directly; and, instead of a common relative pass-through for all goods, we assign each good a formal input share from its supply chain. We find that the VAT on chicken and on purchased food, progressive under the conventional accounting, weighs more on the bottom half of the distribution relative to expenditure. Our second prediction also implies a failure of tax-collection invariance (Slemrod, 2008; Kopczuk et al., 2016): whether a change in the tax reaches informal consumers depends on which transaction it falls on, the sale of the formal input or another one.

[8] They also estimate pass-through after a VAT reform in Mexico, 75 percent in modern stores and 16 percent in traditional ones, a relative pass-through of about one-fifth; show in a supplementary appendix that pass-through to traditional stores should equal the share of their input costs bought from VAT-registered suppliers, about a tenth in Mexico's retail census; argue that pass-through to informal prices is likely to be low because the VAT segments formal and informal supply chains (De Paula and Scheinkman, 2010; Gadenne, Nandi, and Rathelot, 2024); and show that their conclusion survives any common relative pass-through between zero and one, because home production bears no VAT.

Second, we add to the extensive literature on consumption tax pass-through (Carbonnier, 2007; Kosonen, 2015; Gaarder, 2019; Benedek et al., 2020; Benzarti et al., 2020; Benzarti, Garriga, and Tortarolo, forthcoming; Brusco and Velayudhan, 2025; Fuest, Neumeier, and Stöhlker, 2025). These studies measure pass-through at formal sellers, which charge the tax on their own sales; Benzarti, Garriga, and Tortarolo (forthcoming), for instance, study supermarkets in Argentina, where informal retailers account for at most 2 percent of the purchases of the goods they consider. We add stage-by-stage estimates along a supply chain that runs into the informal sector, and show that pass-through to the informal price is δ times the pass-through of its formal input's price. Methodologically, we bring the tools of the vertical pass-through literature in industrial organization, stage-by-stage estimates of pass-through weighted by cost shares (Nakamura and Zerom, 2010; Bonnet et al., 2013; Weyl and Fabinger, 2013), to an incidence question in public finance.

Third, we contribute to the literature on the interaction between the formal and informal sectors. This literature has traditionally focused on supply-side linkages and segmented supply chains (Bose, 1998; De Paula and Scheinkman, 2010). In the case of the VAT, it has shown that the tax on formal inputs acts as a tax on the informal sector and that the credit mechanism segments production by formality (Keen, 2008; De Paula and Scheinkman, 2010; Gadenne, Nandi, and Rathelot, 2024). These studies document how the VAT shapes trade between formal and informal firms, but not how a tax change is transmitted into prices. We observe prices at every stage of supply chains that cross the formal–informal boundary, and we measure how incomplete that segmentation is.

Finally, we contribute to the literature on measuring the informal sector (Restrepo-Echavarria, 2014; Rothenberg et al., 2016; Charman et al., 2017; Brockmeyer et al., 2019). We provide direct evidence on receipt issuance in traditional markets through our own survey.

The remainder of the paper is organized as follows. Section 2 describes the exemption, our survey of market stalls, and the chicken chains. Section 3 presents the model, its two predictions, and where an exemption should sit. Section 4 describes the data. Section 5 presents the pass-through estimates and alternative explanations. Section 6 asks who gained from the exemption and who would gain from a cut in the general rate. Section 7 concludes.

2 Institutional setting

This section describes the exemption and the credit rule that governed it, the market stalls we treat as informal, and the chains through which chicken reaches consumers.

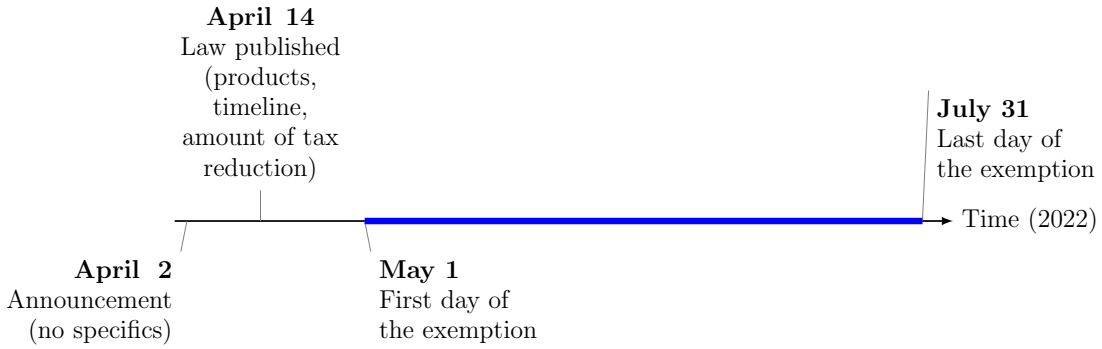
2.1 The exemption and the credit rule

Peru's VAT, set at 18 percent, is levied at every stage on invoiced sales, with a credit for the tax on invoiced purchases. Vendors in the simplified regime (RUS) and unregistered vendors neither charge nor credit it.

Law 31452, published on April 14, 2022, after an announcement on April 2 (Figure 1), exempted five groups of goods from May 1 to July 31, 2022: chicken meat, eggs, sugar, uncooked

pasta and bread.[9] Live birds (heading 0105) were not exempted. We focus on chicken because it is the one exempted good whose supply chain we observe at every stage, from the live bird to the stall and the supermarket; for sugar, pasta and bread we observe neither who sells them to informal vendors nor in what form. Eggs were also exempted, but we leave them to Appendix J: in the first week of June 2022 every egg price series, formal and informal, rose by 15 to 20 log points, a shock specific to eggs that none of our control products shares and that our data cannot separate from the exemption.[10]

Figure 1: Timeline of the policy



Article 3 of the law, implemented by Supreme Decree 083-2022-EF of April 30, let sellers of the exempt goods credit the VAT paid on the principal inputs listed by the decree, which for chicken include live chickens.[11] This matters because it fixes how much tax a formal seller can remove from its price. A standard exemption, unlike a zero rate, denies the credit on inputs and leaves tax embedded in cost (Ebrill et al., 2001; Keen, 2014). With the credit, a formal seller of eviscerated chicken was, for three months, close to zero-rated. An informal seller of the same eviscerated chicken had nothing to recover, because it had never charged output tax and could not file for a credit. A formal seller whose sales were all exempt gained from the credit only if it could use it after July, against taxable sales or as a refund; otherwise the exemption worked as a standard one.[12] We check in Appendix H that the wholesalers we treat as formal behaved as if they kept it.

2.2 Market stalls

Throughout the paper we treat the stalls of traditional markets as informal: they neither charge nor credit the VAT. Some are registered with the tax authority, so the classification

[9] The law identifies the goods by tariff line: fresh, chilled or frozen chicken meat (headings 0207.11 to 0207.14), fresh hen eggs (0407.21), sugar (1701), uncooked pasta (1902) and bread (within heading 1905).

[10] The press attributed the rise to producers cutting their laying flocks after feed costs rose (Avisur, a southern poultry association, in Infobae, May 28, 2022), and the Peruvian Poultry Association (APA) expected production to fall (El Comercio, June 2022); the Lima retail price reached S/ 9.50 a kilo by June 21 (Infobae, June 21, 2022). The Ministry of Agriculture's poultry bulletin (MIDAGRI, 2023) shows laying hens in production rising from May to June and egg production flat; Appendix J gives the details.

[11] Press summaries of the decree's annex list, for chicken, day-old chicks, live chickens, hard yellow maize, maize starch, fish and soybean oils and soybean-meal preparations; the price panel we use flags live chicken accordingly. Ministry of Economy and Finance press note of April 30, 2022, and La República, May 1, 2022.

[12] The credit also covered only the listed principal inputs, not freight, fuel or packaging.

rests on what stalls do, not on their registration. We document it with a survey we conducted in May 2024 in 42 of the traditional markets in our price data, a third of them (30 in Lima and Callao, 8 in Huánuco and 4 in Ucayali), covering 420 stalls. At 318 stalls, enumerators made a test purchase of a randomly drawn amount between 5 and 10 soles, above the value at which a receipt is legally required, and recorded whether a receipt was issued; they also interviewed the vendors about prices, sales, customers and payment methods, and recorded whether other customers received a receipt during the interview. The survey records how stalls sell, not where they buy.

A receipt was issued on 5 of the 318 test purchases (1.6 percent) and to 14 of the 804 other customers observed (1.7 percent); in 88 percent of markets none of our test purchases received a receipt, and in 81 percent none of the customers we observed did. About 1 percent of purchases are paid by card. Official data show 71 percent of these vendors enrolled in the simplified regime (RUS), which neither charges nor credits the VAT. Another 20 percent are in regimes that charge and credit it; the rest pay only income tax. A stall registered for the VAT that issues no receipts could still credit the tax on invoiced purchases, which would lower the tax its price carries. Our test purchases observe sales, not purchases, and the 71 percent is a national figure for the activity, not one for the surveyed stalls. For over three-quarters of vendors the revenue implied by their own reports of customers and purchase sizes exceeds the regime's ceiling (Appendix D). The few receipts we collected point the same way: one vendor registered in October 2022 had reached receipt number 371 fourteen months later (Figure R3a). Figure E1 and Appendix E summarize the survey.

2.3 Two chicken chains

Chicken is produced by a few large, formal, vertically integrated firms, which control genetics, feed, hatching and growing; we call them integrators. The six largest account for more than 60 percent of national output, and the largest alone for about 30 percent. Integrators sell most of their output as live birds: by industry accounts, about 80 percent of chicken is sold live (Agraria.pe, 2020; Pinedo Cavassa, 2021). Supermarket chains buy eviscerated chicken directly from the integrators on invoice; wholesalers buy live chicken mainly from the integrators. For the stalls, what matters for the exemption is the good sold at the last formal sale in their chain: an eviscerated bird (the exempted good), or a live bird (not exempted). In Lima the answer is known: the formal wholesaler sells the stalls a live bird. In the interior the wholesaler sells them an eviscerated bird, and the question is whether that sale is formal.

In Lima, chicken is traded at large scale as live birds. Integrators deliver them to wholesalers that sell them by the crate: in 2022, 274 million live chickens (743,000 metric tons, 36 percent of national broiler output) were sold in Lima and Callao (MIDAGRI, 2023). The price system quotes the live bird daily at Lima's live-poultry wholesale market,[13] which is the live-bird series we use below, and the ministry's wholesale price for Lima is a live-bird price, with the cost of slaughter added downstream (RPP Noticias, 2025). The slaughter happens before dawn, at the stall itself or in informal sites near the collection centers (Poma Fermín, 2021; Pinedo Cavassa, 2021; Zambrano F. et al., 2013), although the law allows it only in plants authorized by SENASA, Peru's agricultural health service.[14] Lima has authorized slaughter

[13] Its name in Spanish, *mercado mayorista de aves vivas*, means wholesale market of live birds.

[14] Article 32 of Supreme Decree 029-2007-AG, the poultry sanitary regulation. SENASA treats slaughter

plants, six within 15 km of its live-poultry market, but this chain does not go through them; if it did, the price system would record eviscerated chicken at Lima's wholesale markets, and it does not (Table 1). The formal chain therefore ends with the sale of the live bird: the stall's formal input is the live bird, and the exempt good, eviscerated chicken, is created afterward, informally.[15]

Table 1: Chicken products quoted in the price panel, by market type (traditional markets, 2021–2022)

	Wholesale markets		Retail markets	
	Eviscerated	Live bird	Eviscerated	Live bird
Lima and Callao	0 of 7	1 of 7	34	0
Interior	27 of 50	0 of 50	27	0

Number of markets with at least one price quote for the product in 2021–2022. The one Lima wholesale market quoting live birds is the live-poultry wholesale market; the other six trade fruit, vegetables and grains. The panel contains no other chicken product.

In the interior, the wholesaler in the city's wholesale market buys live birds from the integrators' regional distribution centers or from local farms, has them slaughtered at night, and sells the eviscerated birds in the early morning to retailers from the smaller markets. Whether the exemption reaches the stall depends on whether that wholesaler is formal, that is, whether it invoices the sale. A formal wholesaler charged VAT on the eviscerated bird and credited the VAT on the live bird before the exemption, and during it sold without VAT and kept the input credit; the stall's formal input is then the eviscerated bird, the exempted good. An informal wholesaler never charged VAT on the eviscerated bird, so it has nothing to pass on, and the last formal sale in the stall's chain is the live bird, as in Lima. We do not observe the wholesalers' tax status; we proxy it by where they can slaughter. Where an authorized slaughter plant is near the wholesale market, the wholesaler can slaughter legally and invoice the carcass; where none is, the bird cannot be slaughtered legally, so its sale is unlikely to be invoiced.

The price system's product lists show what the wholesaler sells (Table 1): of the seven Lima wholesale markets in the panel, none quotes eviscerated chicken, and the only wholesale chicken series in Lima is the live-bird market; of the 50 interior wholesale markets, the 27 that quote any chicken all quote eviscerated chicken, and none quotes live birds.[16] SENASA's registry of authorized slaughter plants gives the proxy for the interior: ten of the 15 interior cities in our estimation sample have an authorized plant within 15 km of the wholesale market, and in the other five the nearest is 77 to 173 km away (Section 5.2.2 and Appendix H).

elsewhere as informal or clandestine (SENASA, 2021; SENASA, 2020).

[15] The industry association described this configuration three days into the exemption: "the chicken sold in markets is the chicken that farms sell live to wholesalers, and it is not on the list of exempted products, so it is still sold with VAT" (APA, May 4, 2022; our translation). The integrators' invoiced sales of eviscerated chicken go mainly to supermarkets; a minority of stalls supplied by distributors of invoiced eviscerated birds cannot be ruled out, and would be a chain inside the Lima sample whose formal input is the exempted good.

[16] An enumerator's product list is a coverage decision, so a missing quote does not by itself prove that a product is not traded; for Lima, the collection-center tonnage and the industry association's statement point the same way.

Throughout, we treat the stalls as informal (Section 2.2) and the integrators, the supermarket chains, Lima's live-bird wholesalers and the interior wholesalers with an authorized plant nearby as formal.[17] The exemption should therefore have lowered the price of the stall's formal input in the interior cities with a plant, but not in Lima or in the interior cities without one.

3 Theoretical framework

We now present a model of the VAT along a supply chain in which each seller can be formal or informal. Section 3.1 defines a statistic that we can measure from prices, the formal input share of the informal price, δ. Section 3.2 shows that pass-through to informal prices depends on it, under two kinds of tax change, one that falls on every formal input and one that falls only on some. Section 3.3 turns that pass-through into incidence across households, and Section 3.4 applies the predictions to the chicken exemption.

3.1 The formal input share, δ

We start with two questions that need no assumption about how sellers set prices. Where in a chain does the VAT become a cost that no buyer can credit? And, when the last seller is an informal vendor, how much of the price that its customers pay, which we call the informal price, goes to the formal seller from which the vendor buys?

Setup. Each good reaches the consumer through a chain of sellers indexed $j = 0, 1, \ldots, N$. Every sale is one transaction with one seller and one buyer; the final consumer buys from seller N and is indexed $j = N + 1$. There is a uniform VAT rate t on all goods. We consider chains that are entirely formal, entirely informal, or formal up to one stage and informal thereafter.[18]

Each transaction has two prices: the producer price q_j of seller j, which it charges before VAT is added to that sale, and the purchase price p_j paid by its buyer after any VAT credit to which the buyer is entitled. The two differ only by the VAT charged on that transaction, but q_j may still include VAT that the seller could not recover on an earlier purchase. Prices at every stage are expressed per unit of final output, incorporating any conversion between stages.

Begin with a chain in which every seller is formal. Each seller charges VAT tq_j on its sale, credits the VAT invoiced by its supplier, if any, and remits the difference. We assume throughout that credits are fully usable, and we follow the principal chain only, leaving aside other taxed inputs, to which Appendix P.1 returns. The consumer sells nothing, so she cannot claim a credit and pays $p_N = q_N(1 + t)$; a formal seller's credit offsets the VAT charged by

[17] Integrators and supermarket chains are large incorporated firms. For the wholesalers the evidence is indirect. In the interior, the plant is a proxy, not proof, because it slaughters for a fee and the seller is the wholesaler (Appendix H). In Lima, we treat the live-bird wholesalers as formal sellers that invoice the tax; if they are not, the last formal sale is the integrators' sale to them, and Lima's δ is lower by their margin.

[18] Appendix A discusses chains that return to a formal seller after an informal link, where Proposition 1 understates the response; a change on a single good then acts at the last formal sale.

its supplier, so its purchase price is $p_{j-1} = q_{j-1}$. Write both cases at once as

$$p_j = q_j(1 + \tau_{j+1}), \qquad \tau_{j+1} \in \{0, t\}, \tag{1}$$

where τ_{j+1}, the VAT wedge on the purchase by stage $j+1$, equals t when seller j charges VAT and the buyer at stage $j+1$ cannot recover it, and zero otherwise.

An informal seller differs on both sides of its business: it does not charge VAT on its sales, and it does not claim credits, so the VAT charged on its own purchase, when its supplier is formal, remains in its cost. The VAT therefore becomes a tax on the inputs that informal sellers buy from formal ones, as in Keen (2008).

The wedge is therefore t on exactly one purchase in each chain: the final sale when every seller is formal and, in a chain that is formal up to some stage and informal from there on, the first informal seller's purchase from the last formal one. We call that sale the *last formal sale* and index the seller at it j^*. The good sold there, g_{j^*}, is the *formal input* of the informal price: what the vendor buys when its supplier is formal, as for chicken, or, with several informal links, the good sold at the last formal sale further up the chain. Appendix B works through a numerical example with three sellers, a farmer, a wholesaler and a retailer (Table B1).

The formal input share. In a chain with a formal input, the informal vendor pays $p_{j^*} = q_{j^*}(1+t)$ at the last formal sale, which includes VAT that it cannot credit. Define

$$\delta = \begin{cases} \dfrac{p_{j^*}}{p_N}, & \text{if the chain has a formal input,} \\ 1, & \text{if every seller is formal,} \\ 0, & \text{if every seller is informal.} \end{cases} \tag{2}$$

For a chain with a formal input, we call δ the *formal input share of the informal price*: the price paid at the last formal sale relative to the informal price. It is one only if the informal vendor resells what it buys from its formal supplier at the price it paid, below one when the vendor adds a margin of its own, and zero only when no formal seller lies upstream, so that the chain is informal from end to end. In an all-formal chain we set $\delta = 1$; this convention makes relative pass-through one there (Section 3.2), and it makes the revenue share below hold in all three cases. δ is an accounting object: it says how much of the informal price is paid for the formal input, not how prices respond when the tax changes. In Table B1, the chain with an informal vendor has $\delta = 7.08/9.08 = 0.78$.

δ also gives the VAT collected along the chain. In a chain with a formal input, the remittances of the formal stages add up to tq_{j^*} (Appendix A), so VAT revenue as a share of the final price is

$$\frac{tq_{j^*}}{p_N} = \frac{t}{1+t}\frac{p_{j^*}}{p_N} = \frac{t}{1+t}\delta. \tag{3}$$

In an all-formal chain they add up to tq_N and the revenue share is $t/(1+t)$, equation (3) with $\delta = 1$; an entirely informal chain generates no VAT revenue on the principal chain, the case $\delta = 0$.

3.2 Pass-through to informal prices

What should a change in the VAT do to the informal price? Beyond Section 3.1, we need two things: how the price of the formal input responds, and how much of that response the informal vendor passes on, which we take to be one sol for one sol.

Pass-through. Let t_k be the VAT rate on good k. Pass-through of a change in t_k to the final price is the elasticity

$$\rho(k) = \frac{d \ln p_N}{d \ln(1 + t_k)}, \tag{4}$$

where $1 + t_k$ is the tax factor, so that one is full pass-through. In an all-formal chain, the final price is a formal retail price; a change in the tax on the final good falls on the final sale, and we write its pass-through as ρ_F, whose size depends on supply, demand and conduct along the formal chain. In a chain with an informal vendor, the final price is the informal price, and we write its pass-through as ρ_{inf}. *Relative pass-through* is ρ_{inf}/ρ_F: pass-through to the informal price relative to the formal retail price of the same good. This is the parameter that matters for incidence, and the one that Bachas, Gadenne, and Jensen (2024) set to zero.

In a chain with a formal input, the tax enters the informal price through the price of that input, p_{j^*}, with pass-through

$$\rho_{j^*} = \frac{d \ln p_{j^*}}{d \ln(1 + t_{g_{j^*}})}. \tag{5}$$

It depends on supply, demand and conduct in that market, and the chain model does not determine it.

The change in the price of the formal input must then pass through the informal stages after the last formal sale. We assume that they add a margin fixed in soles to the price paid there,

$$p_N = p_{j^*} + m, \tag{6}$$

where the margin m does not change with the tax.[19] We call this the one-sol-for-one-sol assumption: a change in the price of the formal input moves the informal price by the same number of soles, and so by δ times as much in proportion: $d \ln p_N = (p_{j^*}/p_N)\, d \ln p_{j^*} = \delta\, d \ln p_{j^*}$. In a chain in which every seller is informal there is no formal input, and equation (6) holds with $p_{j^*} = 0$, consistent with $\delta = 0$ in equation (2): the informal price is the margin alone.

3.2.1 The general VAT rate

A change in the general rate moves the rate on every good taxed at it, all at once: $d \ln(1+t_k) = d \ln(1 + t)$ for each such k. We write pass-through of a change in the general rate to the final

[19] If instead the informal price moves by λ soles per sol of change in the price of the formal input, every pass-through to the informal price is multiplied by λ: below one if the vendor absorbs part of the change, and equal to the vendor's markup if it marks up its costs in proportion, in which case $\delta\lambda$ is the formal share of its costs.

price and to the price of the formal input as

$$\rho^G = \frac{d \ln p_N}{d \ln(1+t)}, \qquad \rho^G_{j^*} = \frac{d \ln p_{j^*}}{d \ln(1+t)}, \tag{7}$$

and ρ^G_F for pass-through to a formal retail price.

Proposition 1 (A change in the general rate)**.** *Suppose that sourcing and formality do not change with the tax,*[20] *that formal sellers' credits are fully usable, and that the informal stages add a margin fixed in soles, as in equation* (6)*. In a chain with a formal input, suppose that the formal input is taxed at the general rate. Then pass-through of a change in the general rate to the final price is*

$$\rho^G = \begin{cases} \rho^G_F, & \text{if every seller is formal,} \\ \delta\, \rho^G_{j^*}, & \text{if the chain has a formal input,} \\ 0, & \text{if every seller is informal.} \end{cases} \tag{8}$$

Proofs of the propositions and the corollary are in Appendix A.

Equation (8) is a standard result in industrial organization (Nakamura and Zerom, 2010; Bonnet et al., 2013; Weyl and Fabinger, 2013): when a seller passes changes in its cost on one sol for one sol, the percentage change in its price is the percentage change in the price of its input times that input's share of its price. What the VAT adds is which input that is: because formal buyers credit the tax, a change in the general rate reaches the informal price only through the formal input, whatever good it is (Panel A of Table B2).

3.2.2 The VAT rate of a single good

Now change the tax on one good only, with each formal seller keeping the credit on its own purchases. Keeping the credit makes the change a zero rate rather than a standard exemption, which would deny it; under the 2022 credit rule, the exemption we study was close to a zero rate (Section 2.1). In a formal chain, a change on a single good passes through only if it falls on the good sold at the final sale, wherever that good was first produced. In a chain with an informal vendor, it passes through only if it falls on the formal input (Panel B of Table B2). Record whether a change on good k falls on the formal input:

$$R_k = \mathbf{1}\{k = g_{j^*}\}. \tag{9}$$

[20] Over longer horizons, the credit mechanism itself shapes who buys from whom, so that formality tends to run along supply chains (De Paula and Scheinkman, 2010). Both propositions hold that structure fixed.

Proposition 2 (A change on a single good)**.** *Consider a change in the VAT on a single good k. Suppose that sourcing and formality do not change with the tax, that formal sellers' credits are fully usable, including when their own sales become exempt, and that the informal stages add a margin fixed in soles, as in equation (6). Suppose also that the change moves the price of the formal input only through the tax on the chain's own sales. Then, in a chain with a formal input, the local response*[21] *of the informal price is*

$$\rho(k) = R_k \delta \rho_{j^*}. \tag{10}$$

In a chain in which every seller is informal, $\rho(k) = 0$.

Corollary 1 (Which changes on a single good pass through)**.** *Under the assumptions of Proposition 2, in a chain with a formal input, a change on good k passes through only if k is the formal input, and then $\rho(k) = \delta\rho_{j^*}$. If k is sold only before the last formal sale, the credits of the formal sellers that follow offset the change. If k is first produced after it, for example by the informal vendor, the change removes a tax that nobody was paying. In both cases $\rho(k) = 0$, even though $\delta > 0$.*

The VAT already in place passes through to the informal price in every chain with a formal seller upstream, whatever the formal input is.

Where an exemption should sit. Corollary 1 also says where in the chain an exemption reaches informal consumers. In a formal chain it does not matter: exempting an intermediate stage changes nothing, because the next taxable stage recoups the tax. With informal vendors it decides the outcome: an exemption written on the final good misses every chain whose last formal sale is of an earlier good. The input exemption, useless in a formal chain, is then the instrument that reaches informal consumers, provided the exempt seller keeps the credit on its own inputs. Under a standard exemption, which denies that credit, the tax on the exempt seller's own inputs cascades through the formal chain, and only part of the tax is removed from the informal chain.[22]

3.3 Redistribution

Who gains from a change in the tax, in the general rate or on a single good? Measuring it requires only the pass-through in each chain and the share of each household's budget spent through each chain.

Let c denote a chain and p_c its final consumer price, denoted p_N above; chains can differ across goods, locations and retail formats. Let x_h denote household h's total expenditure before the change in the tax, and let s_{hc} denote the share of that expenditure spent on purchases through chain c. For a change on good k, the relevant chains are those that sell k, and $\sum_c s_{hc}$ over

[21] A large change, such as an exemption, also moves δ; the exact response is then equation (10) evaluated at a δ between its values before and after the change.

[22] This is a failure of tax-collection invariance seen from the design side. Slemrod (2008) argues that, with evasion, it matters who remits, and Kopczuk et al. (2016) show that moving the collection point of the diesel tax from retailers to prime suppliers raised pass-through because it reduced evasion. Here the failure is binary rather than continuous, because the friction is not a rate of evasion among firms that file but a sector that never files.

them is the household's total budget share for that good. Let $d\ln(1+t)$ denote the change in the tax factor, on good k or in the general rate, and ρ_c the pass-through of that change to the price in chain c: $\rho_c(k) = d\ln p_c / d\ln(1+t_k)$ for a change on good k, as in equation (4), and ρ_c^G for a change in the general rate.

Suppose that households maximize utility subject to their budgets and that preferences do not change with the tax. Hold nominal household income fixed and, for a change on a single good, the prices of other goods. To first order, by Shephard's lemma (Appendix A), the money-metric gain to household h from the price changes, dB_h, is

$$\frac{dB_h}{x_h} = -\sum_c s_{hc}\, d\ln p_c = -\left[\sum_c s_{hc}\rho_c\right] d\ln(1+t). \tag{11}$$

A tax reduction has $d\ln(1+t) < 0$, so positive pass-through generates a consumer gain. The expression uses initial expenditure shares.

Who benefits more? If pass-through were the same in every chain, only budget shares would matter in equation (11), and a food exemption would favor poorer households because they spend more of their budget on food. Pass-through is not the same: it is zero in every informal chain in the conventional accounting of Bachas, Gadenne, and Jensen (2024), and zero for a change on a single good that misses the formal input (Corollary 1). What matters is the product: a change gives a poorer household ℓ a larger gain relative to expenditure than a richer household h only if $\sum_c s_{\ell c}\rho_c > \sum_c s_{hc}\rho_c$, so supply chains can preserve, weaken, or overturn the progressive effect implied by budget shares alone.

3.4 The chicken exemption through the model

What do the two predictions imply for the chicken exemption? Table 2 places each chain of Section 2.3 in the model. The supermarket chain is formal to the end, so its price responds by formal pass-through, ρ_F. In the ten interior cities with an authorized slaughter plant near the wholesale market, the exemption falls on the stall's formal input, the eviscerated bird bought from a wholesaler we treat as formal ($R_k = 1$), so by Proposition 2 the stall's pass-through is δ times the wholesaler's. In Lima, and in the five interior cities without a plant, the last formal sale is the live bird, which the law did not exempt ($R_k = 0$), so the stall price should not move. This does not mean that no VAT passes through to Lima's stall price, whose formal input share is also positive: the tax on the live bird does, but the exemption left it in place. Section 5 first pools the 15 interior cities. In the five without a plant the wholesaler's price should not move either, because its formal input, the live bird, was not exempted, so the pooled stall should still move by about δ times the pooled wholesaler.

Table 2: The chains in the data and what the model predicts for an exemption on eviscerated chicken

Chain	Last formal seller (good sold)	R_k	δ	Prediction
Chicken, supermarkets	supermarket (eviscerated; final sale)	–	1	ρ_F
Chicken, interior stalls, plant nearby	wholesaler (eviscerated)	1	$\delta < 1$	$\delta\rho_{j^*}$
Chicken, interior stalls, no plant nearby	integrator (live bird)	0	$\delta < 1$	0
Chicken, Lima stalls	wholesaler (live bird)	0	$\delta < 1$	0

$\delta = p_{j^*}/p_N$, the formal input share of the stall's price; Section 4 measures it from price levels. In an all-formal chain $\delta = 1$ by the convention of equation (2). Plant nearby: an authorized slaughter plant within 15 km of the wholesale market (Section 2.3).

4 Data

4.1 Prices

Prices come from the Ministry of Agriculture's price system (SISAP), provided on request by the Ministry.[23] Enumerators record the price of each product variety in 57 wholesale markets (a few of them municipal slaughterhouses) and 61 retail markets in Lima and 27 interior cities,[24] and in 67 supermarkets of one retail group (Tottus and its discount format Precio Uno) in Lima and 24 other provinces (counts for 2021–2022). Wholesale prices in Lima and supermarket prices are recorded almost daily, and wholesale prices in the interior and retail prices in traditional markets on Mondays, Wednesdays and Fridays. For each product the enumerator samples three or four stalls, which change from day to day, and reports the average, so a market's series averages over stalls that may buy through different chains (Section 5.2 returns to what this implies). The panel covers four of the five exempted goods in traditional markets (all but bread) and three in supermarkets (chicken, eggs and sugar).

We use the retail markets for the consumer end of each chain and, for chicken in the interior, the wholesale markets that quote the eviscerated bird, which sell the stalls' formal input where the wholesaler is formal (Section 2.3). Each market reports a retail price, a wholesale price, or both. The treated product is eviscerated chicken; the controls are 23 products that the law did not affect (Appendix G); each cell draws its controls from the products quoted at its own stage. Sugar and pasta, the exempted goods in the panel besides chicken and eggs, are not estimated (Section 2.1).[25] The live-bird price is the daily quotation of Lima's live-poultry wholesale market. Table 3 reports median prices by stage in the week before implementation.

[23] Domestic trade statistics of agricultural and agro-industrial food products, compiled by the Ministry's office of economic and statistical studies and its regional agricultural information offices.

[24] The regional capitals Abancay, Arequipa, Ayacucho, Cajamarca, Cerro de Pasco, Chachapoyas, Chiclayo, Cusco, Huaraz, Huancavelica, Huancayo, Huánuco, Ica, Iquitos, Moquegua, Piura, Pucallpa, Puerto Maldonado, Puno, Tacna, Trujillo and Tumbes; the subregional capitals Andahuaylas, Jaén and Chota; and Juliaca and Tarapoto.

[25] There are three further reasons. The panel records their retail and wholesale prices but no producer stage. Sugar's domestic price rose by half just before the exemption while the world price was flat, which defeats a food-control counterfactual. And packaged pasta shows no movement at any stage, including the formal wholesaler, a configuration the model allows ($\rho_{j^*} = 0$) but cannot use.

Table 3: Price levels by stage, week before implementation (April 24–30, 2022)

Product	Stage	Sector	Soles/kg	Markets
Chicken	Live bird, Lima wholesale (live weight)	formal sellers	4.96	1
	eviscerated-equivalent at 0.75 yield		6.61	
	Wholesale markets, interior	likely formal (10 of 15)	8.50	15
	Stalls, interior	informal	9.50	15
	Stalls, Lima	informal	8.38	26
	Supermarkets, Lima	formal	9.10	31

Medians across markets of market medians, in soles per kilogram. Interior wholesale is treated as formal in the 10 cities with an authorized slaughter plant near the wholesale market. δ is the ratio of two medians: $8.50/9.50 = 0.89$ in the interior and $6.61/8.38 = 0.79$ in Lima, where the live bird is converted at a 0.75 dressing yield; the Lima value is an upper bound if the giblets the stall sells cover part of the bird's cost.

4.2 Household expenditure and other sources

Household data come from the 2022 National Household Survey (ENAHO) of the National Institute of Statistics (INEI). Its food module records, for each product code bought in the previous 15 days, the amount spent and the type of store where it was bought, and its summary file gives total household expenditure. We classify supermarkets, bakeries and restaurants as formal stores and street vendors, corner stores, markets, trucks, fairs and other places as informal (Appendix R). Other places are about 1 percent of purchased food spending. Section 5.3 uses the place of purchase to test for switching between sectors, and Section 6 uses it to compute who gained from the exemption (interviews in January–March 2022, before it) and the tax content of food spending (all 2022 interviews); both exclude the five departments wholly in the Amazon VAT regime (Amazonas, Loreto, Madre de Dios, San Martín and Ucayali). The Ministry of Agriculture's monthly poultry bulletin (MIDAGRI, 2023) reports the tonnage of live chicken sold in Lima and Callao and the farm-gate price of live chicken; the wholesale-market administrations report daily intake by product and province of origin; SENASA's registry of authorized slaughter plants, which records each plant's tax identifier, locates the formal slaughter capacity of each interior city (Appendix H). World prices of maize, soybean meal and chicken are from the World Bank's commodity price data, and the exchange rate and Lima's food price index from the Central Reserve Bank of Peru.

5 Pass-through analysis

5.1 Empirical strategy

We build a weekly panel of log prices at the market–product level, keeping the first quote of each week, and treat a market's retail and wholesale quotes as separate series. For each cell we compare the treated series with a synthetic control built from the same stage's series of the control products, using the synthetic difference-in-differences estimator of Arkhangelsky et al. (2021). Unit weights $\hat{\omega}_i$ choose donor series whose pre-period path matches the treated series in trend, since unit fixed effects absorb levels, and time weights $\hat{\lambda}_t$ choose the pre-period

weeks most predictive of the post-period. The estimate solves

$$(\hat{\tau}^{sdid}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \underset{\tau,\mu,\alpha,\beta}{\arg\min} \left\{ \sum_{i=1}^{N} \sum_{t=1}^{T} (Y_{it} - \mu - \alpha_i - \beta_t - W_{it}\tau)^2 \, \hat{\omega}_i^{sdid} \hat{\lambda}_t^{sdid} \right\} \tag{12}$$

where i is a market–product series, Y_{it} its log price in week t and W_{it} the treatment indicator. We use the event-study version of Clarke et al. (2024), which returns a coefficient for each week relative to the weighted pre-period.

Each cell is a stage in a geography: supermarkets, stalls and the live bird in Lima, and wholesalers and stalls in the 15 interior cities whose price series cover the exemption, excluding Iquitos, which is under the Amazon VAT regime (Appendix H). Weeks run Sunday to Saturday, and week 0 is the first week of the exemption (May 1–7, 2022). We use 2022 and code treatment from week −4, the week after the announcement, to absorb anticipation.

Before estimating, we net out the price of the live bird: each cell's log prices are residualized on the log live-bird price in Lima, with the coefficient estimated on the cell's own series over the untreated weeks −13 to −5. The live bird is the right control because the exemption did not touch it: live birds stayed taxed, so their price carries the chain's common costs, such as the April–May oversupply and the feed-cost increase from June, but not the tax. Appendix L.1 discusses this choice and reports every cell under the alternatives.

Intervals come from a joint cluster bootstrap with 200 draws that resamples interior departments and Lima markets with all their series, treated and control, and re-estimates the live-bird slope in each draw; for the live-bird price, a single series, they come from placebo permutation.

What the estimates are in the model. For each cell c the design returns weekly coefficients $\beta_{c,w}$: the difference in log price between the treated series and their synthetic control in week w, net of the pre-period difference. Averaged over a window, $\bar{\beta}_c$ estimates the change in $\ln p_c$ caused by the exemption, and the exemption changed the tax factor on the exempt line by $\Delta \ln(1 + t_k) = -\ln(1.18)$. Dividing one by the other gives the model's elasticity for that price, $\hat{\rho}_c = \bar{\beta}_c / \Delta \ln(1 + t_k)$, which is $\rho(k)$ of equation (4) evaluated at stage c. Table 2 says which object each cell estimates. Two cells are not in it: interior wholesale, the price of the interior stalls' formal input, whose estimate is ρ_{j^*}, and Lima's live bird, the formal input of Lima's stalls, which was not exempted. Tables report $\hat{\rho}_c$ and the event-study figures its weekly counterpart, $-\beta_{c,w}/\ln(1.18)$, so that zero is no response and one is full pass-through. We report two horizons, the first month (weeks 0–3) and the exempt window (weeks 0–12); Appendix I reports the weeks after repeal.

5.2 Results

Table 2 gives the predictions. Supermarkets and the interior wholesaler should fall by their own pass-through, ρ_F and ρ_{j^*}; the interior stalls by $\delta\rho_{j^*}$, the test of the one-sol-for-one-sol assumption; and Lima's stalls not at all.

Table 4 and Figure 2 report pass-through at each stage, with Lima in the left panel of the figure and the interior in the right. We estimated many alternative specifications: other

Table 4: Pass-through by stage, chicken

Stage	Units	First month	Exempt window
Supermarkets, Lima	31	0.79 [0.72, 0.85]	0.52 [0.43, 0.60]
Wholesale, interior	15	0.58 [0.34, 0.87]	0.39 [0.14, 0.63]
Stalls, interior	15	0.56 [0.33, 0.71]	0.38 [0.09, 0.59]
Stalls, Lima	26	0.16 [0.10, 0.22]	0.07 [0.00, 0.16]
Live bird, Lima	1	0.32 [−1.04, 1.68]	−0.38 [−1.89, 1.12]

Pass-through is $-\bar{\beta}/\ln(1.18)$: zero is no response and one is full pass-through. First month: weeks 0–3; exempt window: weeks 0–12. Units are treated market–product series. Brackets: percentile 95 percent intervals of the joint cluster bootstrap; for the live bird, a single series, placebo-permutation intervals.

residualizations, other donor pools, a plain difference-in-differences, and a panel that ends at week 20. The gap between the interior and Lima stalls holds in every one that nets out input costs; Appendix K and Appendix L report each one.

5.2.1 Supermarkets

Supermarkets, at the end of an all-formal chain, passed through 79 percent of the tax in the first month and 52 percent over the exempt window (Table 4). This is what the model predicts for a chain that is formal to the end: the supermarket charged the VAT on its sales, and the exemption removed it. The response was largest in the first month, when the exemption drew the most attention: the government charged the consumer-protection agency, Indecopi, with monitoring the prices of the exempt goods in markets and supermarkets, and the press reported its price checks through the first weeks of May.[26]

5.2.2 Interior stalls

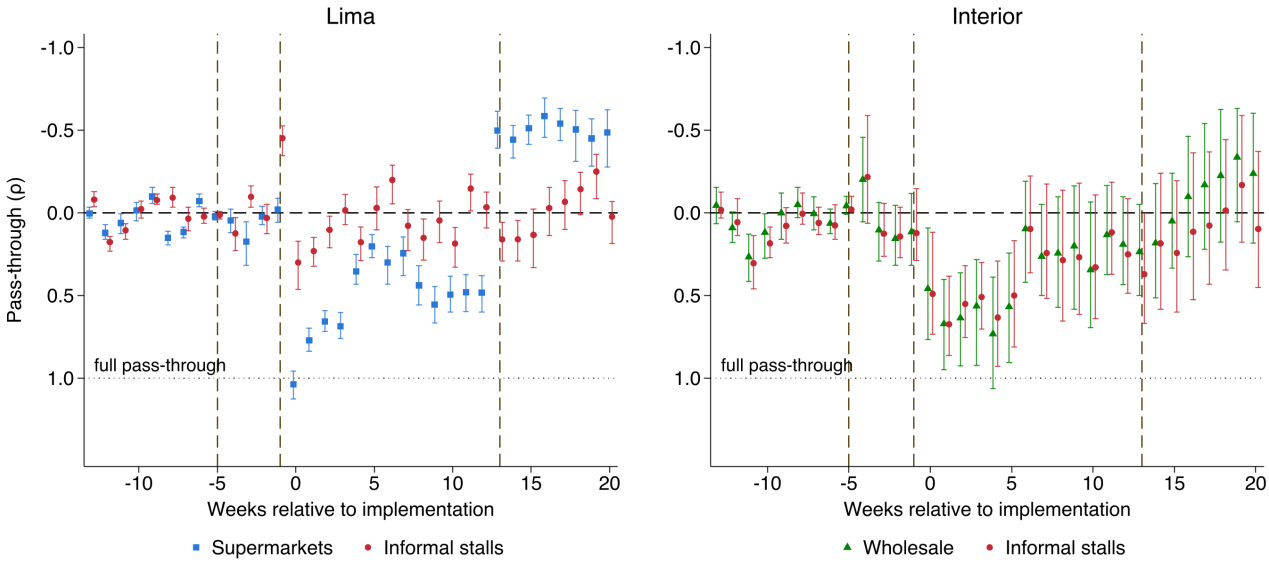
In the interior, the wholesaler, who sells the stalls' formal input where it is formal, passed through 58 percent of the tax in the first month and 39 percent over the exempt window, and gave the cut back after repeal; this is the pass-through of the formal input's price, ρ_{j^*}. It is not statistically different from what supermarkets passed through (95 percent intervals of the difference: [−0.09, 0.47] in the first month and [−0.13, 0.42] over the window).

The interior stalls that the wholesaler supplies passed through 56 percent in the first month and 38 percent over the window (Table 4). This is the model's test. Proposition 2 predicts the stalls' pass-through from two things we observe, the wholesaler's response and the formal input share, $\delta = 0.89$ (Table 3): 0.52 [0.31, 0.77] in the first month and 0.35 [0.12, 0.56] over the window. The prediction and the direct estimates are statistically indistinguishable.[27]

[26] Indecopi's first round covered 86 markets, 30 supermarket stores and 81 bread outlets in Lima and 18 regions (El Peruano, May 2, 2022; Infobae, May 17, 2022).

[27] Because the exemption is a large change, the exact prediction uses a δ between its values before and after it; with the margin fixed in soles and full pass-through to the wholesale price, δ falls from 0.89 to 0.88, and the point predictions change by less than 0.01. Because each market's series averages three or four stalls that rotate (Section 4), the predicted response is $\phi\delta\rho_{j^*}$, where ϕ is the share of the sampled stalls that buy the exempt product from a formal seller; with $\phi < 1$ the prediction lies below these values.

Figure 2: Event study on the price of chicken



Left, Lima: supermarkets, an all-formal chain (estimate ρ_F), and the stalls, whose formal input is the live bird ($R_k = 0$; model predicts 0). Right, interior (all 15 cities; Figure 3 splits them by plant): the wholesale stage, which sells the stalls' formal input, the exempted product, where the wholesaler is formal ($R_k = 1$; estimate ρ_{j^*}), and the stalls it supplies (model predicts $\delta\rho_{j^*}$). Weekly pass-through, $-\beta_{c,w}/\ln(1.18)$, where $\beta_{c,w}$ is the treated-minus-synthetic log price difference, with 95 percent joint cluster bootstrap intervals; zero is no response and one is full pass-through. Vertical lines: the week of the announcement (-5), the last week before the exemption (-1) and the week of repeal (13).

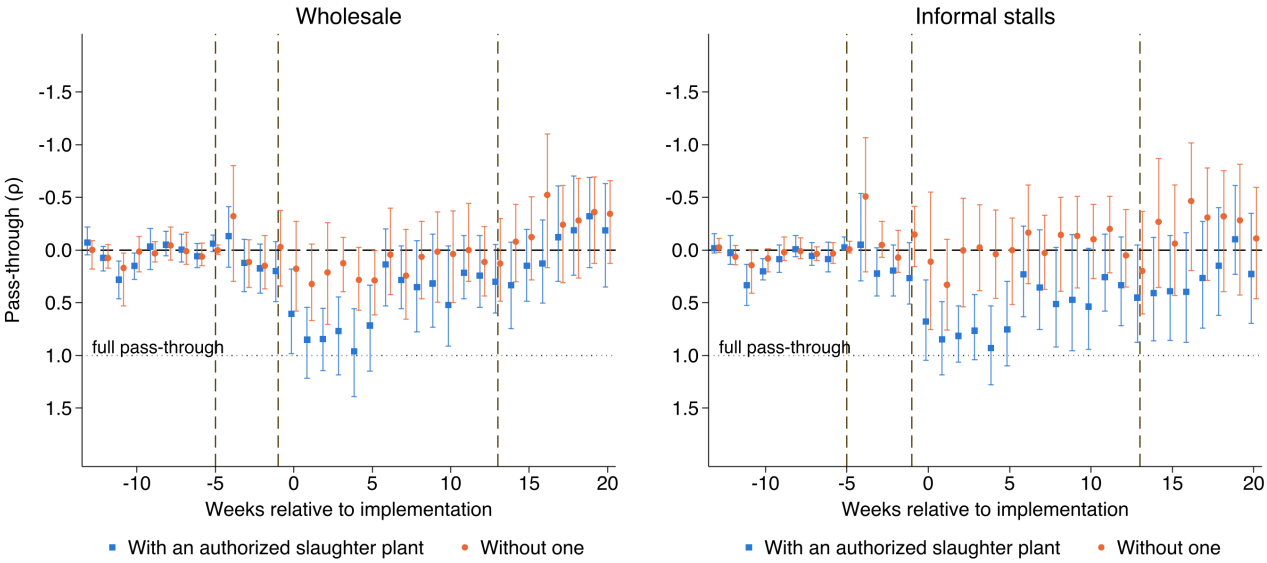
Appendix L.1 reports how these estimates change with the residualization.

Cities with and without an authorized slaughter plant. We split the interior cities by whether an authorized plant is near the wholesale market (Section 2.3); since the plants are either within 15 km or at least 77 km away, any threshold in between gives the same split.[28] In the five cities without one, the bird cannot be slaughtered legally, so the wholesaler's sale is likely informal, and the last formal sale in the stalls' chain is the live bird, as in Lima.

In the ten cities with a plant, the stalls passed through 0.78 [0.46, 1.04] of the tax in the first month and 0.58 [0.19, 0.88] over the window, and the wholesaler 0.77 [0.48, 1.12] and 0.52 [0.19, 0.81] (Figure 3). In the five cities without one, neither stage shows a statistically significant response. A wholesaler trucking in carcasses from a distant authorized plant would have passed on the exemption, so the absence of a response points to an informal wholesaler. The interior result is therefore driven by the cities with a plant, where the wholesaler is likely formal. The stalls' pass-through over the window also falls with distance to the nearest plant (Spearman rank correlation -0.55, $p = 0.021$), and the gap between the two groups' stalls survives controls for city size, coastal location, supermarket spending and distance to the main broiler-producing centers (Appendix H).

[28] We defined the split after seeing the estimates for the interior as a whole; the rule uses only the registry.

Figure 3: Heterogeneity within the interior: cities with and without an authorized slaughter plant



Interior cities with an authorized slaughter plant within 15 km of the wholesale market by 2022 (10) and with none within 77 km (5). The design of Figure 2 estimated separately for each group; weekly pass-through with 95 percent joint cluster bootstrap intervals; zero is no response and one is full pass-through.

5.2.3 Lima stalls

Lima's formal input is the live bird, which was not exempt. Estimated with the same design against Lima wholesale donors, its price shows no statistically significant response to the exemption (Table 4, last row).

Lima stalls, in turn, dipped for two weeks around implementation, passing through 16 percent of the tax in the first month, and only 7 percent over the window, an estimate whose 95 percent interval reaches zero, [0.00, 0.16]. The interior stages, by contrast, sustain the cut through the window. Because the bootstrap draws the interior and Lima series together, the contrast between the two stalls can be tested rather than read off overlapping intervals: the interior stall sits above the Lima stall by 0.31 [0.03, 0.54] over the window (0.39 [0.16, 0.57] in the first month).

Lima is the cell most exposed to the choice of residualization, because its informal price tracks the live bird and the live-bird price dipped around implementation. What survives every residualization that nets out input costs is the contrast: over the window the interior stall sits above the Lima stall by 0.26 to 0.34 of full pass-through (Appendix L.1).

5.3 Alternative explanations

Three stories could lower informal prices without the tax reaching the informal chain: a seasonal fall in chicken prices, customers switching to supermarkets, and stalls cutting their margins to compete with them. The data support none of them.

5.3.1 Seasonality

If chicken prices fell relative to other foods every May, or if the design produced a May decline on its own, the estimates would not measure the tax. Figure 4 runs the same design on 2021, with a fictitious exemption on May 1, 2021, the same donors and the live-bird slope re-estimated on 2021's untreated weeks. In 2021 no line was exempted, so the model predicts no response at any stage. No stage shows a cut: if anything, chicken rose relative to the control products after May 2021 at every stage, and the two-week Lima dip of 2022 does not appear. The same holds with no residualization at all.

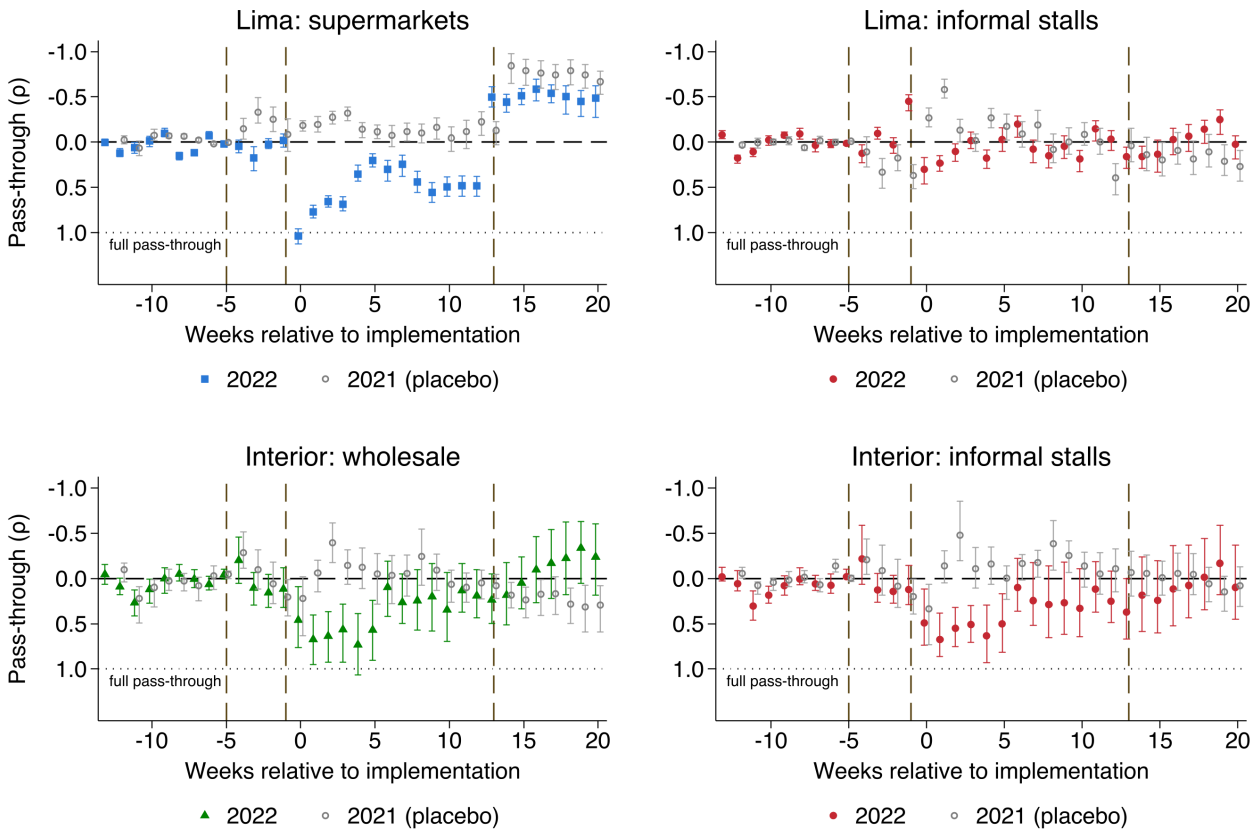
5.3.2 Switching to supermarkets

If informal prices fell because customers moved to supermarkets, the formal share of chicken purchases should have risen during the exemption. In household data it did not change significantly relative to control products: the upper end of the 95 percent interval of its change is 1.5 percentage points, and standard estimates of substitution between store types predict a rise of only 0.3 to 0.5 points. Producing the informal response we measure through switching would require a rise of 3.9 points, if every sol of spending that leaves the stalls goes to supermarkets (Figure 5 and Table F1). Appendix F.1 derives this benchmark and shows that switching is rejected as the full explanation for informal demand elasticities of about one half or more.

5.3.3 Competition with supermarkets

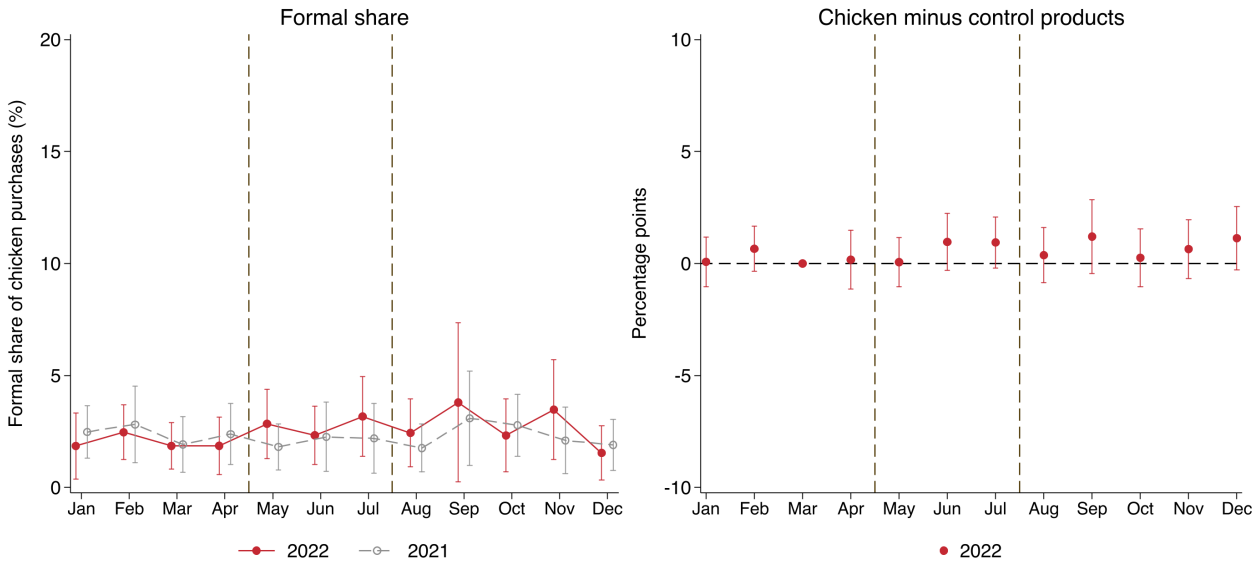
If stalls cut prices to compete with supermarkets, Lima markets close to a supermarket should have responded more than those far from one. They did not: in the first month close and far markets fell by the same amount, 0.15 and 0.16 of the tax, and no difference between them is statistically significant (Figure 6, right panel, and Panel B of Table F1). This rules out competition that decays with distance. It does not rule out a city-wide response to a heavily reported price cut, which may account for Lima's brief dip (Appendix F.2). The test can be run only in Lima, where supermarkets are dense enough to split markets by distance.

Figure 4: Alternative explanation 1: seasonality



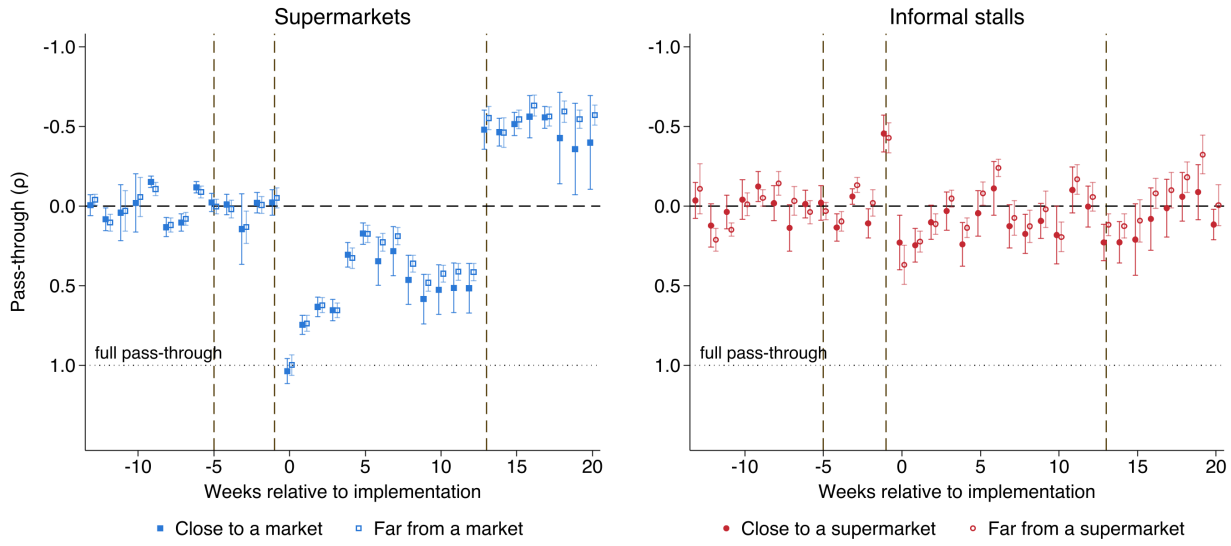
Each stage in 2022 (color, as in Figure 2) and in 2021 (hollow gray), with a fictitious exemption on May 1, 2021, the same donors and the live-bird slope re-estimated on 2021's untreated weeks. Weekly pass-through with 95 percent joint cluster bootstrap intervals; zero is no response and one is full pass-through. Vertical lines: the week of the announcement (−5), the last week before the exemption (−1) and the week of repeal (13), at the 2022 dates.

Figure 5: Alternative explanation 2: switching to supermarkets



Left: formal share of chicken purchases (quantity) by month in 2022 and 2021, with 95 percent confidence intervals. Right: difference between chicken and the 13 control products in the formal share, by month (reference: March), with 95 percent confidence intervals clustered by primary sampling unit. Dashed lines: the exempt window. ENAHO food module (15-day recall), survey weights.

Figure 6: Alternative explanation 3: competition between supermarkets and informal stalls



Left: Lima supermarkets within 2.10 km of the nearest informal market of the price panel (the nearest 40 percent, 13 supermarkets) and beyond (18). Right: Lima informal stalls within 0.46 km of the nearest chain supermarket (10 markets) and beyond (16). Weekly pass-through of the market-level gaps to the shared synthetic control of Figure 2, with 95 percent intervals from the dispersion across markets; zero is no response and one is full pass-through.

6 Redistributive analysis

This section moves from what the exemption measured to what it implies: who gained from the exemption, using measured responses and one prediction of the model, and what it would have done had it also named the live bird (Section 6.1); and who would gain from a cut in the general rate, which rests on the most assumptions (Section 6.2).

Poorer households buy a larger share of their food from informal vendors. In ENAHO's first quarter of 2022, the informal share of purchased food spending falls from 99 percent in the bottom expenditure decile to 75 percent in the top (Figure 7).[29] For chicken, formal stores matter even less: supermarkets account for almost none of the chicken spending of the bottom half of the expenditure distribution and for 11 percent of the top decile's (Table 5). The conventional accounting of Bachas, Gadenne, and Jensen (2024) assigns the tax by store type, so that an informal purchase carries none; we call it the sales-tax view, because it treats the VAT as a retail sales tax collected only by formal stores. We call the accounting that also charges an informal purchase the VAT on its formal input the chain view.

6.1 Who gained from the exemption

Equation (11) turns the estimated price responses into gains by household. To first order, the gain from the exemption relative to household h's total expenditure is

$$\frac{dB_h}{x_h} = \ln(1.18) \sum_c s_{hc}\, \hat{\rho}_c,$$

where c indexes the chains through which households buy chicken, s_{hc} is the household's chicken spending in chain c as a share of its total expenditure, and $\hat{\rho}_c$ is the estimated response of that chain's price. No pricing rule and no counterfactual chain is needed: every $\hat{\rho}_c$ but one is measured, and that one is the model's prediction. We use the estimates of Table 4 and Section 5.2.2: 0.52 at supermarkets, 0.07 for informal retail in Lima, and, outside Lima, 0.58 in provinces with an authorized slaughter plant. In provinces without one we set the response to zero, as the model predicts: their last formal sale is not the eviscerated bird, so the law did not reach it. The estimate there, -0.02 $[-0.27, 0.38]$, is not statistically different from zero.[30] Just over half of informal chicken spending outside Lima is in provinces without a plant.

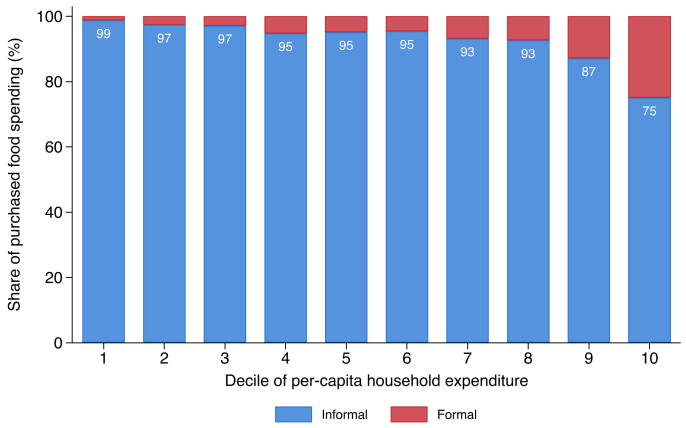
Table 5 reports the result by decile of per-capita household expenditure (Appendix M plots it). Relative to expenditure, the exemption's relief was largest in the middle of the distribution. Over the exempt window the bottom decile gained 0.066 percent of its expenditure, the top decile 0.058 and the second to sixth deciles about 0.11. The bottom half of the distribution received 31 percent of the relief [27, 35]:[31] more than its 26 percent share of total expenditure,

[29] The gradient is steeper for vendors outside our price data, such as street vendors, corner stores and fairs: 64 percent of the bottom decile's food spending and 19 percent of the top decile's.

[30] With that estimate in place of zero, the bottom decile's gain is 1.08 times the top decile's and the bottom half receives 31 percent of the relief, but the interval for that share widens to [6, 38], because these provinces hold 37 percent of the bottom half's chicken spending against 21 percent of the top half's.

[31] Brackets: 95 percent intervals. We resample the January–March households by survey cluster within strata 200 times, recomputing the expenditure deciles each time, and combine each sample with each of 199 draws of the joint bootstrap of the pass-through estimates. For each decile's gain, most of the width comes from the pass-through estimates. For the bottom half's share they matter less, because they raise or lower every decile's

Figure 7: Share of food spending at informal and formal stores, by expenditure decile



Purchased food, by decile of per-capita household expenditure. Formal stores: supermarkets, bakeries and restaurants. ENAHO, January–March 2022, survey weights.

but less than the 50 percent a per-capita transfer of the same cost would give. Under the sales-tax view, only supermarket purchases gain: the bottom decile gets nothing, the top decile 0.013 percent of expenditure, and the bottom half 4 percent of the relief [0, 12] (Table 5, last column), so the exemption would be a transfer to the richest households.

Two features of the chicken budget explain the shape, and they pull in opposite directions. Poorer households spend a larger share of their budget on chicken, about 3.9 percent in the second to fifth deciles against 1.7 percent in the top decile. But less of their chicken is bought where the exemption arrived: 18 percent of the bottom decile's chicken spending is in supermarkets or in provinces with an authorized slaughter plant, against 31 percent of the top decile's, and half of the bottom decile's is in provinces without a plant; per sol of chicken bought, the relief rises from 2.1 percent in the bottom decile to 3.4 in the top (Figure M2).

The bottom decile's gain is 1.14 times the top decile's over the window and 1.09 times in the first month, but neither ratio can be distinguished from one ([0.67, 1.61] and [0.76, 1.48]); the bottom half's share is the more precise measure, and it is 31 percent in the first month too. If rural households, whose chains the no-plant evidence suggests the exemption did not reach, gain nothing, the bottom decile's gain is 0.77 times the top decile's over the window and in the first month. If pass-through were symmetric, a comparable increase in the VAT on chicken would also weigh more, relative to expenditure, on the bottom half of the distribution than on the top half; at the repeal it was not fully symmetric (Appendix I).

Had the law exempted the live bird. Had Law 31452 also exempted live birds, Lima's stalls would have been reached: their formal input would have been exempt, and by Proposition 2 their price would have fallen by $\delta\rho_{j^*}$, about 0.41 of the tax over the window ($\delta = 0.79$, with ρ_{j^*} from the wholesaler in the plant cities), against the 0.07 measured under the law.

gain together: survey sampling alone gives [29, 35], and the pass-through estimates alone [28, 32] (Appendix M).

Table 5: Who gained from the exemption, by decile of per-capita household expenditure

Decile	Chicken, % of expenditure	Chicken spending by chain (%)				Gain, % of expenditure (window)	
		Super-markets	Informal, Lima	Informal, plant	Informal, no plant	Our estimates	Sales-tax view
1	3.2	0.0	30.3	18.5	51.2	0.066	0.000
2	3.9	0.0	31.4	24.6	44.1	0.104	0.000
3	3.9	0.3	40.2	27.6	31.9	0.121	0.001
4	3.9	0.9	46.1	22.3	30.7	0.106	0.003
5	3.8	0.0	40.9	23.5	35.7	0.100	0.000
6	3.6	0.3	43.3	31.7	24.7	0.126	0.001
7	3.2	1.2	47.8	24.3	26.6	0.094	0.003
8	3.2	0.4	55.6	22.5	21.6	0.090	0.001
9	2.5	3.4	52.6	24.6	19.5	0.081	0.007
10	1.7	10.8	54.2	20.1	15.0	0.058	0.013
Bottom/top decile ratio						1.14 [0.67, 1.61]	0.00
Bottom half's share of relief (%)						31 [27, 35]	4 [0, 12]

ENAHO 2022, households interviewed January–March; deciles of per-capita household expenditure. Informal: markets, corner stores, street vendors, trucks, fairs and other places. Plant / no plant: provinces with or without an authorized slaughter plant by 2022. Gain = $\ln(1.18)\sum_c s_{hc}\hat{\rho}_c$ over the exempt window. Our estimates: 0.52 at supermarkets, 0.07 for informal retail in Lima, 0.58 in provinces with a plant and zero in provinces without one, the model's prediction (the estimate there, −0.02, is not statistically different from zero). Sales-tax view: supermarkets at 0.52 and every informal purchase at zero. Brackets: 95 percent intervals combining survey sampling and the sampling error of the pass-through estimates; the sales-tax view's share does not depend on the pass-through, so its bracket reflects survey sampling alone (Section 6.1).

Nothing would change in the interior provinces with a plant, which the law already reached through the eviscerated bird. Whether the extra relief would favor poorer households depends on who buys through the newly reached chains. Lima's informal markets serve rich and poor alike: chicken bought there is 0.9 percent of expenditure in both the bottom and the top decile, so reaching Lima alone would leave the bottom/top ratio near one, as under the law. The provinces without a plant are poorer: chicken bought there is 1.7 percent of the bottom decile's expenditure and 0.3 percent of the top decile's. If their wholesalers buy invoiced live birds, the exemption would reach them too, and the bottom decile's gain would rise to 1.75 times the top decile's; if they buy from farms outside the VAT, nothing changes there, and we do not observe which. Exempting the live bird, with or without the eviscerated bird, would cost 1.6 to 3.4 times the revenue the law gave up (Table C1). What matters for poorer households is therefore not which good is exempted but whether the exemption falls on the last formal sale in the chains that supply the informal stores where they buy.

6.2 Who would gain from a cut in the general rate

A cut in the general VAT rate would lower every price with a formal seller upstream, not only chicken's. By equation (11), who would gain depends on budget shares and on the pass-through in each chain. For the food basket as a whole we need two inputs. The first is the pass-through of formal prices: as Bachas, Gadenne, and Jensen (2024) do, we assume that formal prices, including the price of the formal input, pass the tax through fully. Proposition 1 then gives pass-through δ_g to the informal price of good g, rather than zero. The second is each good's formal input share δ_g, which we assign below. For chicken we can check the result: over the window the interior stalls' relative pass-through was 0.74 [0.17, 1.25], an interval that includes $\delta = 0.89$ and excludes zero.[32] Under these assumptions, equation (11) says that a household's gain from a cut is proportional to its formal budget shares plus δ_g times its informal ones, which is, up to the rate, the VAT that its food purchases carry today. Setting $\delta_g = 0$ for every good gives the sales-tax view; the chain view uses δ_g.

Each good gets its δ_g in Table C2 (Appendix P documents the assignment): the wholesale-to-informal-retail price ratio where the price panel has the good, 0.85 for other goods with a formal establishment upstream, and zero where there is none or we cannot classify the chain. The zero makes each decile's rate a lower bound and, because poorer households buy more of these goods informally, lowers their rates more than the top decile's: giving the unclassified goods, 14 percent of purchased food, the default 0.85 when bought informally moves the top-to-bottom ratio from 0.76 to 0.71 (Appendix P). It therefore works against our finding that the VAT on food is less progressive than the sales-tax view implies. For chicken, δ combines Lima's live bird (0.79), the interior provinces with an authorized slaughter plant (0.89) and those without one, where we convert the farm-gate live-bird price, which the ministry reports only for Lima, to eviscerated weight and divide it by those cities' own stall price (0.74). Each good carries its statutory rate.

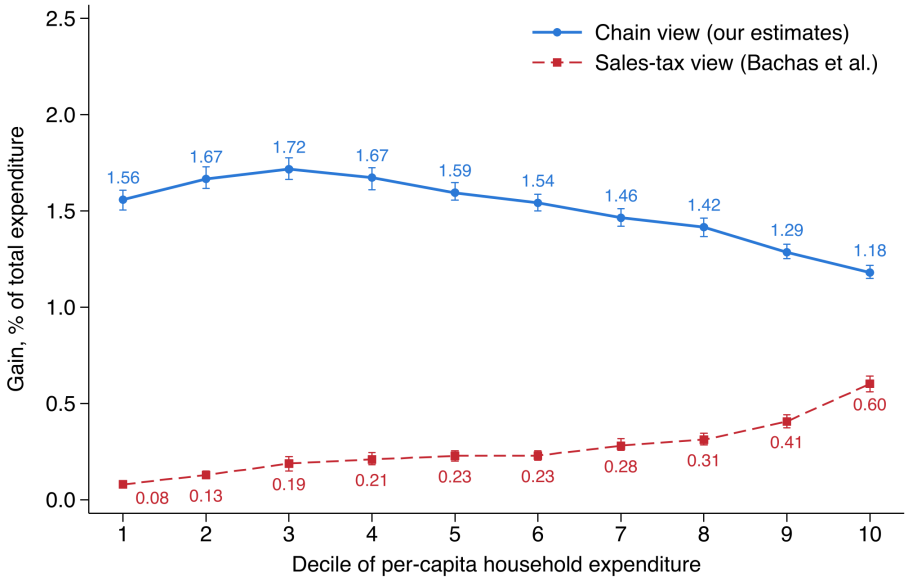
Relative to total expenditure, the VAT carried on purchased food under the chain view is 1.56 percent in the bottom decile [1.51, 1.61] and 1.18 percent in the top [1.15, 1.22] (Figure 8), so the top decile's rate is 0.76 times the bottom decile's [0.73, 0.79].[33] A cut in the general rate would give the bottom half of the distribution 30.7 percent of the relief [30.1, 31.3], more than its 26 percent share of total expenditure. Under the sales-tax view the VAT carried rises from 0.08 percent of expenditure in the bottom decile [0.06, 0.10] to 0.60 in the top [0.56, 0.64], and the bottom half would receive 13 percent of the relief [12, 14].

Two things shape the chain-view profile. Per sol of food purchased, the rate is nearly flat across deciles and rises only in the top one (Figure M3), because an informal purchase carries most of the tax a formal one does, δ between 0.6 and 0.9 wherever a formal seller is upstream. Relative to total expenditure it then falls toward the top of the distribution, because food is a larger share of poorer households' budgets, 31 percent in the bottom decile against 19 in

[32] In the first month it was 0.70 [0.43, 0.93]. The interior wholesaler passed through 0.74 [0.42, 1.12] as much as supermarkets in the first month and 0.76 [0.26, 1.25] over the window, so the stall's ratio is close to δ times the wholesaler's, as the model implies.

[33] Brackets on the food and whole-VAT profiles: 95 percent intervals for survey sampling alone, from 500 samples that resample households by survey cluster within strata and recompute the expenditure deciles. Each good's statutory rate and formal input share (other than chicken's, an average over its chains recomputed in each sample) and full pass-through to formal prices are taken as given; Appendix P varies the formal input shares.

Figure 8: Gain from removing the VAT on food, by expenditure decile



Gain from removing the VAT on purchased food, percent of total household expenditure; with full pass-through it equals the VAT that food purchases carry. Sales-tax view: $\delta_g = 0$ for every good, our construction of the premise of Bachas, Gadenne, and Jensen (2024) with our store classes and deciles (Appendix Q compares it with theirs); chain view: δ_g of Appendix P; each good at its statutory rate (goods on the VAT law's list of exempt goods carry none, and rice carries a 4 percent tax under its own regime instead of the VAT). Bars: 95 percent intervals for survey sampling alone (Section 6.2). ENAHO 2022, survey weights; deciles of per-capita household expenditure, person-weighted.

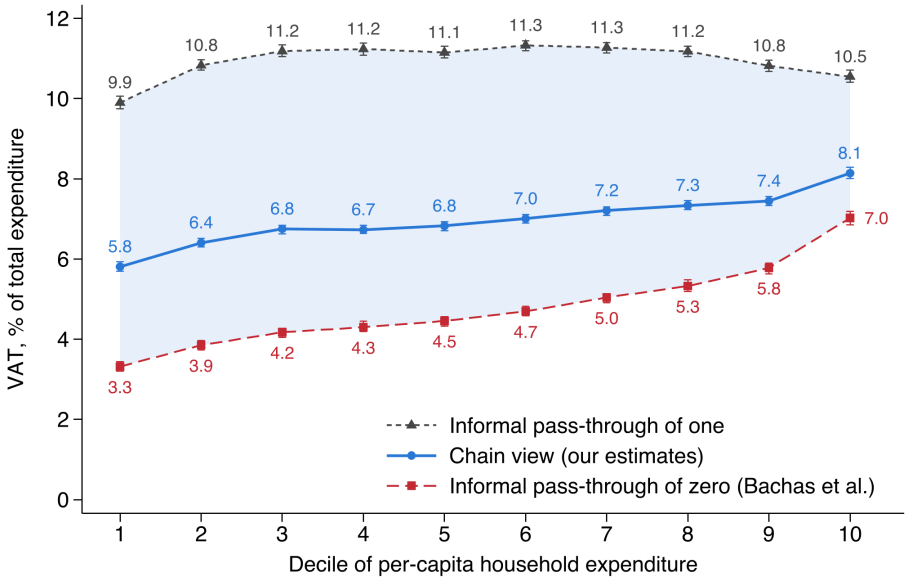
the top. Under the sales-tax view the rate per sol of food rises with the formal share, from 0.26 percent to 3.46, and that outweighs the food share.

For the VAT on all consumption we cannot assign δ_g good by good, but any informal market purchase carries between none and all of the tax, and consumption that households do not buy, such as their own production, carries none. Figure 9 uses this to bound the gain from removing the whole VAT, relative to total expenditure.[34] Under the sales-tax view the top decile's rate is 2.1 times the bottom decile's [2.0, 2.2]. In our central case, with informal food at δ_g and informal manufactured goods at 0.85, it is 1.40 times [1.36, 1.44]. When every informal purchase carries the full tax it is 1.07 [1.04, 1.09], close to proportional, and the remaining gradient comes from consumption that is never taxed: what households do not buy, larger for poorer households, outweighs rent, larger for richer ones. The bottom half of the distribution pays between 20 and 26 percent of the VAT, against 26 percent of total expenditure. The VAT as a whole thus remains mildly progressive, but much less so than the sales-tax view implies.

The food-basket profile rests on many assumptions, which Appendix P varies one at a time

[34] Consumption by place of purchase as constructed by Bachas, Gadenne, and Jensen (2024) for Peru (Appendix Q), at a uniform 18 percent rate. Spending outside their measure other than rent, mostly food eaten away from home, carries the household's own rate, and rent carries none.

Figure 9: Gain from removing the whole VAT, by expenditure decile



Effective VAT rate, percent of total household expenditure; with full pass-through to formal prices, the gain from removing the VAT. Shaded band: every share of the tax an informal market purchase could carry, from none (informal pass-through of zero, the sales-tax view) to all (informal pass-through of one). Chain view: informal food at δ_g, informal manufactured goods at 0.85, services from individuals and non-market consumption at zero. The shaded band is a range of assumptions about informal pass-through, not a confidence interval; the bars on each line are 95 percent intervals for survey sampling alone (Section 6.2). ENAHO 2022; deciles of per-capita household expenditure, person-weighted.

and in combination. Relative to total expenditure, the top decile's rate stays between 0.66 and 0.93 times the bottom decile's across alternative assignments of δ_g, and the bottom half pays 29 to 32 percent of the VAT on purchased food, against 26 percent of expenditure. Only two extreme cases reverse the slope: if chicken were the only good whose informal price carries any VAT (1.66, with the bottom half paying 25 percent), and if three pessimistic assignments hold at once (1.07, with the bottom half still paying 27 percent). Under the sales-tax view the ratio is 7.6 [6.0, 9.4]. With the pass-through we measure for chicken applied to every formal price instead of full pass-through, the gain roughly halves, and the bottom half still receives 31 percent of it. On the object of Bachas, Gadenne, and Jensen (2024), all consumption net of rent, the chain-view correction removes a third to a half of their gradient (Appendix Q). We measure incidence relative to expenditure, as they do, because survey expenditure is better measured than income and closer to lifetime resources; relative to income every profile falls more heavily on poorer households, mostly because many households in the bottom income decile report spending more than their income (Appendix N).

7 Conclusion

Recent work concludes that consumption taxes can be progressive in developing countries because poorer households buy from informal vendors who do not remit the VAT. That conclusion rests on a premise about prices: that the informal purchase carries none of the tax. We show that it carries the tax charged at the last formal sale in its chain, in proportion to the formal input share of the informal price.

Peru's 2022 chicken exemption tests this. Where a formal wholesaler sold the exempt eviscerated bird to informal stalls, as is likely in the interior cities with an authorized slaughter plant, the stall price fell with the wholesale price, by about as much as the formal input share predicts. Where the last formal sale was the live bird, which the law left taxed, as in Lima, the stalls showed no sustained response. Seasonality, switching to supermarkets and competition with them do not explain the difference.

These results change who bears the VAT. The VAT on chicken weighed more, relative to expenditure, on the bottom half of the distribution than on the top half: the bottom half received 31 percent of the exemption's relief [27, 35], against about a quarter of total expenditure and the 50 percent a per-capita transfer would have given; under the premise, it would have received 4 percent. For purchased food, the premise of no pass-through makes the VAT steeply progressive relative to expenditure; with informal prices carrying the tax on their formal inputs, it is mildly regressive. For the VAT on all consumption, which we can only bound, the tax remains mildly progressive: the top decile's rate is 1.4 times the bottom decile's in our central case, against 2.1 under the premise, and close to proportional when every informal purchase carries the full tax.

The same logic says where a change in the tax reaches informal consumers. A change in the general rate reaches every chain with a formal seller upstream; a change on a single good reaches only the chains whose formal input is that good. An exemption written on the final good, as Peru's was, misses chains whose last formal sale is of an earlier good, such as the live bird in Lima; there, only an exemption on the input reaches informal consumers. A tax cut meant to help poorer households should therefore fall on the last formal sale in the chains that supply the informal stores where they shop, which need not be the good they buy. In Peru, exempting the live bird would have reached Lima's stalls, but Lima's informal markets serve rich and poor alike, so it would not have made the relief more pro-poor.

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Appendices

Appendix A Proofs and derivations

Chains that return to a formal seller. In a chain that returns to a formal seller after an informal link, such as a formal producer, an informal trader of live birds and a formal wholesaler that has the bird slaughtered and sells it eviscerated, the formal seller after the informal link charges VAT on a price that already includes tax it could not credit, so the general rate enters twice and Proposition 1 understates the response; a change on a single good then acts at the last formal sale.

Derivation of equation (3). Each formal seller invoices tq_j on its sale. A formal seller at $j \geq 1$ credits the tq_{j-1} invoiced by its supplier and remits the difference. The first seller has no taxed input in the principal chain and remits tq_0. In a chain with a formal input, the remittances of the formal stages therefore telescope, $tq_0 + \sum_{j=1}^{j^*} t\,(q_j - q_{j-1}) = tq_{j^*}$, and dividing by p_N and using $p_{j^*} = q_{j^*}(1+t)$ gives equation (3). In an all-formal chain there is no last formal sale to an informal buyer: VAT becomes unrecoverable at the final consumer, and total remittances equal tq_N. □

Proof of Proposition 1. In a chain with a formal input, every buyer before the last formal sale is formal and credits the VAT on its purchase, so by equation (1) its purchase price is $p_j = q_j$. The rate therefore enters no purchase price before the last formal sale except through the response of producer prices, which $\rho^G_{j^*}$ includes. At the last formal sale, $p_{j^*} = q_{j^*}(1+t)$, and because the general rate applies to every good taxed at it, it applies to g_{j^*}, whatever that good is. After it, no seller charges VAT, so the rate enters p_N only through p_{j^*}, and by equation (6) $dp_N = dp_{j^*}$. Applying the chain rule,

$$\frac{d\ln p_N}{d\ln(1+t)} = \frac{p_{j^*}}{p_N}\,\frac{d\ln p_{j^*}}{d\ln(1+t)},$$

and the two terms are δ and $\rho^G_{j^*}$. In an all-formal chain, p_N is a formal retail price, and its response is ρ^G_F by definition. In an all-informal chain no seller charges VAT, so the rate enters no price in the chain, and with the margin fixed $\rho^G = 0$. □

Proof of Proposition 2. By equation (6), $dp_N = dp_{j^*}$, so applying the chain rule,

$$\frac{d\ln p_N}{d\ln(1+t_k)} = \frac{p_{j^*}}{p_N}\,\frac{d\ln p_{j^*}}{d\ln(1+t_k)}. \tag{13}$$

The ratio is δ. The last term is pass-through to the price of the formal input. If $k = g_{j^*}$, it is ρ_{j^*} by equation (5). If $k \neq g_{j^*}$, it is zero. If k is sold in the chain, either it is sold only before

the last formal sale, or it is first produced after it. In the first case every buyer of k is formal and credits the tax on it, so by equation (1) the tax on k changes neither what its seller keeps nor what its buyer pays after the credit. It therefore changes no producer price up to the last formal sale, and the tax charged there, on g_{j^*}, does not change. In the second case no seller of k charges VAT, because every seller after the last formal sale is informal, so the tax on k enters no price in the chain. If k is not sold in the chain, as with a good that competes with it, the tax on k can move p_{j^*} only indirectly. The proposition's last assumption rules out every such indirect route, in this case and in the other two. Hence the last term is $R_k \rho_{j^*}$, and substitution gives equation (10). In an all-informal chain no seller charges VAT, so with the margin fixed no price in the chain responds. □

Proof of Corollary 1. The case $k = g_{j^*}$ in the proof of Proposition 2 gives $\rho(k) = \delta \rho_{j^*}$, and the two ways in which a good sold in the chain can differ from g_{j^*}, sold only before the last formal sale or first produced after it, give $\rho(k) = 0$. □

Derivation of equation (11). By Shephard's lemma, the change in expenditure needed to keep household h at its initial utility is, to first order, $\sum_c y_{hc}\, dp_c$, where y_{hc} is the quantity it buys through chain c; the gain dB_h is minus that change. Dividing by x_h and writing $p_c y_{hc}/x_h = s_{hc}$ gives the first equality, and $d \ln p_c = \rho_c\, d \ln(1 + t)$ gives the second. □

Appendix B A numerical example: the formal input share and pass-through

This appendix works through the numerical example used in Sections 3.1 and 3.2: the formal input share of the informal price, then pass-through of a change in the general rate and of a change on a single good.

B.1 The chains and the formal input share

Table B1 compares a fully formal chain with a chain whose last seller is an informal vendor. There are three sellers, a farmer, a wholesaler and a retailer, $t = 0.18$, the farmer's producer price is 4, and each downstream seller adds 2 to its purchase cost. For this example, those additions and the farmer's price are held fixed across cases.

In the formal chain, the wholesaler's purchase price after its credit is 4 and the retailer's is 6. The retailer adds 2 and charges VAT on a producer price of 8, so the consumer pays 9.44. The farmer, wholesaler and retailer remit 0.72, 0.36 and 0.36, respectively, for total revenue of 1.44. Now make the retailer an informal vendor, and the unrecovered VAT moves one step up the chain. The wholesaler charges VAT on its producer price of 6, so the vendor pays 7.08 and cannot recover the 1.08. It adds 2 and charges no VAT, so the consumer pays 9.08. The

farmer and wholesaler remit 0.72 and 0.36, and the vendor remits nothing. Total revenue is 1.08.

Under the fixed additions of the example, the informal consumer pays 0.36 less than the formal consumer, because the vendor's own addition is no longer taxed. But the 1.08 collected upstream remains in the vendor's purchase cost. If the entire chain is informal, no VAT is collected and the consumer pays 8.

Table B1: Toy example: a fully formal chain, a chain with an informal vendor, and a fully informal chain

		All formal			Informal vendor			All informal		
Seller	Buyer	q_j	p_j	Remittance	q_j	p_j	Remittance	q_j	p_j	Remittance
Farmer	Wholesaler	4.00	4.00	0.72	4.00	4.00	0.72	4.00	4.00	0
Wholesaler	Retailer	6.00	6.00	0.36	6.00	7.08*	0.36	6.00	6.00	0
Retailer	Consumer	8.00	9.44*	0.36	9.08	9.08	0	8.00	8.00	0
Total VAT revenue				1.44			1.08			0
Revenue / consumer price				15.25%			11.9%			0%

Each row is one sale, from seller j to buyer $j+1$. q_j: the seller's price before VAT on that sale. p_j: the buyer's cost after any available credit. Remittance: the seller's output VAT less its input credit. Asterisk: the purchase on which VAT is not recovered.

In the chain with an informal vendor, the formal input share of the informal price is $\delta = 7.08/9.08 = 0.78$.

B.2 A change in the general rate

Table B2 changes the tax in the three chains of Table B1, holding producer prices and each seller's addition fixed. Start with a change in the general rate that raises the tax factor $1+t$ by 1 percent (Panel A). In the formal chain, the consumer pays $8(1+t)$, so the consumer price rises by the full 1 percent. In the chain with an informal vendor, the vendor pays $6(1+t) = 7.08$ for its formal input. That price also rises by 1 percent, or 0.0708 soles. The vendor adds the same 2 soles, so the informal price rises by the same 0.0708 soles, which is 0.78 percent of 9.08. The informal price passes through 0.78 times what its formal input passes through, and 0.78 is δ, the formal input share of the informal price. In the all-informal chain, no price moves. Relative pass-through, the informal price's response relative to the formal price's, is therefore $0.78/1.00 = 0.78$, which is δ.

B.3 A change on a single good

Panel B of Table B2 removes the tax on one good only, with each formal seller keeping the credit on its own purchases. The positions in the row labels of Panel B refer to the chain with an informal vendor, which resells the wholesaler's good; in the formal chain, the farmer's good is sold before the final sale and the other two are sold at it.

Table B2: Toy example: the consumer price after a change in the tax

	All formal	Informal vendor	All informal
Consumer price before the change	9.44	9.08	8.00
A. A change in the general rate: $1+t$ rises by 1 percent			
Consumer price	9.53	9.15	8.00
Change in the consumer price, percent	1.00	0.78	0
B. The tax on a single good is removed			
The farmer's good (sold before the last formal sale)	9.44	9.08	8.00
The wholesaler's good (the formal input)	8.00	8.00	8.00
A good the last seller makes (after the last formal sale)	8.00	9.08	8.00

Chains, producer prices and additions of Table B1, held fixed; formal sellers keep the credit on their own purchases. Panel A raises $1+t$ from 1.18 to 1.1918. Panel B: the consumer price after the tax on that good alone is removed; positions refer to the chain with an informal vendor.

Take first *the farmer's good*, which is sold only before the last formal sale. The farmer charges no VAT, so the wholesaler pays 4 and has nothing to credit. Its purchase price after the credit was already 4, so its cost does not change. It still charges VAT on its producer price of 6, and it now remits all 1.08 of it instead of 0.36. The informal vendor still pays 7.08, and the consumer still pays 9.08. The credit of the formal seller that follows offsets the change. The same happens in the formal chain, where the consumer still pays 9.44.

Next take *the wholesaler's good*, the formal input. In the formal chain, the wholesaler keeps the credit on its purchase from the farmer and sells without VAT. The formal retailer buys the good for 6, adds 2 and also sells it exempt, so, with producer prices fixed, the consumer pays 8 rather than 9.44. In the chain with an informal vendor, the wholesaler keeps its 0.72 input credit and no longer charges VAT on its sale, so the vendor's purchase price falls from 7.08 to 6. If the vendor still adds 2, the informal price falls from 9.08 to 8, a fall of 1.08. The change falls on the formal input, and the informal price moves with the price of that input.

Last, take *a good the last seller makes itself*, first produced after the last formal sale. Up to the last seller, the chain sells a different, taxable input. Suppose first that the last seller is a formal retailer. It buys the input at a cost of 6 after its credit, adds 2 and charges VAT on a producer price of 8, so the consumer pays 9.44. If its good becomes exempt and it keeps the credit on its input, its cost stays at 6, it adds the same 2 and it no longer charges VAT, so the consumer price falls from 9.44 to 8, a fall of 1.44. Now make the last seller an informal vendor. It pays 7.08 for the taxable input, because it cannot claim a credit, adds 2 and charges no VAT, so the consumer pays 9.08. Exempting its good changes nothing in this chain. The vendor was already charging no VAT on its sales, and it still cannot recover the VAT on its input. The change removes a tax that nobody was paying.

Appendix C Additional tables

This appendix collects two tables that Section 6 relies on.

Table C1: Where the exemption sits: the law against an exemption on the live bird (exempt window)

Exempt good, and the chains it reaches	Gain, % of expenditure: Bottom decile	Gain, % of expenditure: Top decile	Bottom/top decile ratio	Bottom half (%)	Revenue given up (law = 1)	Reaching consumers (%)
A. Eviscerated bird (the law)	0.066	0.058	1.14	31	1.0	65
B. Eviscerated and live bird						
Lima and no-plant provinces	0.227	0.130	1.75	34	3.4	58
Lima only	0.119	0.112	1.06	30	2.6	59
C. Live bird only						
Lima and no-plant provinces	0.171	0.083	2.07	36	2.4	55
Lima only	0.063	0.065	0.98	29	1.6	55

Equation (11), exempt window; ENAHO 2022, households interviewed January–March. A: responses of Table 5. "Lima and no-plant provinces": no-plant provinces' wholesalers buy the live bird on invoice; "Lima only": they do not. Bottom half: percent of the gain to the five lowest deciles. Appendix O defines the last two columns and reports variants.

Table C2: The food basket and the formal input share of each good (ENAHO 2022, purchased food)

Good	Share of food	Bought formally	Statutory VAT	δ_g
A formal establishment creates the good				
Chicken (eviscerated, with giblets, offal)	11.2%	2.6%	18%	0.80
Bread (all varieties)	5.0%	44.4%	18%	0.90
Evaporated and powdered milk	3.9%	16.7%	18%	0.91
Bottled vegetable oil	2.4%	10.0%	18%	0.89
Sugar	2.4%	5.6%	18%	0.82
Packaged pasta	2.1%	5.2%	18%	0.80
Bottled soft drinks and water	1.2%	6.2%	18%	0.85
Yogurt	1.1%	23.3%	18%	0.85
Canned fish, oats, coffee, biscuits, butter, margarine, salt, chocolate, beer, flour	5.8%	13.4%	18%	0.85
No formal establishment upstream, or exempt				
Fresh produce: potatoes, vegetables, fruit	24.9%	1.7%	exempt	0
Fresh meat, fish, artisanal cheese	10.3%	3.8%	fish exempt	0
Rice	5.3%	5.6%	4% (IVAP)	0
Prepared food away from home	3.8%	68.8%	18%	0
Eggs	3.1%	5.3%	18%	0
Dried legumes (lentils, beans, peas)	2.2%	3.8%	exempt	0
All remaining codes	15.1%	8.5%	mixed	0

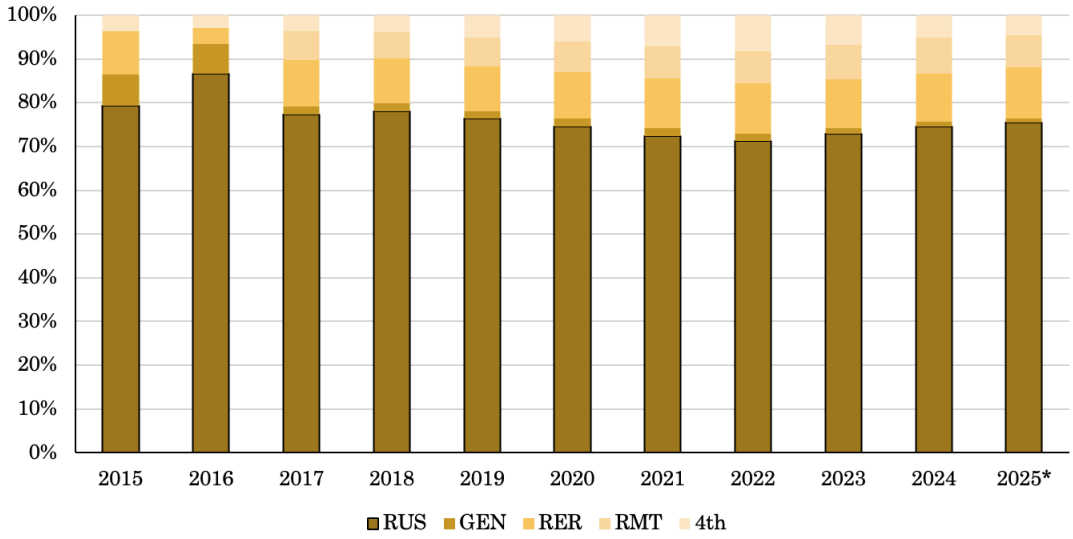
Share of food: percent of purchased food spending, survey weights. Bought formally: percent of the good's spending at a supermarket, bakery or restaurant. IVAP: the 4 percent tax that milled rice pays instead of the VAT. δ_g: formal input share of the good's informal price, assigned in Appendix P.

Appendix D Simplified tax regime

Peru offers a special, simplified tax regime (*RUS*, by its Spanish acronym) designed for small merchants and producers, including many of the vendors in our data. Vendors enrolled in the RUS do not pay the standard value-added tax (known as IGV in Peru) or income tax. Instead, they pay a low, fixed monthly fee based on their sales volume.[35] A critical consequence of this special status is that these vendors cannot claim tax credits for the VAT they pay on inputs purchased from formal suppliers in higher tax categories. Therefore, the VAT included in those purchases becomes a non-recoverable cost and can affect the final price of informal vendors, even if they do not remit VAT on their final sales.

The administrative requirements for RUS participants are minimal. While they are exempt from keeping formal accounting books, the tax authority does require them to retain official receipts for their business purchases and to issue them for sales. Issuing a receipt is mandatory for any sale over 5 soles (about US$1.45) and is also required for smaller amounts if the customer requests one. At the end of each day, vendors must issue a single, consolidated receipt that summarizes all sales for which individual receipts were not provided.

Figure D1: Tax regimes of retail market vendors



Source: authors' calculations from SUNAT data (retrieved 2025). Notes: RUS: simplified single regime; RER: special income-tax regime; RMT: micro and small enterprise regime; GEN: general regime; 4th: others.

Figure D1 presents a breakdown by tax regime for taxpayers whose economic activity is classified as "Retail sale in stalls and markets", a category that corresponds to the traditional markets in our study. According to this official data, 71 percent of these taxpayers were

[35] Revenue must be below 8,000 Peruvian soles a month (US$2,328) as of September 2025, and the fixed monthly payment is 50 soles (US$14.5).

enrolled in the simplified RUS category in 2022, the year of the policy we analyze.[36]

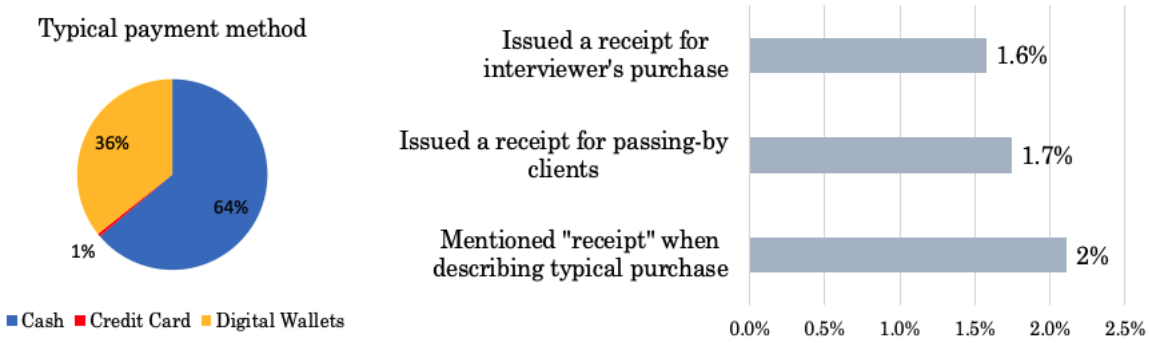
However, our own fieldwork suggests a significant discrepancy with this official classification. By estimating monthly revenue from the average number of daily clients and typical purchase amounts declared in our survey, we find that over 77 percent of our survey respondents earn above the revenue cutoff that legally requires them to switch to a different tax regime. This finding points to widespread tax evasion and is strong evidence that these markets operate in a largely informal way, even when their owners are formally registered taxpayers.

[36] Of the other categories, RER, RMT and GEN (20 percent of these taxpayers in 2022) charge and credit the value-added tax; the rest ("4th" in the figure) pay only income tax.

Appendix E Our survey of market stalls

Figure E1 and Tables E1 and E2 summarize the survey of Section 2.2.

Figure E1: Survey of market stalls: summary of results



Source: authors' calculations from our survey of market stalls, May 2024.

Table E1: Survey of market stalls: summary statistics

			Type of stand			
		All	Groceries	Animals and Eggs	Sausages & similar	Fruits and Vegetables
Share surveyed stands		100%	24%	32%	15%	29%
Average number of workers		1.60	1.51	1.81	1.53	1.46
Average number of customers in a day						
	1-15	3%	1%	4%	0%	5%
	16-30	27%	32%	26%	25%	24%
	31-50	38%	33%	43%	39%	37%
	51-100	22%	21%	16%	27%	26%
	101+	11%	14%	11%	9%	7%
Average purchase value (in soles)						
	s/1-5	3%	0%	2%	3%	5%
	s/6-10	23%	17%	18%	20%	33%
	s/11-20	42%	37%	34%	48%	51%
	s/21-40	25%	39%	33%	17%	7%
	s/41-60	4%	7%	4%	6%	1%
	s/61+	4%	0%	8%	5%	2%
Mentioned "receipt" while describing a typical purchase		2%	4%	2%	3%	0%
Typical payment method						
	Cash	64%	64%	64%	60%	66%
	Credit Card	1%	0%	1%	1%	0%
	Yape or Plin	36%	36%	35%	39%	34%
Made a purchase here		74%	84%	71%	80%	67%
% that were given a receipt (without asking for one)		1.6%	4.7%	0.0%	0.0%	1.2%
N other customers buying during survey		804	179	214	149	259
% other customers that were given a receipt		1.7%	2.2%	2.3%	0.0%	0.8%

Table E2: Survey of market stalls: what sets prices, by product

	Relevance of each factor on determining final price (0=nothing, 1=little, 2=relevant, 3=very relevant)																				Level of competition		
	Cost + Margin				Demand				Decide alone, looking at other stands in the market				Decide alone, looking at supermarkets				Decide together with other stands in the market						
Product	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3	None	Little	High
apple	0%	28%	26%	47%	88%	9%	2%	0%	93%	7%	0%	0%	84%	12%	5%	0%	36%	36%	21%	7%	0%	30%	70%
avocado	2%	16%	41%	41%	91%	9%	0%	0%	93%	5%	0%	2%	95%	5%	0%	0%	30%	43%	20%	7%	0%	23%	77%
banana	0%	23%	38%	38%	83%	13%	4%	0%	94%	4%	2%	0%	85%	9%	7%	0%	32%	38%	17%	13%	0%	13%	87%
beef	0%	0%	100%	0%	100%	0%	0%	0%	100%	0%	0%	0%	100%	0%	0%	0%	20%	0%	80%	0%	13%	38%	50%
cheese	0%	21%	44%	35%	90%	7%	2%	2%	91%	5%	3%	0%	83%	16%	2%	0%	31%	43%	7%	19%	3%	21%	76%
chicken	1%	20%	39%	39%	82%	15%	2%	0%	92%	5%	2%	1%	88%	7%	5%	0%	32%	19%	14%	35%	4%	10%	87%
eggs	1%	18%	38%	43%	83%	9%	6%	2%	89%	10%	1%	0%	87%	13%	0%	0%	42%	22%	16%	20%	0%	12%	88%
fish	0%	0%	100%	0%	100%	0%	0%	0%	100%	0%	0%	0%	100%	0%	0%	0%	100%	0%	0%	0%	0%	17%	83%
honey	0%	0%	100%	0%	100%	0%	0%	0%	100%	0%	0%	0%	100%	0%	0%	0%	100%	0%	0%	0%	0%	100%	0%
legumes	0%	0%	0%	100%	100%	0%	0%	0%	100%	0%	0%	0%	100%	0%	0%	0%	0%	0%	50%	50%	0%	0%	100%
noodles	2%	28%	39%	31%	36%	30%	20%	14%	85%	12%	3%	0%	94%	6%	0%	0%	88%	8%	3%	1%	1%	15%	84%
olives	0%	11%	50%	39%	67%	28%	6%	0%	94%	6%	0%	0%	78%	17%	6%	0%	17%	28%	17%	39%	0%	22%	78%
onion	0%	24%	44%	32%	84%	12%	0%	4%	88%	12%	0%	0%	80%	20%	0%	0%	16%	44%	20%	20%	0%	20%	80%
pork	0%	0%	33%	67%	67%	33%	0%	0%	100%	0%	0%	0%	50%	33%	17%	0%	0%	100%	0%	0%	17%	33%	50%
potato	0%	19%	43%	38%	81%	10%	5%	5%	95%	5%	0%	0%	90%	10%	0%	0%	14%	38%	24%	24%	0%	10%	90%
rice	2%	21%	51%	26%	83%	11%	5%	1%	91%	6%	3%	0%	90%	9%	1%	0%	35%	36%	11%	17%	0%	17%	83%
sausage	2%	13%	44%	40%	77%	17%	2%	4%	88%	10%	0%	2%	83%	15%	2%	0%	38%	33%	15%	13%	4%	27%	69%
sugar	1%	47%	28%	24%	87%	3%	10%	0%	93%	0%	7%	0%	93%	1%	6%	0%	35%	12%	36%	17%	2%	22%	76%
tomato	0%	26%	43%	30%	70%	30%	0%	0%	87%	9%	4%	0%	70%	22%	9%	0%	13%	48%	30%	9%	0%	22%	78%
turkey	0%	0%	100%	0%	-	-	-	-	100%	0%	0%	0%	100%	0%	0%	0%	0%	0%	100%	0%	0%	100%	0%
average	**1%**	**24%**	**40%**	**35%**	**78%**	**13%**	**6%**	**3%**	**91%**	**6%**	**2%**	**0%**	**88%**	**10%**	**3%**	**0%**	**39%**	**28%**	**17%**	**16%**	**2%**	**19%**	**80%**

treated (2022 policy)

Table F1: Switching and competition

Panel A. Switching between sectors (ENAHO household DiD against control products)

Outcome	Announcement	Exempt window	Post-repeal
Chicken, formal share of quantity	0.002 (0.007)	0.006 (0.005)	0.008 (0.005)
Chicken, any formal purchase	−0.001 (0.007)	0.005 (0.005)	0.008 (0.005)
Chicken, log informal quantity	0.018 (0.040)	−0.019 (0.034)	−0.013 (0.034)
Chicken, log informal spending	−0.011 (0.035)	−0.037 (0.028)	−0.029 (0.028)
Same window of 2021 (placebo), chicken share	−0.002 (0.005)	0.001 (0.004)	−0.001 (0.003)
Predicted by a switching model, chicken share		+0.039	
Predicted by store substitution, chicken share		+0.003 to +0.005	

Panel B. Competition: Lima traditional markets by distance to a chain supermarket

	Close (≤ 0.46 km)	Far	All
Gap to synthetic control, first month	−0.025 (0.004)	−0.027 (0.004)	−0.026 (0.003)
Gap to synthetic control, exempt window	−0.016 (0.007)	−0.008 (0.004)	−0.011 (0.004)
Gap to synthetic control, month after repeal	−0.031 (0.013)	−0.011 (0.007)	−0.018 (0.007)

Panel A: coefficients on treated × window, chicken against 13 control products, survey-weighted; standard errors clustered by primary sampling unit in parentheses. Predicted rows: the change in the formal share implied by the window pass-through estimates (Appendix F.1). Panel B: gaps to the shared synthetic control, headline specification, in log points; 10 close and 16 far markets; weeks 0–3, 0–12 and 13–16; standard errors from the dispersion across markets.

Appendix F Alternative explanations

This appendix gives the details behind two of the alternative explanations of Section 5.3, switching and competition. Table F1 reports the household estimates of switching (Panel A) and the market estimates of competition (Panel B). Panel A compares chicken with the 13 control products that have an ENAHO product code, in 210,677 household–product observations.

F.1 Switching to supermarkets

This subsection derives the benchmark of Section 5.3: how far the formal share of chicken purchases would have had to rise for consumer switching alone to produce the informal price response we measure.

In a switching model the informal seller neither charges nor pays VAT on anything, so a change in the tax reaches the informal price only through demand: the formal price falls by $\rho_F \ln(1.18)$, some consumers move to formal stores, informal demand shifts down, and the informal price falls along the informal supply curve. Let informal supply have elasticity $\varepsilon^S \geq 0$. If the informal price falls by $\rho_{\text{inf}} \ln(1.18)$, informal quantity changes by $-\varepsilon^S \rho_{\text{inf}} \ln(1.18)$ and informal spending falls to

$$X_I' = X_I \exp\big(-(1+\varepsilon^S)\,\rho_{\text{inf}} \ln(1.18)\big).$$

Hold chicken spending fixed over the window. The soles that leave the informal sector then go to formal stores,

$$X'_F = X_F + X_I - X'_I,$$

and in quantities at pre-period prices the formal share becomes

$$s'_F = \frac{X'_F/p_F}{X'_F/p_F + X'_I/p_I}, \qquad p_c = \exp\bigl(-\rho_c \ln(1.18)\bigr).$$

The shift in s_F rises with ε^S, so vertical supply, $\varepsilon^S = 0$, gives the smallest shift consistent with a measured ρ_{inf}: informal quantity is fixed and informal spending falls by the price change alone. To first order the implied change in formal spending is $d \ln X_F = \sigma(1+\varepsilon^S)\rho_{\text{inf}} \ln(1.18)$ with $\sigma = X_I/X_F$, which is why the benchmark is large where formal purchases are rare: at $\sigma = 40$, each point of informal price response requires formal spending to absorb 40 times as large a share of its own base. We evaluate the level expression, not the first-order one, because the first-order elasticity compounds when applied to a change of this size. The expression above equates the soles that leave the informal sector with the fall in informal spending, which amounts to a unit elasticity of informal demand ε. With vertical supply, what has to leave at the old price for the informal price to fall by $x = \rho_{\text{inf}} \ln(1.18)$ is demand worth $\varepsilon\,(1 - e^{-x})X_I$, so $X'_F = X_F + \varepsilon(X_I - X'_I)$ in general. Shares are in pre-period value units, and the observed change is in the formal share of quantity, which starts at 2.1 rather than 2.4 percent; the comparison is between changes.

The inputs are the January–March 2022 chicken spending of the household sample of Section 6.1 ($X_F/(X_F + X_I) = 2.4$ percent, $\sigma = 40.5$) and the pass-through estimates of Table 4: ρ_F from Lima supermarkets, and ρ_{inf} the spending-weighted average of the Lima and interior informal responses. Over the exempt window ($\rho_F = 0.52$, $\rho_{\text{inf}} = 0.23$) the formal share rises from 2.4 to 6.3 percent at $\varepsilon = 1$, 4.5 percent at $\varepsilon = 0.5$ and 3.6 percent at $\varepsilon = 0.25$; in the first month ($\rho_F = 0.79$, $\rho_{\text{inf}} = 0.36$) to 8.6, 5.8 and 4.3 percent. The observed change over May–July is +0.6 points (SE 0.5). The switching row of Table F1, +0.039, is this benchmark over the window with vertical supply and $\varepsilon = 1$; with the first-month estimates it is +0.062.

The benchmark asks how much switching would be needed to produce the informal response. A second calculation asks how much switching the observed price changes would produce. Let demand across store types be CES with elasticity of substitution η, so that $q_F/q_I \propto (p_F/p_I)^{-\eta}$. The exemption changes the ratio of formal to informal quantity by

$$\Delta \ln \frac{q_F}{q_I} = \eta\,(\rho_F - \rho_{\text{inf}}) \ln(1.18),$$

and the formal share of quantity at pre-period prices moves from $r_0/(1+r_0)$ to $r_1/(1+r_1)$, with $r_0 = X_F/X_I$ and $r_1 = r_0 \exp\bigl(\eta(\rho_F - \rho_{\text{inf}}) \ln(1.18)\bigr)$. Atkin, Faber, and Gonzalez-Navarro (2018) estimate average elasticities of substitution across retail outlets between 2.28 and 4.36.

Over the exempt window, the formal share then rises from 2.4 to 2.7 percent at $\eta = 2.28$, 2.8 percent at $\eta = 3$ and 3.0 percent at $\eta = 4.36$ (increases of 0.27, 0.36 and 0.54 points); in the first month, to 2.8, 3.0 and 3.3 percent (0.41, 0.55 and 0.84 points).

Further evidence on switching. In the household data of Section 5.3, using ENAHO's place of purchase and the exact 15-day recall window, the formal share of chicken quantity was 2.1 percent in January–March 2022 and rose by 0.6 percentage points (SE 0.5) in May–July relative to control products, with a similar estimate for the extensive margin; the same window in 2021 shows 0.1 (SE 0.4). In Figure 5 the observed series is flat at 2 to 3 percent through the exempt window, well below the benchmark, and the month-by-month differences against control products are statistically indistinguishable from zero throughout. Aggregate informal chicken spending fell by 5 percent (CI 1 to 9; the household-level estimate in Table F1 is 4 percent, SE 3) and informal quantity did not change detectably (-0.019, SE 0.034), which cannot distinguish a price cut on a fixed quantity from the rise of about 0.038 that unit-elastic demand would produce.[37] Table F1 reports the estimates behind both panels, together with the 2021 placebo. The data reject switching as the explanation but do not show that it is negligible: at the 95 percent upper bound it accounts for about two-fifths of the window response, and for about a sixth at the point estimate. The benchmark also assumes that every sol of spending that leaves the stalls goes to formal stores, which amounts to a unit elasticity of informal demand; with less elastic demand the required shift is smaller, 2.1 points at an elasticity of one half, still above the upper bound, and 1.2 points at one quarter, inside it (above). At one quarter, the point estimate would account for about half of the window response and the upper bound for all of it, so switching is rejected as the full explanation only for elasticities of about one half or more. Lima itself argues against switching as the source of the interior response: supermarkets there passed on 79 percent of the tax in the first month, a price fall of about 13 percent, and the stalls around them moved by about a fifth as much (0.15 to 0.16 of the tax; Section 5.3).

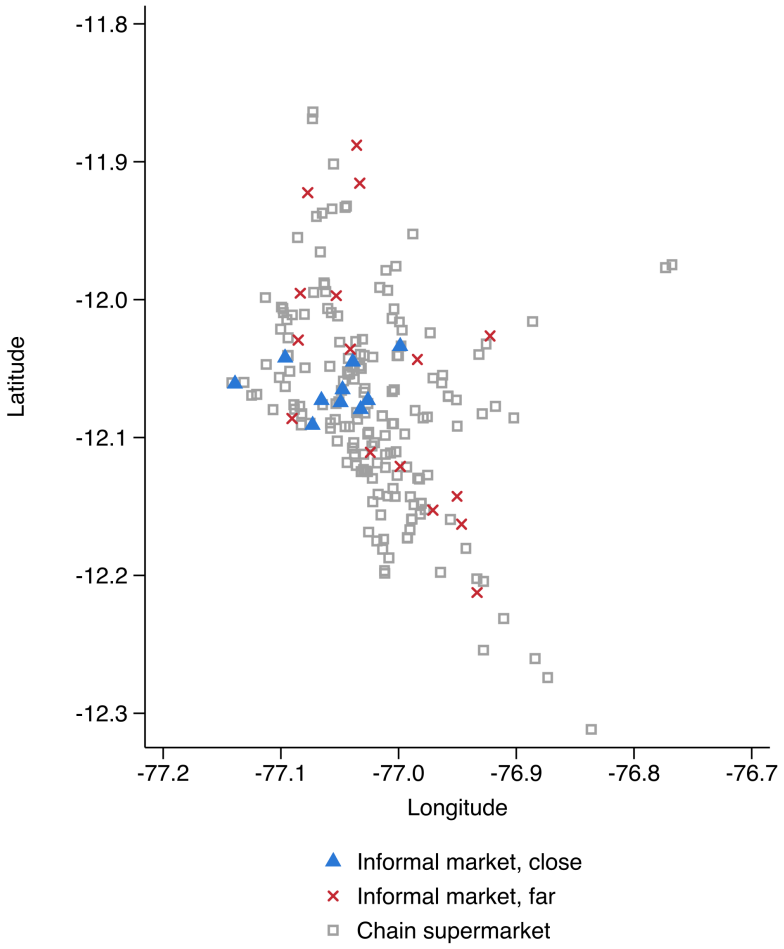
F.2 Competition with supermarkets

This subsection gives the details of the competition test of Section 5.3, on the design of Table F1: each market's weekly gap to the shared synthetic control of the headline specification, averaged over the markets close to and far from a chain supermarket.

Close and far markets. With market-level gaps to the shared synthetic control on the headline specification, close and far Lima markets fall by the same amount in the first month

[37] ENAHO unit values for chicken bought at retail markets fall by 0.8 percent (SE 1.5) in May–July net of control products, and at corner stores by 3.8 percent (SE 1.4). The informal price responses of Table 4, averaged over the exempt window and weighted by where households buy chicken, imply a fall of 3.7 percent. Unit values within a product code also move with the mix of cuts and package sizes, so we read them only as a coarse check.

Figure F1: Lima informal markets by distance to a chain supermarket



Informal markets of the competition test (Table F1, Panel B). Blue triangles: within 0.46 km of the nearest chain supermarket (10 markets); red crosses: farther (16). Gray squares: the chain supermarkets in OpenStreetMap to which the distance is measured; 175 of the 184 fall inside the map. Lima and Callao.

(−0.025 and −0.027 log points, 0.15 and 0.16 of the tax). A small gradient appears later: close markets hold a cut of −0.016 over the window against −0.008 in far markets, and remain lower in the month after repeal (−0.031 against −0.011). None of these differences is statistically significant: in pass-through, close minus far markets is −0.01 [−0.09, 0.06] in the first month, 0.05 [−0.06, 0.15] over the window and 0.12 [−0.07, 0.31] in the month after repeal. In log points, with Welch standard errors, the differences are +0.002 (0.006), −0.008 (0.008) and −0.020 (0.015). All markets are measured against the same synthetic control, whose error therefore cancels in the difference, and the intervals are Welch intervals from the dispersion across the 10 close and 16 far markets. Because the headline nets out the live-bird price,

these are movements in the stall's margin (Panel B of Table F1). The identical first-month response, including in markets 0.4 km from a supermarket, rules out competition that decays with distance. It does not rule out stalls responding to a city-wide, heavily reported price, and in Lima, where the model predicts no response, that is the part of the Lima dip the model does not explain. The test can be run only in Lima, where supermarkets are dense enough to split markets by distance, so for the interior the evidence is indirect. A city-wide response of Lima's size would be between a quarter and a third of the interior stall's first-month response, and netting it out (0.16 of the tax in the first month, 0.07 over the window) would lower the stall-to-wholesaler ratio to 0.67 and 0.78. The cities without a plant (Section 5.2.2), exposed to the same publicity, show no statistically significant response, but their intervals are too wide to rule out one of Lima's size. Over the window, the interval for close minus far markets reaches 0.15 of the tax: at most a small competition effect on the level of the price and, on the Lima evidence, not the source of the interior cut.

Figure 6 shows the underlying event paths, and Figure F1 where the markets are. Formal supermarkets respond identically whether or not a traditional market is close, so the close/far split is not picking up a center–periphery trend common to all stores; among traditional markets, close and far fall together in the first month and separate only afterward, which is the timing of the margin rather than the existence of the cut.

Appendix G Treatment and control groups

Table G1 lists the treated and control products.

Table G1: Treated and control products

Treated (exempt from May 1 to July 31, 2022)		
Estimated: eviscerated chicken		
Estimated in Appendix J: pink eggs		
In the panel, not estimated: sugar; pasta		
Input, not exempt: live chicken, Lima live-poultry market; used to residualize		
Controls (23)		
Vegetable oil	Red onion	Beef, steak
Rice, superior	Tomato	Beef, stewing cut
Rice, extra	Squash	Pork
Lentils	Yungay potato	Bonito (fish)
Canary beans	Banana	Evaporated milk
Black-eyed peas	Apple	Hard yellow maize
Lima beans	Papaya	
Chickpeas	Pineapple	
	Avocado	

Each cell's donor pool is the same stage's series of the 23 control products.

Appendix H City-by-city evidence on the interior chain

The headline interior cells pool 15 cities; Iquitos is excluded because Loreto is under the Amazon VAT regime (Law 27037), although the price system codes it as outside the Amazon. Proposition 2 needs the stalls' formal input to be the exempted product ($R_k = 1$), which takes two things in each of them: that the wholesaler sells the stalls the eviscerated bird, and that it is formal, charging VAT on that sale and crediting the live bird. The price panel's product lists establish the first for every interior city (Table 1): all 27 interior wholesale markets that quote chicken quote it eviscerated, and none quotes live birds. This appendix looks at the second, which the price panel cannot show; where the wholesaler is informal, the last formal sale in the chain is the live bird, as in Lima.

Data. SENASA publishes the registry of establishments authorized for poultry slaughter: 406 establishments, each with its tax identifier, district and date of authorization. We keep those authorized by 2022 and locate each at its district centroid; street addresses are available for fewer than half. We geocode each city's wholesale price point and compute the distance to the nearest authorized plant. For each city we compute the gap between its wholesale and its informal retail series and the shared synthetic control of the headline specification, demeaned over the weeks before announcement, and express the average over the first month and over the exempt window as pass-through. Six cities with poor price coverage are excluded here, as in the headline: Juliaca (no wholesale quote after week -4), Cusco (two to four distinct prices in 30 weeks), Huaraz, Andahuaylas and Tacna (fewer than 10 of the 13 exempt weeks quoted), and Tarapoto (a 40 percent level break in the wholesale series two weeks before implementation, constant for nine weeks, which looks like a change of variety or unit).

What the cities show. The registry splits the interior sharply. Every city with an authorized slaughter plant has one within 15 km of its wholesale market; in the others the nearest is 77 to 178 km away (77 to 173 km among the 15 cities with good coverage; the 178 is Juliaca). Among the 15 cities with good coverage, those with a plant (10) show the chain response and those without (5) do not (Figure 3). Estimated separately with the headline design, informal-retail pass-through is 0.78 [0.46, 1.04] in the first month and 0.58 [0.19, 0.88] over the window in the cities with a plant, against 0.11 [−0.30, 0.54] and −0.02 [−0.27, 0.38] in the cities without one; at wholesale the figures are 0.77 [0.48, 1.12] and 0.52 [0.19, 0.81] against 0.21 [−0.09, 0.48] and 0.15 [−0.12, 0.39]. Intervals are from the joint cluster bootstrap run separately for each group; one of the 200 draws for the cities without a plant drew too few treated series to estimate and is dropped. City by city, median informal-retail pass-through is 0.52 against 0.05 in the first month (one-sided permutation $p = 0.016$) and 0.35 against 0.11 over the window ($p = 0.049$); at wholesale it is 0.70 against 0.32 in the first month ($p = 0.028$) and 0.34 against 0.15 over the window ($p = 0.087$). Pass-through falls with distance to the nearest plant (Figure H1; Spearman correlation −0.55 for informal retail over the window,

$p = 0.021$). The cities without a plant still sell eviscerated chicken wholesale, so the product sold at wholesale is the exempted one; what they appear to lack is a formal seller of it, which would make the live bird the last formal sale in their chain and put them with Lima rather than with the rest of the interior. Table H1 summarizes the three configurations.

Table H1: Three configurations of the chicken chain

	Product sold at wholesale	Wholesaler formal?	Last formal sale	Informal pass-through
Interior, plant within 15 km (10 cities)	eviscerated	likely	eviscerated (exempt)	0.58
Interior, nearest plant 77–173 km (5 cities)	eviscerated	probably not	live bird (taxed)	−0.02
Lima	live bird	yes	live bird (taxed)	0.07

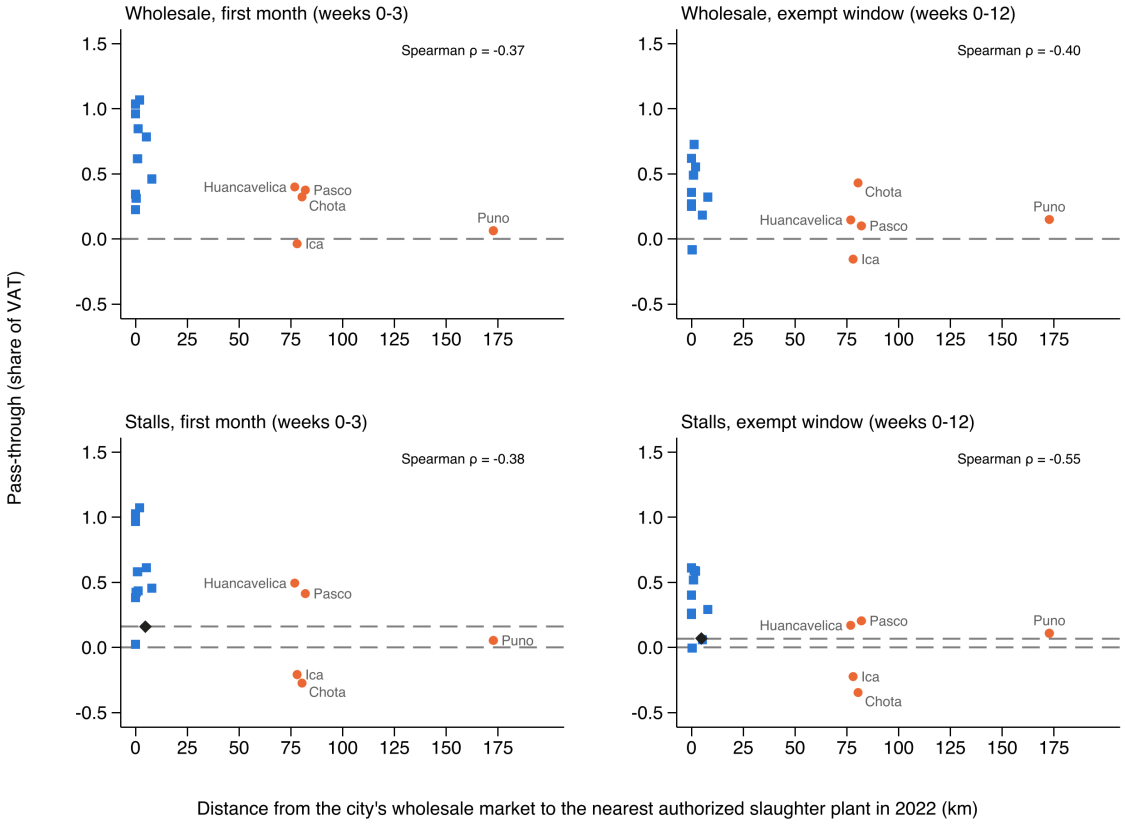
Informal pass-through over the exempt window. Interior rows: headline design estimated separately for each group of cities. Lima: headline estimate, Table 4. For the interior rows, "wholesaler formal" is inferred from an authorized slaughter plant near the wholesale market, not observed; Lima's live-bird wholesalers are treated as formal (Section 2.3).

More on the plant split. In the joint bootstrap restricted to the plant cities, the ratio of the stall's response to the wholesaler's is 1.01 [0.74, 1.23] in the first month and 1.10 [0.55, 1.56] over the window, and the stall sits above Lima's by 0.61 [0.28, 0.90] and 0.50 [0.11, 0.84]. If the wholesalers of the cities without a plant sell the eviscerated bird informally, the last formal sale in their chain is the live bird, so $R_k = 0$ as in Lima, and the model predicts no response; the estimates are consistent with that. A second reading fits the same numbers. A formal wholesaler that bought the bird on invoice but could not use the credit could have cut its price by at most 0.21 of the tax (below); the no-plant wholesale estimates, 0.21 in the first month and 0.15 over the window, cannot tell that seller from an informal one, while in the plant cities the wholesale response, 0.77 [0.48, 1.12], rules it out. Either way, little of the exemption could have reached the stalls of the cities without a plant. The repeal, a check the split was not built on, does not sharpen it (Appendix I).

The input credit. In the cities with a plant, where the wholesaler is likely formal, its first-month cut, 0.77 of the tax (0.58 in all 15 interior cities, Table 4), also shows that it kept the credit on the live bird (Section 2.1). Without it, the exemption would have worked as a standard one, and the wholesaler could have cut its price by at most 0.21 of the tax, the tax on its own value added.[38] That cut is more than three times the ceiling (the cut in all 15 interior cities nearly three times), unless the wholesaler cut its margin or bought the bird without an invoice.

[38] With the live bird at S/ 6.61 of a S/ 8.50 wholesale price (Table 3), the net-of-tax price is $8.50/1.18 = 7.20$ and the unrecoverable input tax is $6.61 \times 0.18/1.18 = 1.01$, so the lowest price the wholesaler could post is 8.21, a fall of 3.4 percent, or 0.21 of full pass-through in logs.

Figure H1: Pass-through by city and distance to the nearest authorized slaughter plant



Fifteen interior cities with good price coverage. Blue squares: the ten cities with an authorized slaughter plant near the wholesale market; labeled orange circles: the five without one. Black diamond and dashed line: Lima's stalls (Table 4), placed at the median distance from Lima's informal markets to a plant, for reference.

Limits. We read this as heterogeneity consistent with the mechanism, not as a test of it. First, the registry measures where formal slaughter capacity exists, not where stalls buy. Lima shows the difference: its informal markets are a median 4.8 km from an authorized plant, most of them companies that must charge VAT, yet Lima's formal chain ends with the sale of live birds, and its stalls show no sustained response. Second, the registry is a 2026 snapshot; plants that closed before then are missing, and a sanitary authorization is not a tax registration, and the plant's registration is not the wholesaler's: in 7 of the 10 cities with a plant at least one is a company, which cannot be in the simplified regime (the others have only individuals or single-owner firms, whose regime we cannot observe), but the plant slaughters for a fee, so it shows that the wholesaler can slaughter legally and invoice the carcass, not that it does. Third, plant cities are disproportionately coastal and larger. Within the highlands alone (six cities with a plant, four without) every comparison keeps its sign but none is significant (p between 0.08 and 0.18). Table H2 compares the two groups and re-estimates the gap with each characteristic as a control. Plant cities are larger, more often coastal and spend more at

supermarkets, but they are no closer to the main broiler-producing centers. Controlling for any one of these characteristics, or for coastal location and size together, the stall's gap stays between 0.33 and 0.51 of full pass-through, with permutation p-values between 0.009 and 0.035. City size absorbs the most, and the wholesale gap over the window loses significance once size is controlled for (p between 0.11 and 0.13). Fourth, the classification was built after the estimates were in hand, although the rule uses no price data, and each city is a single noisy market pair. Direct evidence on where stalls in plant and no-plant cities buy their chicken, and whether they receive an invoice, is what would turn this into a test.

In Table H2, the permutation p-values come from 5,000 reassignments of the plant indicator. The supermarket share is supermarket purchases as a share of the purchased food of urban households in the province, or in its department where the province has fewer than 50 urban households. Distance is to the nearest of the three largest broiler-producing centers, Lima, Trujillo and Arequipa, which stand in for the integrators' distribution centers, whose locations we do not observe.

Table H2: Is the plant split a proxy for city characteristics?

Panel A. City characteristics, means	Plant (10)	No plant (5)
Coastal	0.40	0.20
Province population (thousands)	584	225
Supermarket share of urban food spending (%)	2.4	1.3
Distance to nearest broiler center (km)	218	206

Panel B. Plant minus no-plant gap in city pass-through, controlling for	Stall		Wholesale	
	First month	Window	First month	Window
No controls	0.50 (0.014)	0.37 (0.010)	0.44 (0.009)	0.23 (0.045)
Coastal	0.49 (0.019)	0.35 (0.019)	0.42 (0.013)	0.21 (0.072)
Log province population	0.44 (0.022)	0.33 (0.018)	0.33 (0.028)	0.17 (0.111)
Supermarket share	0.48 (0.023)	0.39 (0.012)	0.42 (0.020)	0.25 (0.047)
Distance to nearest broiler center	0.51 (0.009)	0.37 (0.013)	0.45 (0.007)	0.23 (0.053)
Coastal and log population	0.44 (0.035)	0.33 (0.026)	0.33 (0.041)	0.17 (0.131)

Fifteen interior cities with good price coverage. Panel B: coefficient on the plant indicator in an OLS regression of city-level pass-through on the indicator and the listed characteristic; one-sided permutation p-values in parentheses. Population: INEI, 2020. Supermarket share: ENAHO 2022. Distance: straight line to the nearest broiler center.

Appendix I The repeal

The exemption ended on July 31, and the reversal is a second experiment on the same chains, run in the opposite direction. It matters for two reasons. Extrapolating anything here to a general rate change requires that ρ be the same going up as coming down, since the formal input share δ is a price ratio and carries no direction. And a repeal that restored the cut in proportion to δ would recover it on data that played no part in estimating it. Table I1 does both tests. It leaves out week 13, the transition. A price that fell by x and rose back by x would show the same pass-through of the cut and of the repeal.

Table I1: Passing the tax on, and giving it back

	Treated minus synthetic (logs)			Pass-through	
Stage	Weeks 0–12	Weeks 14–20	Swing	of the cut	of the repeal
Supermarkets, Lima	−0.085	+0.083	0.169	0.52	**1.02**
Wholesale, interior	−0.065	+0.020	0.085	0.39	0.51
Stalls, interior	−0.063	−0.013	0.050	0.38	0.30
Stalls, Lima	−0.011	+0.004	0.015	0.07	0.09
Wholesale, plant cities	−0.087	+0.005	0.092	0.52	0.55
Stalls, plant cities	−0.095	−0.041	0.054	0.58	0.33
Wholesale, no-plant cities	−0.025	+0.046	0.071	0.15	0.43
Stalls, no-plant cities	+0.003	+0.043	0.040	−0.02	0.24

Headline specification. Swing: change in the treated-minus-synthetic difference from weeks 0–12 to weeks 14–20. Pass-through of the cut: minus the weeks 0–12 difference over ln(1.18); of the repeal: the swing over ln(1.18). Interior: 15 cities; plant cities: the 10 with an authorized slaughter plant; no-plant cities: the other 5.

The answer to the symmetry question is no at the formal retailer and only approximately yes further up. Supermarkets passed on 52 percent of the tax when it was removed and all of it (1.02) when it came back, so the shelf price ended the episode half the tax above where the donor products imply it should have been. That is the asymmetry Benzarti et al. (2020) document for VAT changes in Europe, appearing here at the one stage of the chain that is unambiguously formal and price-setting. The interior wholesaler is closer to symmetric: 0.39 on removal, 0.51 on repeal, and in the ten cities with an authorized slaughter plant, where the wholesaler is most likely the formal seller of the formal input, 0.52 and 0.55. The interior stall gives back 30 percent against the 38 it passed on, so its price was still slightly below the counterfactual seven weeks after repeal. Lima's stalls, which received almost nothing, gave back almost nothing. The cities without an authorized plant, which the exemption likely did not reach, should not have responded to the repeal either; their prices nonetheless rose by 0.43 of the tax at wholesale and 0.24 at the stall and ended about a quarter of the tax above their counterfactual (0.28 and 0.26, with intervals from about −0.19 to 0.71, wide enough to include both zero and the plant cities' response). That is either a shock common to the interior after July that the donors miss, or a response to the repeal where the exemption had likely not reached; five cities cannot tell the two apart.

Two consequences. The repeal recovers the formal input share δ only partly: the interior stall's reversal is 0.60 of the wholesaler's (0.30 against 0.51; 0.59 in the plant cities), against 0.97 for the cut over the exempt window, 0.95 in its first month and 0.89 from price levels. The window is seven weeks and the intervals are wide, and a stall slow to raise a price it has advertised would produce this pattern, but we cannot separate that from a stall that, in the upward direction, passes on less than one sol per sol of its cost, against the one-sol-for-one-sol assumption. And any extrapolation from these estimates to a general rate change inherits the asymmetry: under Assumption 1 (Appendix P.4) the incidence of a general rise is $\delta\rho^{\uparrow}_{j^*}$, and $\rho^{\uparrow}$ is twice $\rho^{\downarrow}$ at the formal retailer, while at the interior wholesaler, the node whose pass-through is ρ_{j^*}, the two directions differ by 0.12, and by 0.03 in the plant cities. Both comparisons set a seven-week repeal average against a thirteen-week window average; over the last seven weeks of the window the removal figures are lower (0.43 at supermarkets, 0.21 at the wholesaler, 0.23 at the stall), so part of the gap is the decay of the cut within the window rather than asymmetry. The differences themselves do not depend on that choice: each equals where the price stood seven weeks after repeal relative to the counterfactual, 0.50 of the tax at supermarkets, 0.12 at the interior wholesaler and 0.03 in the plant cities. Measured from the level in the last seven weeks of the window, the repeal-to-removal ratios are 2.2 at supermarkets, 1.6 at the interior wholesaler and 1.1 in the plant cities (whose wholesale removal over those weeks is 0.30); setting the whole-window repeal figure against the last-seven-week removal figure would mix the two bases.

Appendix J Eggs

The law exempted eggs together with chicken. This appendix explains why eggs are not in the main analysis and then reports everything we have on them: the chain, what the model predicts, the data, the estimates and the household data.

J.1 Why eggs are not in the main analysis

In the first week of June 2022 every egg price series in our data, formal and informal, rose by 15 to 20 log points from week 5, and from week 7 at supermarkets (Figure J1 shows the monthly jump). The rise is specific to eggs: none of the control products shares it, and our data cannot separate it from the exemption.

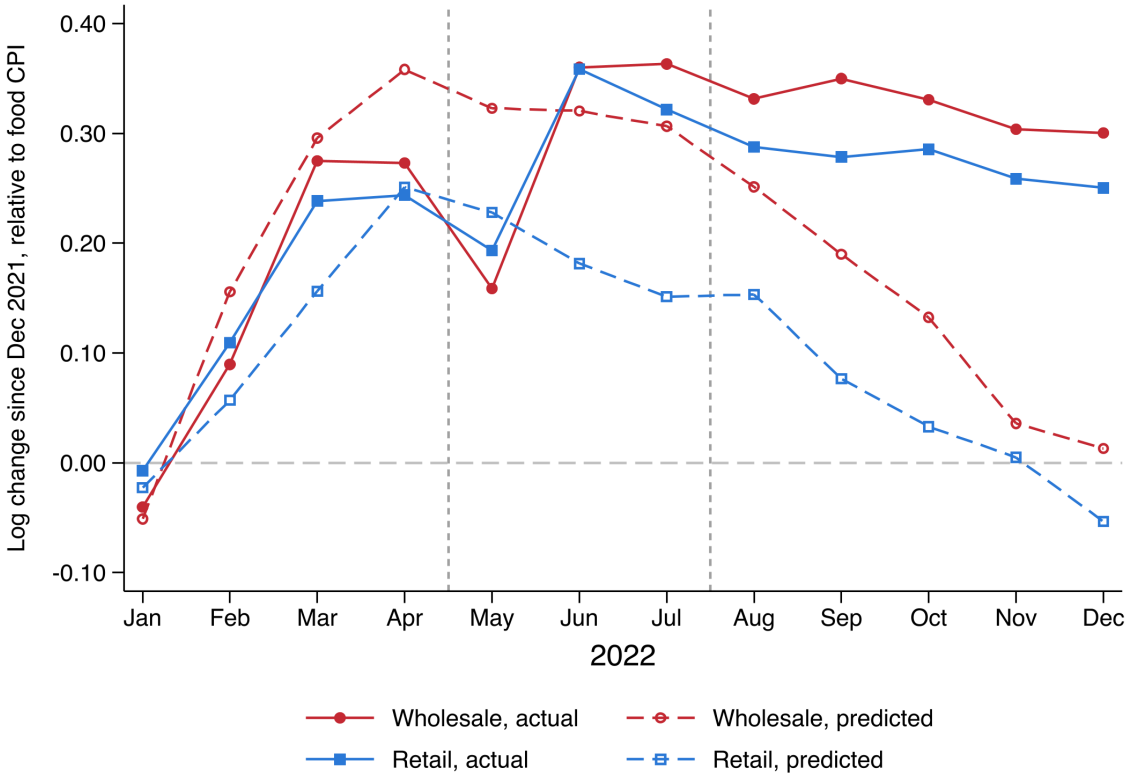
The shock is dated independently of the price panel. On May 28, 2022 the southern poultry association (Avisur) warned that egg prices would rise from June because producers had cut laying flocks by close to half after feed costs rose by more than 50 percent (Infobae, May 28, 2022). The Peruvian Poultry Association (APA) put the expected fall in production at 5 percent (El Comercio, June 2022). The Lima retail price reached S/ 9.50 a kilo by June 21 (Infobae, June 21, 2022).

The Ministry of Agriculture's poultry bulletin does not show a smaller flock (MIDAGRI, 2023). Laying hens in production rose from 29.6 million in May 2022 to 29.8 million in June, and egg production was flat, at 42.1 and 42.2 thousand metric tons a month. What fell were placements of replacement pullets: 1.97 million in May 2022 and 1.78 million in June, against 2.64 and 2.60 million in the same months of 2021. Fewer pullets lower later supply, not June's.

Feed costs do not explain the June jump either. We regress the monthly change in the log real egg price (the egg price over Lima's food CPI) on the current change and K lagged changes in the log feed cost, plus calendar-month dummies. The feed cost is 0.65 times the world maize price plus 0.35 times the world soybean-meal price, in soles. We estimate the model on every month outside 2022, choose K by BIC, and predict 2022 out of sample. BIC picks $K = 0$ for the Santa Anita wholesale price and $K = 2$ for INEI's Lima retail price of bulk eggs. Feed costs account for most of the rise in the wholesale price from January to June (actual +0.40 log points, predicted +0.37) but not for the jump from May to June (actual +0.20, predicted 0.00). For the retail price, the rise from January to June is +0.37 against a predicted +0.20, and the change from May to June +0.17 against −0.05 (Figure J1).

The cause of the June shock is therefore not established by these data. Whatever its cause, it hits every egg series, formal and informal, no donor shares it, and it does not recur in the 2021 placebo year (Figure J3). Weeks 0 to 4 are the only window the shock leaves clean. Window averages for eggs are dominated by it and are not informative about the tax.

Figure J1: Egg prices and feed costs in 2022



Log change since December 2021 in the egg price relative to Lima's food CPI, actual (solid) and predicted from world feed costs (dashed), for the Santa Anita wholesale price and INEI's Lima retail price of bulk eggs. Vertical short-dashed lines: the exemption, May to July. Sources: MIDAGRI poultry bulletin, BCRP (exchange rate, food CPI), World Bank (maize, soybean meal).

J.2 The egg chain

Eggs are the same good at every stage, so the exemption applies from the farm. Egg production is much less concentrated than chicken production: the largest producer accounts for about 20 percent of output and supplies almost all supermarket eggs; the sanitary authority registers 491 laying farms, a sanitary rather than a tax classification, 88 percent of them small, and industry estimates put half of producers and 60 to 65 percent of output outside the formal sector. The Santa Anita producers' wholesale market in Lima received 14,600 metric tons of eggs in 2022, 99.96 percent of it from farms in Lima province and none from the largest producer's farm in Chincha; that is about 1.3 kg per person in Lima and Callao, a small part of what the city consumes, so most of Lima's eggs reach stalls through channels these data do not show. Formal producers other than the largest, some 15 to 20 percent of output, sell into channels we do not observe; where they supply stalls the exempted good is the stall's formal input for that share, as in the interior chicken chain; Appendix J.3 gives the prediction. Most

eggs therefore reach stalls without ever having been invoiced at the exempted line, though their informal price still carries VAT on purchased feed and chicks.

J.3 What the model predicts

Table J1 places the egg chains in the model, in the format of Table 2. Supermarkets buy eggs on invoice from the largest producer and invoice the exempt good at the register, so the chain is formal to the end and the response of the consumer price is formal pass-through, ρ_F. Most eggs reach stalls from farms that neither charge nor credit VAT. Their formal inputs are feed and chicks, which the law did not exempt, so the informal egg price carries VAT ($\delta > 0$); but the eggs are first produced after the last formal sale, so by Corollary 1 the exemption cannot reach the stall price ($R_k = 0$). In the design of Section 5, the supermarket cell therefore estimates ρ_F, and for stalls supplied by unregistered farms the model predicts zero.

Table J1: The egg chains and what the model predicts for the exemption

Chain	Last formal seller (good sold)	R_k	δ	Prediction
Eggs, supermarkets	supermarket (eggs; final sale)	–	1	ρ_F
Eggs, stalls	feed mill, hatchery (feed, chicks)	0	$\delta > 0$	0

Conventions as in Table 2. The stall row covers the stalls supplied by unregistered farms; the stalls supplied by formal producers belong to a chain whose formal input is the exempted good and are discussed in the text.

Stalls supplied by formal producers are the exception. Where formal producers other than the largest supply the stalls, a share s of roughly 0.19 to 0.25 (their share of output, Appendix J.2, over the 80 percent that does not go to supermarkets), the stalls are in a chain whose formal input is the exempted good, as in the interior chicken chain, so the prediction is $s\delta\rho_{j^*}$: with $\delta = 12.11/13.88 = 0.87$ (Table J2) and chicken's ρ_{j^*} of 0.5 to 0.6, about 0.08 to 0.13 of the tax.

The interior wholesale prediction depends on the registration of the interior egg wholesalers, which we do not observe. A formal wholesaler would sell the exempt good as the stalls' formal input and should have fallen, and the interior stall would then have fallen by $\delta\rho_{j^*}$. Both interior estimates are therefore uninformative until that registration is known. Only Lima stalls have a prediction that does not depend on that registration.

J.4 Data

The treated egg product in the price panel is pink eggs, converted to 30-egg trays. The panel quotes eggs at wholesale in Lima (Santa Anita) and the interior, at stalls in Lima and the interior, and at Lima supermarkets. Table J2 reports median prices by stage in the week before implementation.

The egg series are residualized on world maize, soybean-meal and chicken prices, with coeffi-

Table J2: Egg price levels by stage, week before implementation (April 24–30, 2022)

Stage	Sector	Soles per 30-egg tray	Markets
Wholesale, Lima (Santa Anita)	not established	12.62	1
Wholesale, interior	not established	12.11	20
Stalls, interior	informal	13.88	21
Stalls, Lima	informal	14.13	26
Supermarkets, Lima	formal	16.70	27

Medians across markets of market medians, in soles per 30-egg tray. The formal input share of Appendix J.3 is the ratio of the two cross-market medians for the interior, pooled separately rather than matched within market, as in Table 3.

cients estimated on pre-2022 data, rather than on the live-bird price. This is the commodity residualization used for chicken in Tables K1 and L1.

J.5 Estimates

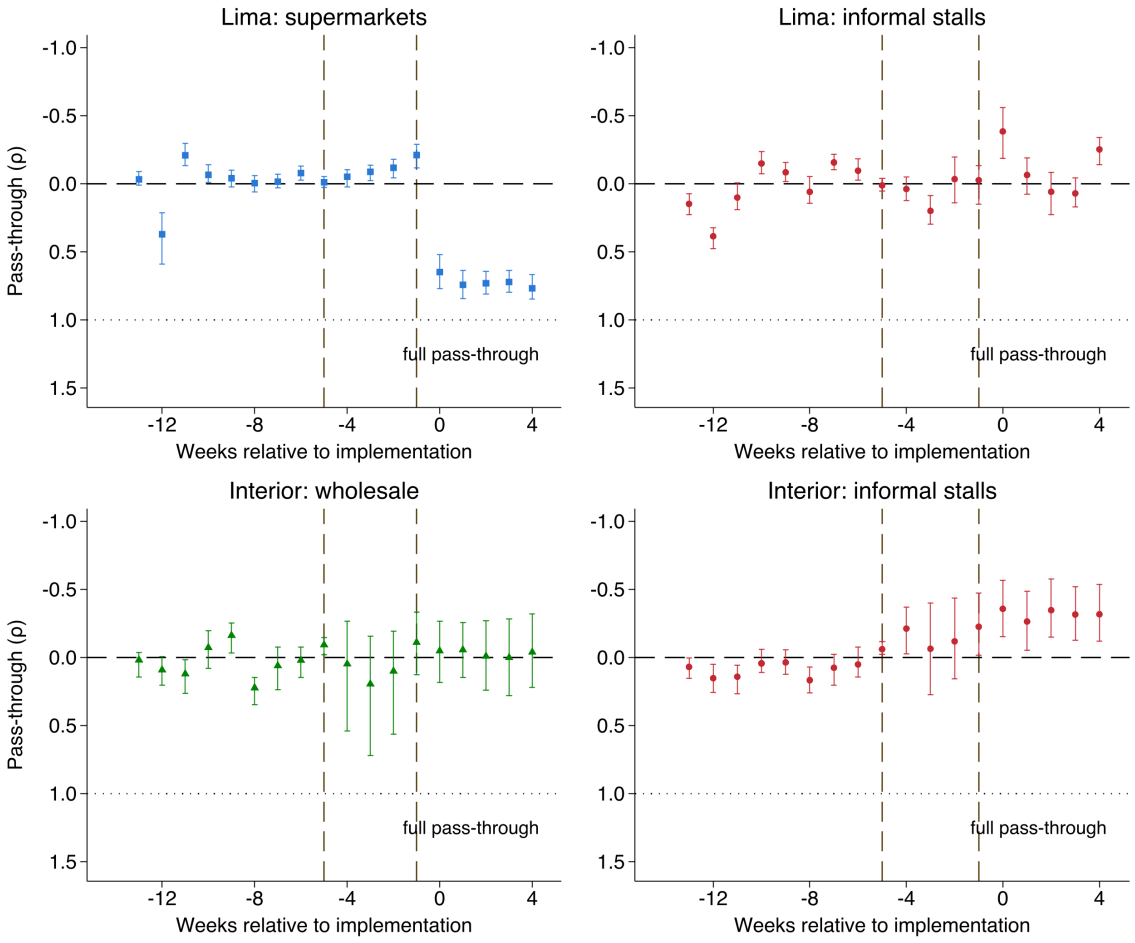
Figure J2 reports the egg stages: Lima supermarkets and informal retail, and interior wholesale and informal retail. Over weeks 0 to 4, supermarkets cut their prices, as the model predicts: −0.120 log points [−0.135, −0.104], 72 percent of the tax. No informal or wholesale stage fell: Lima informal retail +0.019 [−0.007, +0.045], interior informal retail +0.053 [+0.016, +0.089] and interior wholesale +0.005 [−0.044, +0.054].

The interior stall's estimate is wrong-signed, a third of the tax in size (−0.32 in pass-through units), and its interval excludes zero. In a quarter of rising feed costs we read it as a limit of the food-donor counterfactual rather than as evidence about the chain. The same reading has to apply to every egg series: if the food donors miss a feed-cost rise common to all egg chains, removing it would turn the informal and wholesale nulls into declines of a fifth to three-tenths of the tax and put the supermarket cut near full pass-through.

The 2021 placebo of Section 5.3.1, run on the egg cells (Figure J3), puts the zeros in context. With a fictitious exemption on May 1, 2021, supermarket prices rose by 0.22 [0.11, 0.34] of the tax over weeks 0 to 4, against the 2022 cut above, so the supermarket cut is not something egg prices do every May. The informal cells rose relative to the food donors in May of both years: Lima informal retail, the one egg cell whose prediction does not depend on registration (Appendix J.3), by 0.51 [0.42, 0.60] of the tax in 2021 against 0.11 in 2022, and interior informal retail by 0.16 [0.01, 0.31], half its 2022 rise. The one cell with a spurious cut in the placebo is interior wholesale, +0.46 [0.26, 0.67], a fall that starts in week −4 and lasts the whole window, with a pre-period fit as good as in 2022 (RMSPE 0.013 against 0.016). It is the leg of the egg case that carries the least weight, since its prediction depends on the registration of the interior egg wholesalers (Appendix J.3).

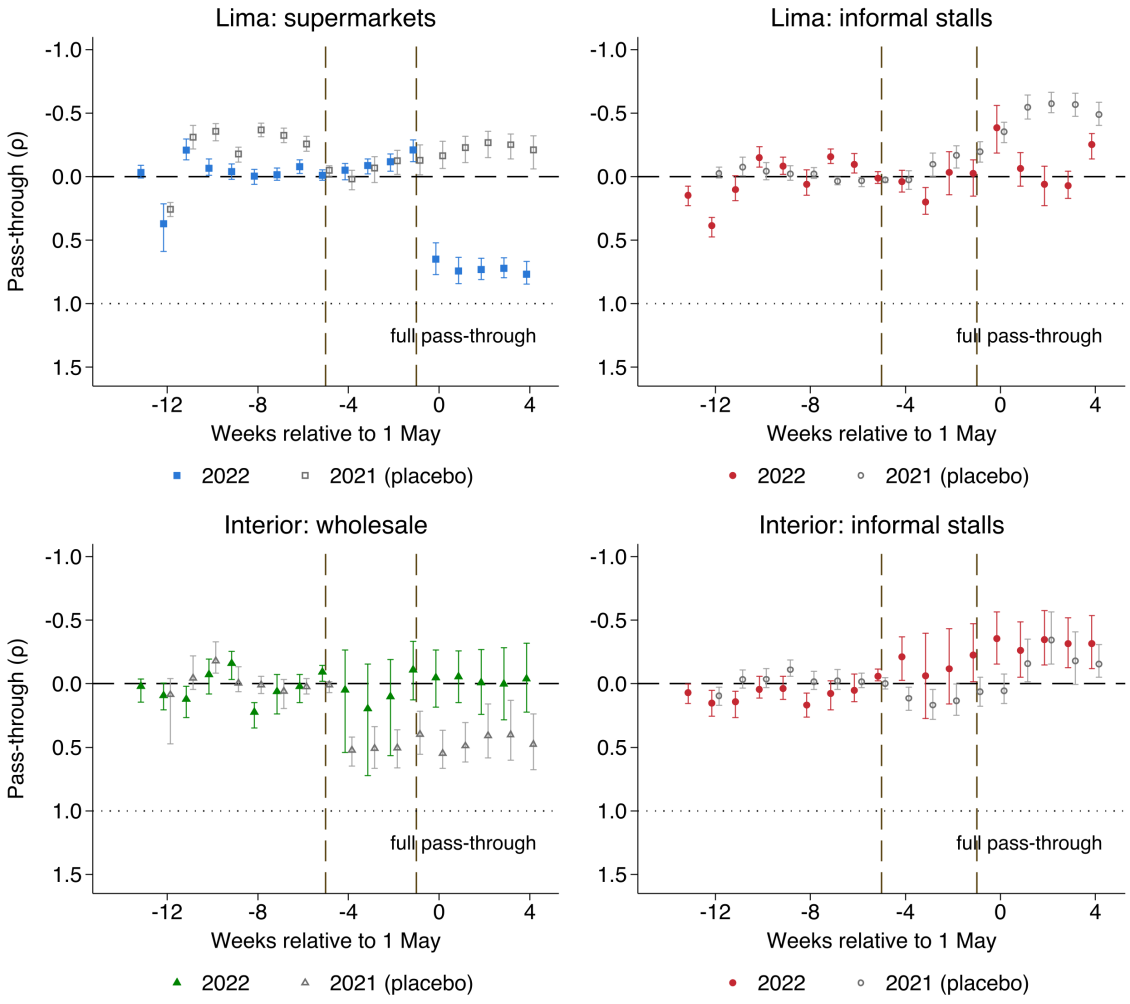
Over a five-week window, and with food donors that may miss a feed-cost rise common to all egg chains, the egg estimates cannot separate zero from the small cut that Appendix J.3

Figure J2: Eggs along the chain, 2022



Lima supermarkets and informal stalls, and interior wholesale and informal stalls; log prices net of world maize, soybean-meal and chicken prices. Weekly pass-through with 95 percent joint cluster bootstrap intervals; zero is no response and one is full pass-through. The figure ends at week 4, before the June 2022 egg-price shock (Appendix J.1).

Figure J3: Placebo for eggs



The design of Figure J2 run on 2021 with a fictitious exemption on May 1, 2021 (hollow gray), against 2022 (color), with the same donors, residualization and bootstrap. The June 2022 shock does not recur in 2021.

predicts for stalls partly supplied by formal producers, so they confirm the model only weakly. Eggs matter for the argument less as a second test than as the boundary case: most eggs sold at stalls come from farms that neither charge nor credit VAT, so they are first produced after the last formal sale, the purchase of feed and chicks, and the exemption could not reach their price (Corollary 1), although that price still carries the VAT on feed and chicks. For the exemption, the sales-tax view is close to right for eggs; for the tax, it is not.

J.6 Household data

In the household difference-in-differences of Panel A of Table F1, run on eggs against the same 13 control products (207,922 household–product observations), the formal share of egg quantity changed by −0.009 (0.008) in the announcement window, 0.005 (0.006) over the exempt window and −0.002 (0.006) after the repeal. None of the three differs from zero. Egg unit values rose at the main informal store types, by 9.7 percent at retail markets and 7.7 percent at corner stores net of control products, in line with the June shock.

Table J3: Share of households buying eggs at a formal store, by expenditure decile (ENAHO, January–March 2022)

Decile	1	2	3	4	5	6	7	8	9	10
Eggs	0.000	0.003	0.006	0.016	0.000	0.008	0.015	0.033	0.046	0.168

Formal = supermarket, bakery, restaurant (Table R1). Deciles of per-capita household expenditure, person-weighted, as in Section 6; shares weighted by household. Households interviewed in January–March that bought eggs in the 15-day recall window: 4,880.

Few households buy eggs at a formal store outside the top deciles; in the top decile the share reaches 17 percent (Table J3). Under the sales-tax view this makes the VAT on eggs progressive per sol of eggs bought. Under the chain view (equation (14)) it stays progressive because informal eggs are set to $\delta = 0$, a lower bound (Table C2).

Appendix K Robustness to the donor pool

Table K1: Chicken pass-through with 23 and 17 control products as donors (first month / exempt window)

Stage	Donors	Live bird (headline)	World commodities	None
Supermarkets, Lima	23	0.79 / 0.52	0.86 / 0.60	0.74 / 0.50
		[0.72, 0.86] / [0.41, 0.62]	[0.78, 0.94] / [0.50, 0.71]	[0.66, 0.82] / [0.39, 0.60]
	17	0.77 / 0.49	0.81 / 0.55	0.73 / 0.48
		[0.69, 0.85] / [0.39, 0.60]	[0.73, 0.90] / [0.44, 0.67]	[0.65, 0.81] / [0.37, 0.59]
Stalls, Lima	23	0.16 / 0.07	0.54 / 0.18	0.49 / 0.15
		[0.04, 0.27] / [−0.07, 0.20]	[0.34, 0.73] / [−0.02, 0.38]	[0.30, 0.68] / [−0.03, 0.34]
	17	0.14 / 0.05	0.52 / 0.16	0.48 / 0.15
		[0.03, 0.26] / [−0.09, 0.19]	[0.32, 0.71] / [−0.04, 0.37]	[0.29, 0.66] / [−0.04, 0.34]
Wholesale, interior	23	0.58 / 0.39	0.74 / 0.40	0.71 / 0.38
		[0.24, 0.92] / [0.08, 0.70]	[0.42, 1.07] / [0.09, 0.70]	[0.37, 1.04] / [0.07, 0.69]
	17	0.55 / 0.35	0.72 / 0.36	0.68 / 0.34
		[0.22, 0.89] / [0.04, 0.66]	[0.39, 1.05] / [0.05, 0.67]	[0.35, 1.02] / [0.02, 0.66]
Stalls, interior	23	0.56 / 0.38	0.81 / 0.44	0.58 / 0.23
		[0.22, 0.89] / [0.07, 0.70]	[0.49, 1.13] / [0.14, 0.74]	[0.26, 0.89] / [−0.07, 0.53]
	17	0.55 / 0.37	0.80 / 0.43	0.58 / 0.23
		[0.21, 0.90] / [0.04, 0.70]	[0.47, 1.13] / [0.11, 0.75]	[0.25, 0.91] / [−0.08, 0.55]

SDID pass-through, weeks 0–3 / 0–12. Brackets: 95 percent intervals from the bootstrap of Arkhangelsky et al. (2021) (500 replications), resampling treated market–product series only, not the joint bootstrap. Donors: the 23 headline control products, or 17 (see text). Columns: log prices net of the Lima live-bird price, of world maize, soybean-meal and chicken prices, or not residualized.

The headline donors are the 23 control products of Table G1; sugar and pasta are excluded because both were exempted by the same statute and neither can serve as a control. Table K1 re-estimates every chicken cell with the 17 that remain after dropping pork, a beef cut, lima beans, chickpeas, black-eyed peas and extra rice. No estimate moves by more than 0.05 of the tax, and almost all move slightly down: the headline interior wholesaler goes from 0.58 to 0.55 in the first month and from 0.39 to 0.35 over the window, the interior stall stays within 0.01 of the headline, and the interior stall's lead over the Lima stall is 0.41 in the first month and 0.32 over the window, against 0.40 and 0.32 with 23 donors. The pre-period fit is slightly worse with the smaller pool in every cell.

Appendix L Pass-through across counterfactuals

Threats to identification. The design assumes that, absent the exemption, each treated series would have followed its synthetic control; the pre-treatment fit (Figures 2 and 3, and Table L1 for Lima) and the placebo year (Section 5.3.1) support it. Two threats are specific to the setting. The exemption responded to food inflation, but the control products are other foods exposed to the same inflation. And if households substituted toward exempt chicken, control prices would fall, which biases the estimated cuts toward zero: the interior cuts are then conservative, but the zero predictions for Lima and the cities without a plant become easier to confirm. Controls linked to chicken output, such as hard yellow maize, carry 2 to 11 percent of the synthetic control's weight (Table L2), and dropping them moves no estimate by more than 0.07: with the headline residualization and without hard yellow maize, beef, pork and fish among the donors, the estimates are 0.86 / 0.54 for supermarkets, 0.58 / 0.39 for interior wholesale, 0.56 / 0.37 for interior stalls and 0.14 / 0.03 for Lima stalls, first month / window. Section 5.3 takes up the behavioral alternatives.

Table L1: Lima and the interior under three residualizations (pass-through, first month / exempt window)

Stage	Live bird (headline)	World commodities	Both
Supermarkets, Lima	0.79 / 0.52	0.86 / 0.60	0.89 / 0.61
	[0.72, 0.85] / [0.43, 0.60]	[0.80, 0.92] / [0.53, 0.68]	[0.82, 0.95] / [0.54, 0.69]
Wholesale, interior	0.58 / 0.39	0.74 / 0.40	0.61 / 0.40
	[0.34, 0.87] / [0.14, 0.63]	[0.45, 1.09] / [0.12, 0.66]	[0.39, 0.92] / [0.16, 0.65]
Stalls, interior	0.56 / 0.38	0.81 / 0.44	0.80 / 0.59
	[0.33, 0.71] / [0.09, 0.59]	[0.58, 1.03] / [0.22, 0.65]	[0.58, 1.00] / [0.29, 0.84]
Stalls, Lima	0.16 / 0.07	0.54 / 0.18	0.38 / 0.25
	[0.10, 0.22] / [0.00, 0.16]	[0.41, 0.67] / [0.05, 0.33]	[0.33, 0.45] / [0.18, 0.34]
Interior − Lima, stalls	0.39 / 0.31	0.27 / 0.26	0.42 / 0.34
	[0.16, 0.57] / [0.03, 0.54]	[−0.01, 0.53] / [0.00, 0.50]	[0.15, 0.62] / [0.04, 0.59]
RMSPE, Lima stalls	0.014	0.053	0.013

Pass-through, first month (weeks 0–3) / exempt window (weeks 0–12). Brackets: 95 percent joint cluster bootstrap intervals, as in Table 4. Columns: log prices net of the Lima live-bird price, of world maize, soybean-meal and chicken prices, or of both. Interior: 15 cities. RMSPE: pre-treatment root mean squared prediction error over the untreated weeks.

L.1 What the residualization does

This subsection complements Section 5. Figure L1 shows Lima's informal retail price under the four treatments of Tables L1 and L4: the live-bird price netted out (the headline), world commodity prices, both, and none. Without the live-bird price netted out, Lima's price falls from week −3 and drops by 0.20 to 0.22 log points in week 0, more than the whole tax, before returning to about zero by week 2; the pre-period fit is poor, with deviations of up to 0.14 in weeks −13 to −9. Netting out the live-bird price tightens the pre-period fit, over the untreated weeks, from about 0.053 to 0.014 and removes most of that drop. What remains is

Table L2: Which control products carry the synthetic control: the five heaviest donors by cell (percent of the SDID unit weight)

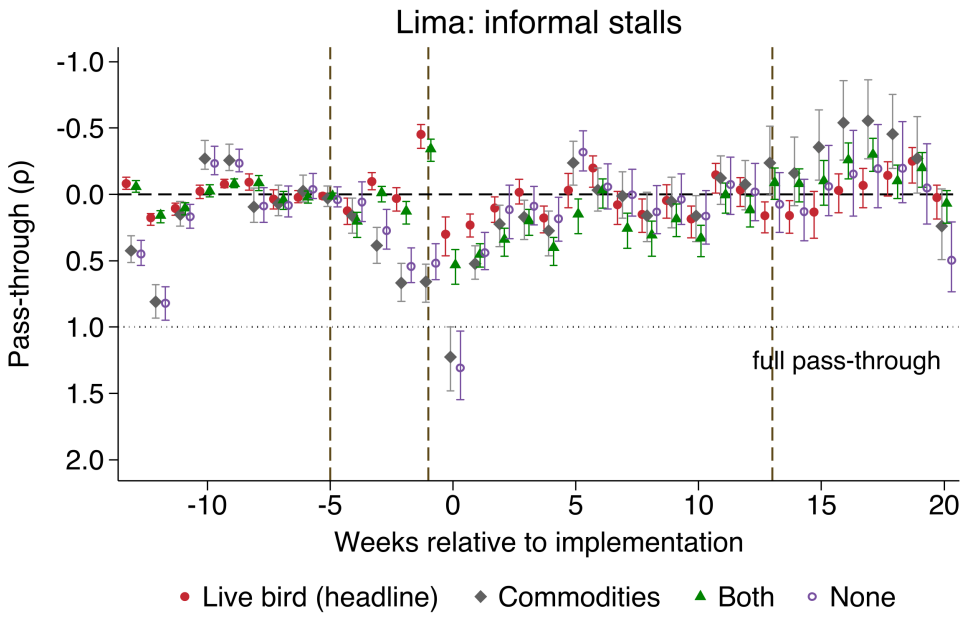
	Supermarkets Lima	Stalls Lima	Wholesale interior	Stalls interior
1	Papaya (11.9)	Banana (9.1)	Red onion (7.7)	Red onion (11.4)
2	Red onion (11.0)	Avocado (8.4)	Apple (6.7)	Squash (8.8)
3	Vegetable oil (6.7)	Canary beans (7.3)	Chickpeas (6.0)	Apple (8.7)
4	Yungay potato (6.0)	Chickpeas (6.6)	Superior rice (6.0)	Tomato (8.0)
5	Canary beans (6.0)	Superior rice (6.5)	Pork (5.8)	Superior rice (7.6)
Maize, meat and fish	9.6	11.3	5.8	2.5

Headline specification. SDID unit weights of the donor series, summed by control product; the last row sums hard yellow maize, beef (two cuts), pork and bonito.

a rise in week -1, when the live-bird price fell and the stall price had not yet followed, and declines of -0.05 and -0.04 in weeks 0 and 1, followed by noise around zero over weeks 2 to 12. Lima's informal price tracks the live bird with a short lag, with a two-week response on top of it. In the interior the choice matters less for wholesale than for the stall: over the window, wholesale is 0.38 to 0.40 under every treatment and the interior stall 0.23 to 0.59, the low end without residualization (Tables L1 and L4); with the headline residualization they are 39 and 38 percent. The ordering that matters, supermarkets and the interior above Lima, holds under all four treatments, but without residualization the interior stall is only 0.09 above Lima's in the first month and 0.08 over the window, against 0.39 and 0.31 with the headline residualization: Lima's raw price carries the drop of weeks -3 to 1 that netting out the live-bird price removes, and its pre-period fit is four times worse. The coefficients on the live-bird price are themselves informative: 0.57 for Lima stalls, 0.22 and 0.23 for interior wholesalers and retailers, which is what one expects if the live bird in that series is what Lima's stalls buy, while interior wholesalers buy their birds locally, at prices that need not track Lima's, and sell the stalls the eviscerated bird.

The choice of residualization. In the robustness columns we residualize on world maize, soybean-meal and chicken prices, with coefficients estimated on pre-2022 data, in place of the live-bird price or together with it. When the two are combined, the live-bird price is netted out of the commodity residual with each cell's own slope: 0.58 for Lima stalls, 0.23 for interior stalls, 0.21 for interior wholesale and -0.04 for supermarkets. Two caveats. The live-bird price is an endogenous price of the same chain. If integrators had lowered it in response to the exemption, to keep their collection-center customers competitive with exempt eviscerated chicken, the residualization would remove part of the treatment effect, and that response would itself be a route by which the exemption reached Lima's informal consumers even though $R_k = 0$ there; if total chicken demand rose with the exemption and pulled the live-bird price up, it would enlarge the estimated cut. The evidence on which of these happened is weak. The farm-gate price rose 4 percent from April to May, which argues

Figure L1: Lima informal stalls under each residualization



The live-bird price (the headline), world commodity prices, both, and none. Weekly pass-through with 95 percent joint cluster bootstrap intervals; zero is no response and one is full pass-through.

against integrators cutting their prices, but the live-bird event study, a first-month point estimate of 0.32 with a placebo interval of $[-1.04, 1.68]$ from a single treated series, cannot bound the concern in either direction; live-bird or farm-gate prices by department would. Because the headline cannot rule the concern out, Tables L1 and L4 and Figure L1 report Lima under every alternative, including no residualization at all. Only the treated series are residualized, not the donors, so at that point estimate the residualization removes about $0.57 \times 0.32 = 0.18$ of the full pass-through from the Lima stall figure. And the coefficient is estimated contemporaneously on nine pre-announcement weeks that include a February oversupply episode; the estimate of 0.57 for Lima, on Lima's own series, sits below Lima's formal input share, $\delta = 0.79$, because transmission from the collection center to the stall takes several weeks; a distributed lag over four weeks gives a long-run coefficient of 0.73, close to that share, and its estimates are reported below as a lower bound. We therefore treat the live-bird residualization as the headline, because its regressor is the price of the transaction the exemption did not touch and is therefore the cleanest available control for the chain's common cost, and report the commodity residualization, whose regressors are world prices, and the two combined as robustness; the contrast between the Lima spike and the interior step does not depend on any of the three (Table L1).

The timing of the interior wholesaler's cut depends on the residualization in the same way.

Under the commodity-only residualization the decline is significantly negative from week −3, the week in which the law was published on April 14, before it took effect; netting out the live-bird price, which itself dipped in exactly those weeks, moves the first significant week to week 0.

Lima and the other cells under each residualization. Table L1 and Figure L2 re-estimate every cell with world commodity prices in place of the live bird, and with both. Interior wholesale barely moves over the window (0.39 or 0.40 under every specification), and supermarkets rise, from 0.52 to 0.60. Over the window the interior stall moves by about as much as Lima across specifications (0.38 to 0.59, against 0.07 to 0.25) but stays well above it; the first month is where Lima is most sensitive, 0.16 with the live bird netted out, 0.38 with both, 0.54 with commodities alone and 0.49 with no residualization at all, the last two with a pre-period fit almost four times as poor. Without residualization, Lima's stalls are at 0.49 [0.35, 0.59] in the first month and 0.15 [−0.00, 0.29] over the window, with joint bootstrap intervals in brackets, and the pre-treatment RMSPE is 0.053; Table L4 gives the other cells. Under the two residualizations with world commodity prices, the joint bootstrap's own point estimates differ from those of Table L1 by at most 0.01. Part of the headline's fit is mechanical, because its slope is fitted on the same pre-period weeks; fitted on weeks −8 to −5 alone, the live-bird residualization still brings Lima's misfit over weeks −14 to −9 from 0.068, with commodities or with no residualization, to 0.025, against 0.017 when those weeks are used to fit it. Lima's window estimate is significantly above zero once world commodity prices are netted out, alone or with the live bird, and at the edge of significance under the headline. The headline residualizes only the treated series; residualizing the donors on the live-bird price as well, each control product with its own slope, so that treated and control series are measured alike, moves Lima further toward zero, 0.08 in the first month and 0.02 over the window, and leaves the interior stages within 0.03 of the headline (0.61 / 0.39 at wholesale and 0.56 / 0.35 at the stall, first month / window); supermarkets fall to 0.67 / 0.44. Each control product's slope is estimated over weeks −13 to −5. Nor does Lima move before its input: its first significant week under the headline, −4, is a single deviation of −0.021 log points that reverses the next week and is matched in size by pre-announcement weeks, while under the commodity residualization, which leaves the live bird in, Lima's price falls from week −4, as the live-bird price dipped (Figure L1). The ratio of the interior stall's response to the wholesaler's, the test of Proposition 2, is less stable than the gap between the interior and Lima stalls. The ratio leaves out the bootstrap draws in which the wholesaler's response is 0.05 or less, at most three of 200, because it is unstable there. In the joint bootstrap it is 1.08 [0.80, 1.54] in the first month and 1.09 [0.62, 2.05] over the window under the commodity residualization, 1.30 [1.03, 1.67] and 1.47 [1.04, 2.33] with both, and 0.82 [0.57, 1.19] and 0.61 [−0.04, 1.25] with no residualization, against 0.95 [0.70, 1.26] and 0.97 [0.49, 1.42] under the headline; it is about 2 under the distributed lag below, where the wholesale estimate is small and not significant. Only the double residualization excludes one, and over the window

only it excludes 0.89, the value under the one-sol-for-one-sol assumption. The spread comes mostly from the synthetic control rather than from the chain (Table L3). A distributed lag of the live-bird price, with lags 0 to 4 and coefficients estimated on every untreated week since January 2021, nets out more of the chain's common cost and lowers every cell, supermarkets included (0.34 in the first month): interior wholesale falls to 0.13, the interior stall to 0.28 and Lima to 0.02, with interior wholesale no longer significant. Over the window the four cells are at 0.30 (supermarkets), 0.19 (interior wholesale), 0.34 [0.04, 0.65] (interior stalls) and 0.01 (Lima stalls), and the long-run coefficients on the live-bird price are 0.67 for supermarkets, 0.73 for Lima stalls and 0.55 at both interior stages. We read it as a lower bound; even there the interior stall sits above Lima by 0.26 in the first month and 0.34 over the window. The residualization decides whether Lima's sustained response is zero or small, and how much of its first-month spike is the live bird. The model predicts no direct effect in Lima; the spike, common to markets near and far from a supermarket (Section 5.3), is the part it does not explain.

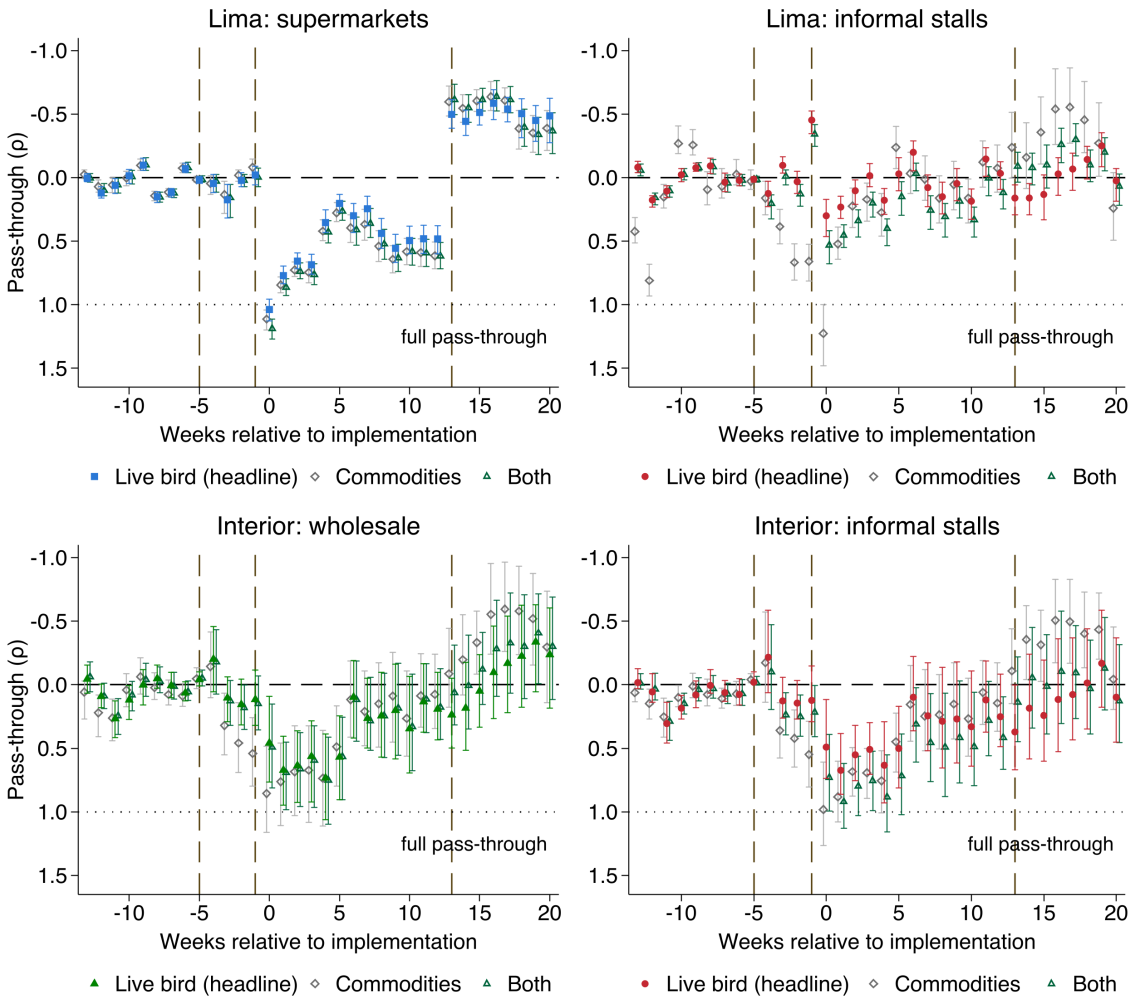
Table L3: What netting out the live-bird price does to the interior window estimates (pass-through, weeks 0–12)

	Wholesale, interior		Stalls, interior	
Live bird netted out of	raw prices	commodity residual	raw prices	commodity residual
Slope on the live-bird price	0.22	0.21	0.23	0.23
Window estimate before	0.38	0.41	0.23	0.44
Window estimate after	0.39	0.41	0.38	0.59
Change	0.01	0.00	0.15	0.15
equal pre-period weights	0.15	0.14	0.16	0.16
SDID time weights instead	−0.12	−0.12	−0.07	−0.07
new unit weights	−0.01	−0.03	0.06	0.06
From week −14: before / after	0.30 / 0.40	0.33 / 0.41		

Joint-bootstrap sample ($b = 0$), which reproduces Table 4. Before / after: without / with the live bird netted out. Components: change at fixed weights with an equal-weighted pre-period; correction for the SDID time weights; change in the unit weights. Last row: wholesale panel from week −14, not −17. Components may not add up because of rounding.

Why the residualization moves the stall and not the wholesaler. The SDID estimate for week ℓ is the treated series minus the synthetic control in week ℓ, less the same difference averaged over the pre-period with the time weights λ_t. At fixed unit and time weights it is linear in the outcome, so the change from one residualization to another splits exactly into the change at the old weights and the change in the weights (Table L3). Netting out the live bird subtracts $\beta \ln p_t^{LB}$ from the treated series only, so at fixed weights the change in the window estimate is β times the rise of the log live-bird price from the pre-period to the window. With the same slope at both interior stages, 0.22 and 0.23, an equal-weighted pre-period would raise both window estimates by 0.15, because the live bird was on average dearer during the window than before it. The time weights take back 0.12 at wholesale and 0.07 at the stall.

Figure L2: Chicken by stage under three residualizations



The live-bird price (the headline), world commodity prices and both; SDID with the 23 control products as donors. Weekly pass-through with 95 percent joint cluster bootstrap intervals; zero is no response and one is full pass-through.

The wholesale panel begins in week −17 and the stall's in week −14, the first week in which every series of the cell is quoted, and the weights put about a fifth of the wholesaler's baseline on week −17, when the live bird was dearer than during the window, and a third of the stall's on weeks −14 and −12, when it was 20 to 30 percent cheaper. Measured from its own baseline, the live bird rose 1.7 percent into the window for the wholesaler and 6.5 percent for the stall. The time weights do not change with the adjustment, which leaves the donors untouched, but the unit weights do: the adjusted stall series is matched by donors whose prices rose about one percent more over the window, which adds 0.06. Started in week −14 like the stall, the wholesale panel responds to the live bird as the stall does, 0.08 to 0.10 at the old weights, and its headline window estimate is 0.40 against 0.39, while its estimates that leave the live bird in fall to 0.30 without residualization and 0.33 with commodities: the headline is the specification that does not depend on where the pre-period starts. The commodity residualization changes every series, each product and stage with its own coefficients on world prices. It raises the stall's window estimate by 0.21 and the wholesaler's by 0.03, mostly through the donors: removing the control products' fitted world-price components moves the synthetic control by 0.13 at retail and 0.09 at wholesale, chicken's own fit moves the stall by 0.02 and the wholesaler by −0.05, and re-weighting adds 0.06 at the stall. In the first month the live-bird step works in reverse, because the live bird dipped in weeks 0 to 3: it lowers the wholesaler's first-month estimate from 0.71 to 0.58 and leaves the stall's at 0.58 to 0.56.

L.2 Designs and inference

Bootstrap and placebo-permutation intervals agree on the sign of every estimate and on the ordering of the stages. All event studies are estimated on the full 2022 panel but plotted and read to week 20. Beyond that point the intervals widen sharply, because the donor units are no longer anchored by the pre-treatment fit and drift apart, and the domestic poultry market is hit by the feed-cost increase and the avian-influenza outbreak of late 2022; nothing in the paper's argument reads weeks after 20. Those weeks still enter the synthetic control's time weights, but ending the panel at week 20 changes no estimate by more than 0.02: with the headline residualization it gives 0.78 / 0.51 for supermarkets, 0.59 / 0.41 for interior wholesale, 0.56 / 0.39 for interior stalls and 0.16 / 0.07 for Lima stalls, first month / window.

The paper's estimates use the synthetic difference-in-differences design of Section 5. Table L4 reports, for each chicken stage, the first-month and exempt-window pass-through under that design with live-bird residualization (headline), with commodity residualization, with both, without residualization, and under a plain difference-in-differences that uses no estimator: the log price net of the live-bird price, with each cell's headline slope, measured from its value in week −5, the last week before the announcement, minus the same change for the unweighted mean of the 23 control products in the same market and stage, averaged across markets, with confidence intervals from the dispersion across markets. The plain design keeps the headline's residualization and drops only the weighting: SDID weights the control products to match

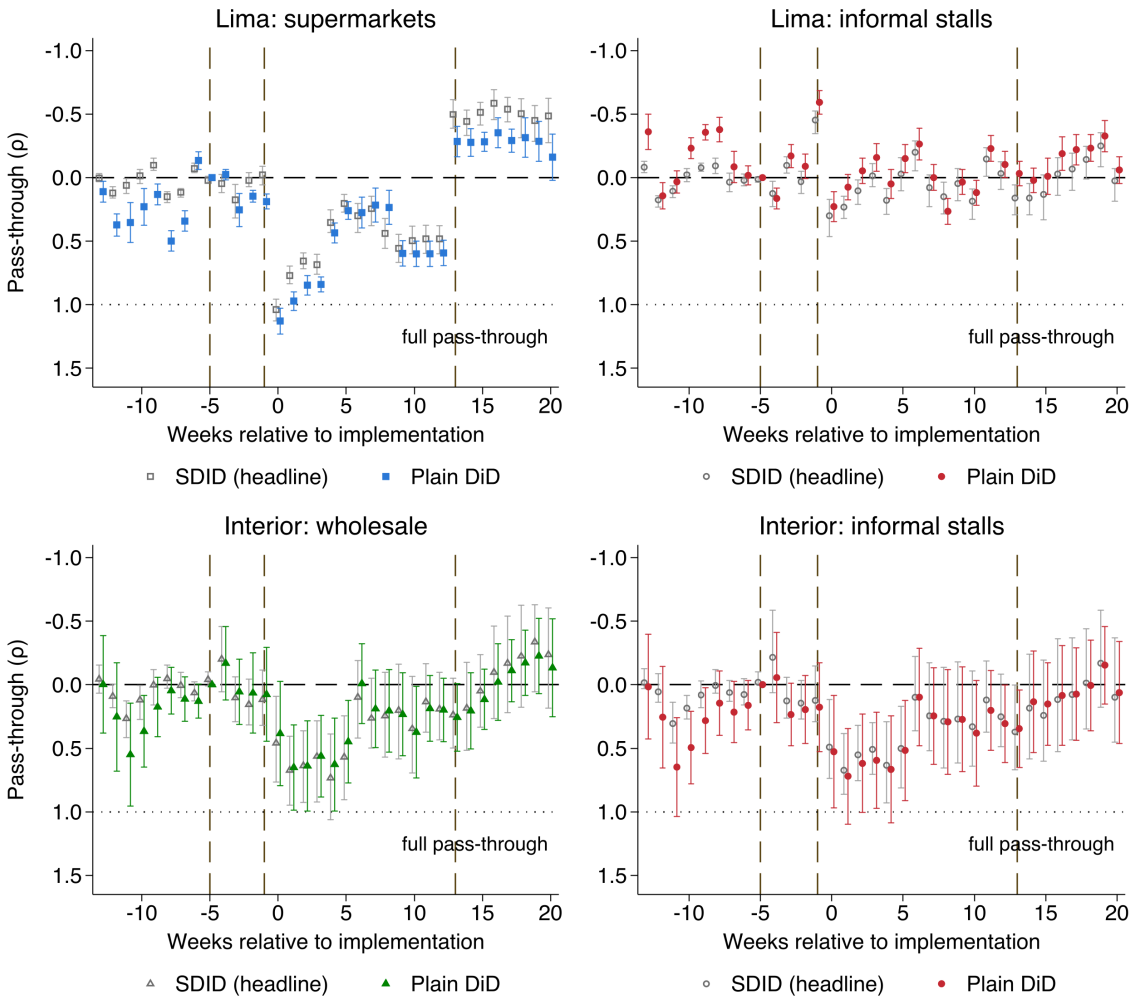
the treated product's pre-period path and the pre-period weeks to predict the post-period, while the plain design weights all control products equally and uses a single base week, the one on which SDID's time weights put most of the baseline. The two designs agree in the interior (Figure L3): the plain design gives 0.56 in the first month and 0.36 over the window at wholesale and 0.62 and 0.42 at the stall, against 0.58 and 0.39 and 0.56 and 0.38 under the headline, a stall-to-wholesaler ratio of 1.10 and 1.16, and the interior decline is significant from week 0 (the stall) or week 1 (wholesale) to week 5. They differ at the two Lima formats, where the weighting matters: the plain design puts supermarkets higher, 0.95 and 0.58, because its first weeks are deeper, and Lima's stalls at zero, 0.02 and −0.01. The ranking, supermarkets first, the interior stages next and Lima's stalls last, holds under both. The base week matters for the plain design and hardly for SDID: measured from the mean of weeks −13 to −5 instead, the plain design's interior windows fall to 0.18 and 0.17, because chicken rose against the other foods over those weeks even net of the live bird, and without the live-bird adjustment they fall to 0.04 and 0.03; SDID measured from week −5 alone moves no estimate by more than 0.04. Market-clustered intervals are narrower than placebo intervals because they measure dispersion across markets, not uncertainty about the counterfactual, and they do not detect a national shock common to all markets. Inference in the text uses the joint cluster bootstrap of Section 5, which resamples interior departments and Lima markets with all their series. Placebo-permutation intervals for the same estimates are wider than those of the bootstrap of Arkhangelsky et al. (2021), which resamples market–product series: two to three and a half times as wide in Lima and about a sixth wider in the interior in the first month. Under placebo intervals the headline estimates (first month / window) are 0.79 [0.54, 1.04] / 0.52 [0.24, 0.79] for supermarkets, 0.58 [0.19, 0.98] / 0.39 [−0.07, 0.85] for interior wholesale, 0.56 [0.17, 0.95] / 0.38 [−0.07, 0.83] for interior informal retail, 0.57 [0.30, 0.83] / 0.39 [0.06, 0.71] for wholesale and stalls estimated jointly, and 0.16 [−0.11, 0.42] / 0.07 [−0.23, 0.36] for Lima informal retail. The point estimates are identical by construction; with placebo intervals the separate interior window estimates just include zero, while the joint estimate does not.

Table L4: Chicken pass-through by stage and counterfactual (first month / exempt window)

	SDID, log price net of				Plain DiD, live bird,
Stage	live bird (headline)	commodities	both	nothing	from week −5
Supermarkets, Lima	0.79 / 0.52	0.86 / 0.60	0.89 / 0.61	0.74 / 0.50	0.95 / 0.58
Wholesale, interior	0.58 / 0.39	0.74 / 0.40	0.61 / 0.40	0.71 / 0.38	0.56 / 0.36
Stalls, interior	0.56 / 0.38	0.81 / 0.44	0.80 / 0.59	0.58 / 0.23	0.62 / 0.42
wholesale and stalls, jointly	0.57 / 0.39	–	–	–	0.56 / 0.37
Stalls, Lima	0.16 / 0.07	0.54 / 0.18	0.38 / 0.25	0.49 / 0.15	0.02 / −0.01

Pass-through, first month (weeks 0–3) / exempt window (weeks 0–12). SDID as in Table L1; nothing: not residualized. Plain DiD: live-bird-adjusted change from week −5 minus that of the unweighted mean of the 23 control products, averaged across markets. Jointly: interior wholesale and stalls as one cell (plain DiD: their differences averaged by city and week).

Figure L3: SDID and a plain difference-in-differences

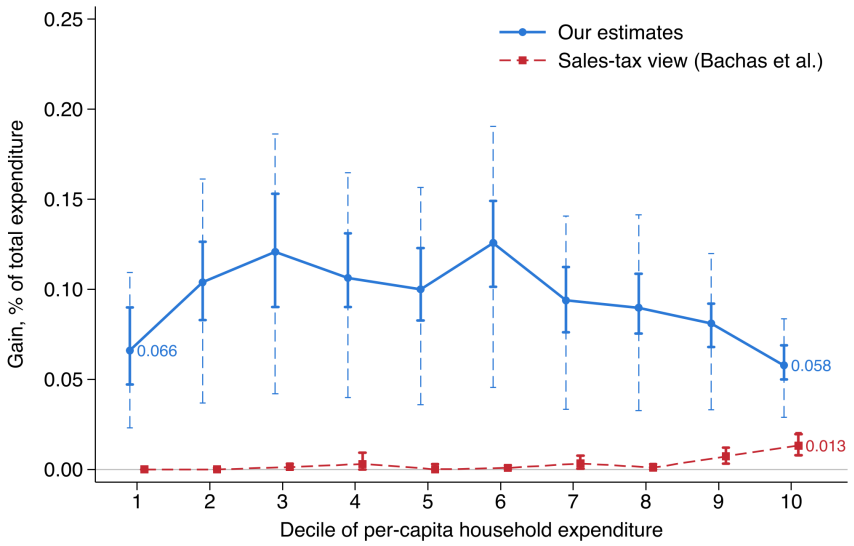


Weekly pass-through; zero is no response and one is full pass-through. Hollow gray: headline SDID, 95 percent joint cluster bootstrap intervals. Color: plain difference-in-differences from week −5 with the same live-bird adjustment (Table L4), 95 percent intervals from the dispersion across markets.

Appendix M Gains by decile

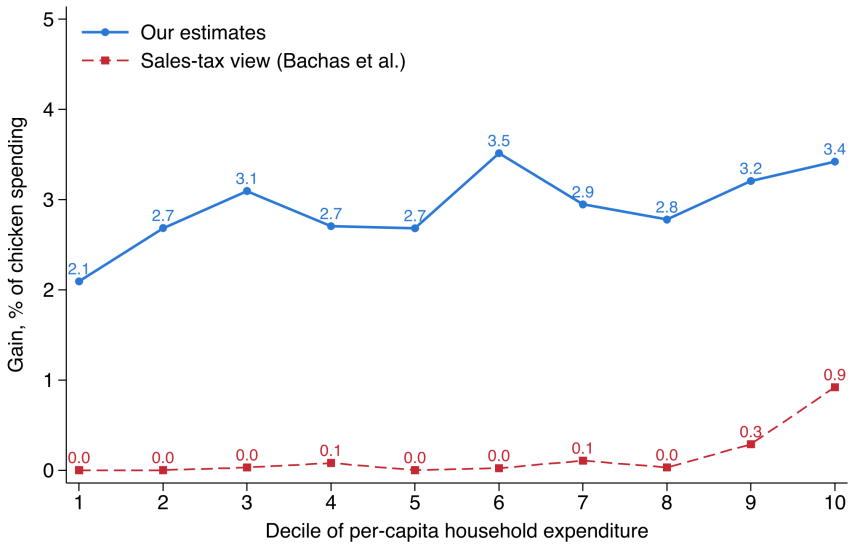
Figure M1 plots Table 5 with two 95 percent intervals for each decile. The solid bars show survey sampling alone: they hold the pass-through estimates at their point values and resample households by survey cluster within strata, recomputing the deciles each time. The dashed bars add the sampling error of the pass-through estimates, combining each survey sample with each draw of their joint bootstrap. For our estimates the solid bars are short and the dashed bars long: most of the uncertainty about how much each decile gained comes from the pass-through estimates, mostly the stall response in the cities with a plant, which weighs most in the middle deciles, where chicken bought in those provinces is the largest share of expenditure. Because those estimates move every decile together, they matter less for how the relief is shared: for the bottom half's share, survey sampling alone gives [29, 35] and the pass-through estimates alone [28, 32]. Figure M2 expresses the chicken gain per sol of chicken bought, and Figure M3 the VAT on purchased food per sol of food purchased, rather than relative to total expenditure: they show whether a purchase carries the tax, not who bears it.

Figure M1: Gain from the chicken exemption, by expenditure decile



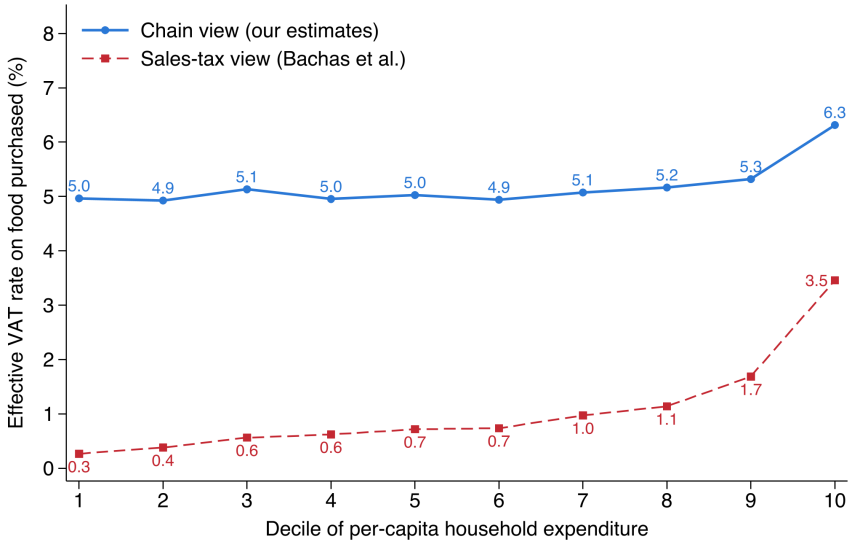
First-order gain over the exempt window, percent of total household expenditure; our estimates and the sales-tax view as in Table 5. Solid bars: 95 percent intervals for survey sampling alone; dashed bars: 95 percent intervals that also include the sampling error of the pass-through estimates. ENAHO 2022, households interviewed January–March; deciles of per-capita household expenditure.

Figure M2: Gain from the chicken exemption per sol of chicken bought, by expenditure decile



The gain of Figure M1 divided by each decile's chicken spending: ln(1.18) times the decile's shares of chicken spending by chain, each weighted by the chain's response in Table 5 (zero in provinces without a plant).

Figure M3: VAT per sol of food purchased, by expenditure decile



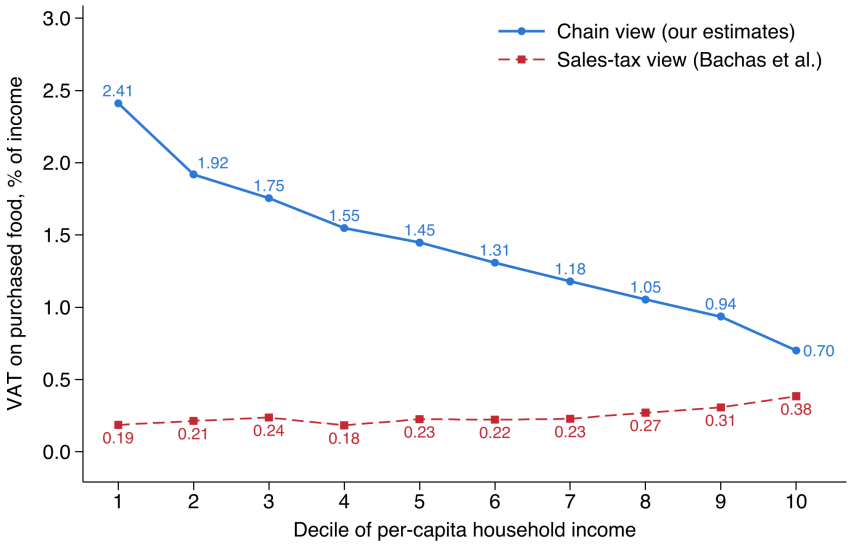
Equation (14) by decile: the VAT carried by purchased food per sol of food purchased, under the chain view and the sales-tax view. ENAHO 2022, survey weights; deciles of per-capita household expenditure, person-weighted.

Appendix N Income instead of expenditure

Section 6 ranks households by per-capita expenditure and measures the VAT relative to expenditure, as Bachas, Gadenne, and Jensen (2024) do. Figures N1 and N2 rank them instead by per-capita net household income and divide by it. Every profile then falls more heavily on poorer households. On purchased food, the chain view falls from 2.4 percent of income in the bottom decile to 0.7 in the top, and the sales-tax view rises from 0.19 to 0.38; the bottom half pays 35 and 17 percent of the VAT on food, against 23 percent of income. For the whole VAT, even the sales-tax view is regressive relative to income: the top decile's rate is 0.65 times the bottom decile's, against 0.44 in the central case and 0.33 at the upper bound.

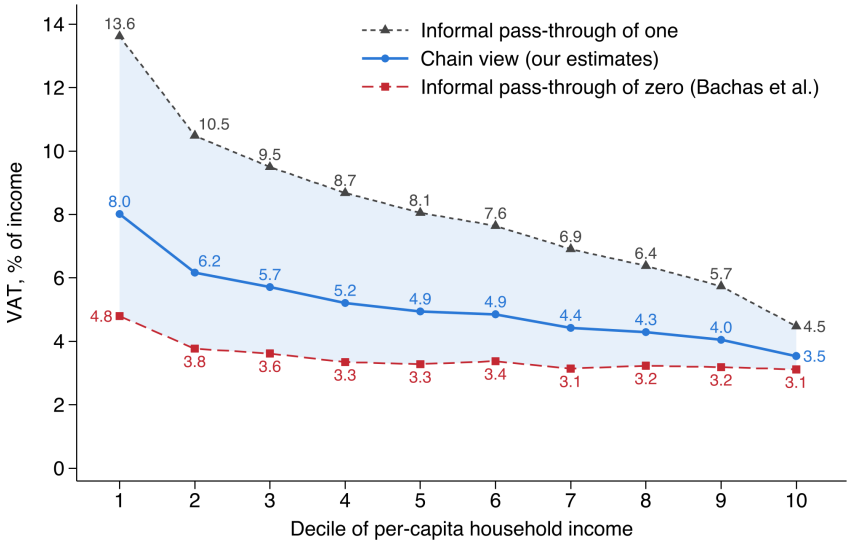
Most of the shift comes from the bottom income decile, where 76 percent of households report spending more than their income; without it, the sales-tax view of the whole VAT is flat at 3.1 to 3.8 percent of income from the second decile to the tenth. Ranking by expenditure but dividing by income, the central case remains mildly progressive (a top-to-bottom ratio of 1.34) and the upper bound is proportional (0.99). Gross income gives the same picture.

Figure N1: VAT on purchased food, percent of income, by income decile



Ratios of decile totals. ENAHO 2022; deciles of per-capita net household income, person-weighted.

Figure N2: The whole VAT, percent of income, by income decile



Band and lines as in Figure 9; ratios of decile totals. ENAHO 2022; deciles of per-capita net household income, person-weighted.

Appendix O Had the law exempted the live bird

Table C1 carries the counterfactual of Section 6.1 through the calculation of Table 5. Under B and C, a chain that buys the live bird from a taxed seller responds by $\delta\rho_{j^*}$, with $\rho_{j^*} = 0.52$ of the wholesaler in the plant cities (Section 5.2.2): $0.79 \times 0.52 = 0.41$ in Lima (Table 3) and $0.74 \times 0.52 = 0.39$ in the provinces without a plant (δ as in Appendix P). Under C the eviscerated bird stays taxed, so supermarkets and the stalls of the plant provinces do not respond. "Lima and no-plant provinces" assumes that the wholesalers of the provinces without a plant buy the live bird from invoicing integrators; "Lima only" assumes they buy from farms outside the VAT, so those provinces get no response under B, as under the law, or under C. Revenue given up is the VAT on households' chicken purchases, relative to the law's: the VAT on supermarket purchases and on the plant-province wholesaler's sale of the eviscerated bird, and on the live-bird sale to the chains reached (0.1525×0.79 of their spending in Lima and 0.1525×0.74 in the provinces without a plant); live birds sold to formal buyers were already credited. Reaching consumers is the gain in the chains with an invoiced sale of the exempt good, valued in soles at the quantities bought, as a percent of that revenue.

With no response for rural households, the bottom/top ratios are 0.77 (A), 1.12 and 0.87 (B), and 1.34 and 0.98 (C). In the first month (weeks 0–3; $\rho_{j^*} = 0.77$, Lima response 0.61) they are 1.09 (A), 1.74 and 1.04 (B), and 2.07 and 0.98 (C). With ρ_{j^*} from the wholesaler in all 15 interior cities (0.39; Lima response 0.31), a lower bound, since it includes the no-plant cities, where the wholesaler is likely informal, they are 1.69 and 1.08 (B) and 2.07 and 0.98 (C), the gains are 14 to 25 percent lower, and 49 to 51 percent (B) and 42 percent (C) of the revenue given up reaches consumers. Under C the ratio does not depend on ρ_{j^*}, and depends on δ only through the ratio of the no-plant value to Lima's.

Appendix P The food-basket profile

This appendix documents the assignment of δ_g behind Section 6.2, varies it, and reports the profile with measured rather than full pass-through. Per sol that household h spends on food, the VAT its purchases carry is

$$\tau_h = \sum_g \frac{t_g}{1+t_g} w_{hg} \Big[s^F_{hg} + (1 - s^F_{hg}) \delta_g \Big], \tag{14}$$

where w_{hg} is the share of good g in the household's food spending, s^F_{hg} the share of its spending on g made at formal stores, δ_g the formal input share of the informal price of g, averaged over its chains with their spending weights, and t_g the statutory rate of good g. Figure M3 plots it by decile; Figure 8 multiplies it by the decile's food budget share. Setting $\delta_g = 0$ for every good gives the sales-tax view.

P.1 The assignment

The chicken chain is the one we can observe at every node. This subsection asks what the correction implies for food spending as a whole. We evaluate equation (14) for each decile d of per-capita household expenditure, with the decile's budget shares w_{dg} and formal shares s^F_{dg} from ENAHO and an assignment of δ_g to every good. Each good carries its statutory rate: goods on the VAT law's list of exempt goods (most fresh produce) carry none at any stage, and rice pays a 4 percent tax under its own regime (IVAP) rather than the VAT; restaurant purchases carry the general rate throughout, although the rate on small restaurants was cut in September 2022 (Law 31556).[39] The budget shares use all 2022 interviews, including the exempt months, and restricting them to January–March gives a ratio of 1.20 (0.71 relative to total expenditure).

It is worth being precise about what δ_g is, because it is easy to read it as an index of how formal a chain is, and it is not. It is a share of the final price, and it factors into a binary part and a continuous one: whether a formal establishment creates the good at all, and how much value is added after the good leaves formal hands. A chain that is mostly formal by value added can still deliver a small δ if the good leaves formal hands early and several informal links add margin after it. Interior chicken has a large δ for the opposite reason: one formal node and one informal link, which adds 11 percent of the retail price. Empirically, though, the continuous part barely varies. Across the products of the price panel quoted at both wholesale and informal retail, the wholesale-to-informal-retail ratio runs from 0.60 to 0.91 with a mean of 0.77, so once a formal remitter exists, δ is between roughly 0.6 and 0.9 whatever the good. Nearly all the useful variation is in the binary part.

[39] Where a formal store buys from an informal supplier whose price already embeds upstream VAT, the tax cascades and the purchase carries more than the full rate, which would raise the top decile's rate.

Table C2 assigns δ_g to the basket. δ_g is the wholesale-to-informal-retail price ratio from the price panel where the good appears in it (milk, sugar, oil, pasta) and 0.85 otherwise. For chicken it is the spending-weighted average within informal purchases of Lima's live-bird value (0.79, Table 3), the interior value in provinces with an authorized slaughter plant (0.89) and, in provinces without one, the farm-gate live-bird price at the 0.75 yield, which the ministry reports only for Metropolitan Lima, over those cities' own stall price ($7.32/9.90 = 0.74$). For bread it is 0.90, on the assumption that both the mill and the bakery remit; where the informal bread purchase is from a bakery in the simplified regime the formal input is the flour and δ is the flour share of the loaf, nearer 0.3, so bread is the assignment most exposed to this choice. The binary part of the assignment is ours and rests on the industrial organization of each chain, not on the price data; rice is treated as informal because milling is largely in the simplified regime. Eggs are set to $\delta = 0$, a lower bound, since their informal price still carries the VAT on purchased feed and chicks (Appendix J). Rows (12)–(13) of Table P1 vary the value for provinces without a plant. The assignment is conservative by construction, since a code we have not classified contributes its spending to the denominator and nothing to the numerator; goods with a positive δ are 33 to 38 percent of purchased food depending on the decile.[40] We set every unclassified code to zero for that reason: it makes each decile's rate a lower bound and, because poorer households buy more informally, lowers their rates more than the top decile's, which works against our conclusion. Giving the unclassified codes, the taxed goods in the last row of Table C2 and 14 percent of purchased food, the default 0.85 when bought informally raises the VAT on purchased food from 1.56 to 2.04 percent of expenditure in the bottom decile and from 1.18 to 1.46 in the top, so the top-to-bottom ratio falls from 0.76 to 0.71; per sol of food purchased, it falls from 1.27 to 1.18.

Per 100 soles that Peruvian households spend on chicken, 2.6 go through formal stores, 46.5 through informal retailers in Lima, 23.9 through informal retailers in interior provinces with an authorized slaughter plant and 27.0 in provinces without one (ENAHO 2022). The VAT embedded in those 100 soles is $0.1525 \times (2.6 \times 1 + 46.5 \times 0.79 + 23.9 \times 0.89 + 27.0 \times 0.74) = 12.3$ soles under the chain view and 0.39 under the sales-tax view. Both figures are equation (14) for chicken alone: they are the statutory tax content of 100 soles of spending, which is the incidence only if the VAT passes through in full.[41]

[40] Informal vendors also buy other inputs from formal sellers, such as packaging, fuel and electricity, and cannot credit the VAT on them either. Each such input adds its own share of the informal price, times its pass-through, to the informal price's response, so following only the principal chain understates both the VAT in the informal price and its response to a change in the general rate.

[41] They measure the VAT already in place, not what the exemption could remove. With $R_k\delta_g$ in place of δ_g, counting only supermarkets and the provinces with an authorized slaughter plant, the exemption could remove at most 3.7 soles per 100 (January–March sample of Section 6.1).

P.2 All food consumed

Dividing by all food consumed answers an obvious objection. ENAHO values food the household did not buy, from its own production, its own business, in kind or from programs, and that food carries no tax by any route; it is 37 percent of the bottom decile's food and 9 percent of the top's. Dividing by all food consumed rather than by purchases, the chain-view profile is no longer flat: 3.1 percent at the bottom against 5.8 at the top, a factor of 1.8. The VAT on food is then mildly progressive, against the factor of 19 the sales-tax view reports on the same denominator. Both are per-sol measures: they show whether a purchase carries the tax, not who bears it, which Section 6.2 measures relative to total expenditure. Per sol spent in the market, the tax is nearly uniform on the baseline assignment and at most 2.8 times heavier at the top on any assignment we consider; per sol of food consumed, it is progressive but an order of magnitude less so than under the sales-tax view. Neither reading supports the claim that the informal purchase escapes the tax.

P.3 Alternative assignments

Only chicken's δ is backed by a causal estimate; for every other good the assignment in Table C2 rests on the organization of its chain and on price ratios. Suppose, then, that chicken were the only good whose informal price carries any VAT, the one good for which the experiment shows that it does, and that every other good were treated as the sales-tax view treats it. The top-to-bottom ratio of the effective rate on purchased food would still be 2.8, against 13 under the sales-tax view. Table P1 recomputes the profile under this and other alternative assignments, grouped by kind. The first group contains every change we can think of that would make the VAT on food more progressive: lowering δ to 0.60, the bottom of the range observed in the price panel, for every good with a formal establishment gives a ratio of 1.52; making flour bread's formal input gives 1.35; both together give 1.59. The second group (rows 5 to 9) contains refinements; three flatten the profile (rows 6, 7 and 9), and decile-specific chicken δ and prepared food leave it essentially unchanged (1.28 against 1.27). The last group (rows 12 to 15) varies what the paper's own evidence leaves open. Chicken bought informally in the provinces without an authorized slaughter plant may come from formal wholesalers after all, at the interior value of 0.89 (ratio 1.25), or carry no VAT at all if the birds come from farms outside it (1.45); formal sellers upstream may remit less than the statutory tax, which we represent by scaling every informal δ_g by three-quarters (1.48); and those two with flour as bread's formal input give 1.82. The profile therefore lies between 1.1 and 2.8 across everything we have tried: much less progressive than the sales-tax view's 13 on any assignment, and close to flat only on the baseline and the refinements that flatten it further.

Some rows of Table P1 need a definition. Row 5 applies each decile's own mix of Lima, plant-province and no-plant-province purchases to chicken, so its δ runs from 0.783 in the

first decile to 0.803 in the tenth. Row 6 scales δ_{dg} by the ratio of the code's median unit value to the decile's, capped at one. Row 7 counts only supermarkets as formal stores, so bakery and restaurant purchases carry δ_g, with bread at $\delta = 0.90$ and prepared food as in row 8. Row 8 sets δ for prepared food away from home to 0.4, an ingredient share of the meal price, times the decile's spending-weighted δ of informal at-home food purchases. Rows 12 to 15 first assign chicken line by line (Lima 0.79, provinces with an authorized slaughter plant 0.89, provinces without one 0.74), which gives 1.28 on its own, as row 5; giving the provinces without a plant Lima's 0.79 instead gives 1.27 (0.76 relative to total expenditure). Applying 18 percent to every good, ignoring the statutory exemptions, gives a ratio of 1.34 under the chain view and 14.9 under the sales-tax view (0.79 and 8.5 relative to total expenditure).

Table P1: The food-basket profile under alternative assignments of δ_g

	Per sol of food purchased			All food	Total expenditure	
	Decile 1	Decile 10	D10/D1	D10/D1	D10/D1	Bottom half (%)
Benchmarks						
Sales-tax view ($\delta_g = 0$ for all g)	0.26	3.46	13.1	18.9	7.6	13.0
Chain view, baseline (Table C2)	4.97	6.32	1.27	1.84	0.76	30.7
Assignments that push toward progressivity						
(1) $\delta > 0$ for chicken only	1.59	4.41	2.77	4.01	1.66	24.6
(2) $\delta = 0.60$ for all goods with a formal establishment	3.62	5.50	1.52	2.19	0.89	29.0
(3) Bread's formal input is flour ($\delta = 0.30$)	4.56	6.17	1.35	1.96	0.80	30.0
(4) (2) and (3) together	3.41	5.42	1.59	2.30	0.93	28.5
Refinements						
(5) Decile-specific δ for chicken	4.94	6.32	1.28	1.85	0.76	30.6
(6) δ_{dg} scaled by the decile's unit value	5.06	6.22	1.23	1.77	0.73	30.9
(7) Bakeries and restaurants informal	4.89	5.40	1.11	1.60	0.66	32.1
(8) Prepared food carries its inputs' δ	4.97	6.35	1.28	1.84	0.76	30.6
(9) Rice milled formally ($\delta = 0.85$ at the 4 percent IVAP)	5.24	6.39	1.22	1.76	0.73	31.1
Burden of a change: each term of (14) times its price response (uses Assumption 1)						
(10) First-month pass-through	3.82	4.92	1.29	1.86	0.77	30.5
(11) Exempt-window pass-through	2.60	3.28	1.26	1.82	0.75	30.7
(10c) First month, one pass-through at every formal node	3.91	4.98	1.27	1.84	0.76	30.7
(11c) Window, one pass-through at every formal node	2.56	3.26	1.27	1.84	0.76	30.7
What the evidence leaves open						
(12) No-plant provinces' chicken at the interior value (0.89)	5.08	6.35	1.25	1.81	0.75	30.8
(13) No-plant provinces' chicken carries no VAT	4.26	6.17	1.45	2.09	0.86	29.4
(14) Upstream sellers remit three-quarters of the tax	3.79	5.60	1.48	2.13	0.87	29.2
(15) (3), (13) and (14) together	2.95	5.39	1.82	2.63	1.07	27.2

Notes: Effective VAT rate on food (equation (14)), percent, at statutory rates; ENAHO 2022, survey weights, deciles of per-capita household expenditure. All food: divides by all food consumed, purchased or not. Total expenditure: VAT on purchased food over total expenditure, as in Figure 8; bottom half: percent of it paid by the five lowest deciles. D10/D1: ratio of the deciles' mean rates. (10c)–(11c): the supermarket's pass-through at every formal node. The text defines the other rows.

P.4 Measured pass-through

Equation (14) is the tax content of what households buy, the burden of the VAT if it passes through to prices in full. Who would gain from a cut in the general rate depends also on pass-through: the burden of a change multiplies the formal term of (14) by the response of the formal retail price, ρ_F^G, and the informal term, δ_g, by the response of the formal input's

price, $\rho^G_{j^*g}$ (Proposition 1), so that with every ρ equal to one the burden is the tax content. The expression uses the one-sol-for-one-sol assumption. In the interior, where both ends of the chain are observed, the stalls passed through 38 percent against the 35 percent that the assumption predicts, a difference that is not statistically significant (Section 5.2). For a change in the general rate, the expression also requires an assumption, because our estimates come from a change on a single good:

Assumption 1 (Formal-input pass-through invariance)**.** *The formal retail price and the price of the formal input respond the same to a change in the general rate as to a change on the good alone:* $\rho^G_F = \rho_F$ *and* $\rho^G_{j^*} = \rho_{j^*}$.[42]

We measure $\rho_F = 0.79$ at supermarkets in the first month and 0.52 over the window, and take ρ_{j^*} from the wholesaler in the cities with an authorized slaughter plant, where the seller of the formal input is most likely formal: 0.77 and 0.52. We apply both to every good, since chicken is the only one whose pass-through at a formal node we measure; with the pass-through of every price that would move, this step would need neither δ nor the model, only spending shares and measured responses, as in Section 6.1. Applying them to Figure M3 scales both profiles down without reordering them: the chain-view rate falls from 5.0 to 3.8 percent in the bottom decile and from 6.3 to 4.9 in the top in the first month, and the top-to-bottom ratio stays close to that of the tax content, 1.29 in the first month and 1.26 over the window (Table P1), because the supermarket and the plant-city wholesaler pass on almost the same share of the tax: the joint bootstrap puts the supermarket 0.02 [−0.34, 0.32] above the wholesaler in the first month and 0.01 [−0.33, 0.34] below it over the window. A cut in the general rate would therefore lower food prices roughly in proportion to food spending across the distribution, whereas under the sales-tax view, with the same pass-through, it would go 13 times more, relative to food spending, to the top decile than to the bottom. Relative to total expenditure, the gain over the window is about half the full-pass-through gain, and the bottom half still receives 31 percent of it. Because one pass-through at every formal node leaves the ratio at 1.27 (rows 10c and 11c), measured pass-through lowers the gain without changing who gets it, which is why Section 6.2 can normalize it to one.

[42] The assumption can fail for two reasons. A general change also changes the tax on goods that compete with the formal input, which a change on the formal input alone does not; and pass-through need not be the same for increases and decreases (Benzarti et al., 2020), although Appendix I shows that the asymmetry is smaller at the sale of the formal input than at formal retail. Only the extrapolation from the exemption to a change in the general rate uses the assumption, and that extrapolation also assumes that the vendor's margin does not move with a general change, which the exemption cannot check.

Appendix Q Bachas, Gadenne and Jensen for Peru

This appendix reproduces the Peru estimate of Bachas, Gadenne, and Jensen (2024) and reconciles it, step by step, with the sales-tax profile per sol of purchased food (Figure M3). Their Peru replication code and crosswalk of places of purchase are public (Zenodo 10.5281/zenodo.8025277 and the authors' GitHub repository); we run their module-by-module construction of household consumption on ENAHO 2017, their survey year, and 2022, ours, changing only file paths and dropping the group-subtotal rows of the food module, which their code keeps for own production and gifts and so counts that food twice. Consumption is every expenditure module net of rent, actual and imputed, including own production, payments in kind and gifts, which they classify as non-market and informal. Places of purchase map into their store classes: non-market, no store front, corner stores, services from individuals and informal entertainment are informal; specialized and large stores, institutions, entertainment, pharmacies and spending with no recorded place, which is mostly utilities, meals away from home, household services and durables, are formal. Households are ranked by household-weighted deciles of per-capita consumption net of rent. Under their scenario 1, a uniform rate on formal consumption that raises 10 percent of consumption, a decile's effective rate is that rate times its formal share, so the ratios in Table Q1 are ratios of formal shares.

In short, the sales-tax profile of Figure M3 is our construction of their premise on purchased food, not their estimate for Peru. On their own object, a uniform VAT on formal consumption has an effective rate 2.46 times higher in the top decile than in the bottom in their data and 2.77 in our reproduction. Letting informal food purchases carry δ_g lowers that factor to 2.12, and letting manufactured non-food goods bought informally carry $\delta = 0.85$ as well lowers it to 1.89: the correction removes a third to a half of their gradient. The VAT remains progressive on their object because it also includes consumption that bears no VAT by any route, such as home production and food received in kind, a larger share of poorer households' budgets. On purchased food, their construction and ours differ even before δ enters: in 2022, with their store classes and deciles, the factor is 4.1; our classes, which put wholesale grocers and other places with the informal stores, raise it to 8.9, person-weighted deciles of per-capita household spending to 13.4, and excluding the Amazon departments, applying statutory rates and using ENAHO's imputed purchase values bring it to the 13.1 of Figure M3 (Table Q1).

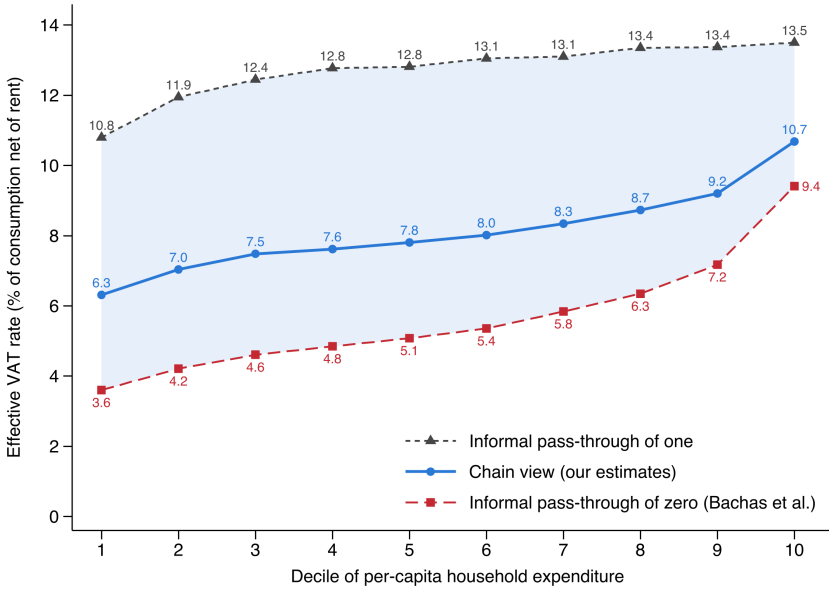
Two data limits apply. INEI's public release has no separate personal-care module, which their code reads, and our reproduction leaves it out. Their published Peru inputs (their Supplementary Appendix and replication files) put the formal share at 32.2 percent in the bottom decile and 79.0 in the top. Our formal share matches theirs in the top decile (79 percent) and is about four points below it in the bottom decile (28 against 32 percent), which the missing module may explain in part. And their durables code values items bought since 1999 with price multipliers that end in 2010, so in 2022 it drops most durables, which are formal and concentrated at the top; the rows that include durables are reported for 2017 only,

and the 2022 band of Figures 9 and Q1 uses INEI's durables aggregate and person-weighted deciles, so it is not the construction of Table Q1.

The chain view on their object keeps their classes, deciles and uniform rate, and changes one thing: an informal purchase carries δ_g of the tax instead of none. Food bought at a store with no front or a corner store takes the assignment of Table C2 at the uniform rate; in row (1b), manufactured non-food goods bought there (household non-durables, leisure goods, clothing and footwear, other goods, and fuel from informal pumps) take 0.85, the default for a good with a formal establishment upstream; services from individuals, informal entertainment, furniture and non-market consumption carry none.

On their own object, consumption net of rent, the band of Figure 9 is steeper (Figure Q1): in 2022, the top decile's rate is 2.61 times the bottom decile's under the sales-tax view, 1.69 in the central case and 1.25 at the upper bound, and the bottom half pays 21, 25 and 28 percent of the VAT, against 30 percent of consumption. About three-quarters of the sales-tax gradient in logs comes from the premise of no pass-through and the rest from consumption that is never bought. Figure 9 divides by total expenditure instead, which also includes rent, untaxed, and spending their measure does not collect, mostly food eaten away from home, to which it gives the household's own rate.

Figure Q1: The whole VAT on consumption net of rent, by expenditure decile



Effective VAT rate at a uniform 18 percent, percent of consumption net of rent as constructed by Bachas, Gadenne, and Jensen (2024); band and lines as in Figure 9. ENAHO 2022; deciles of per-capita household expenditure, person-weighted.

Table Q1: Bachas, Gadenne and Jensen's Peru estimate and the sales-tax profile per sol of purchased food (Figure M3)

	ENAHO 2017			ENAHO 2022		
	Decile 1	Decile 10	D10/D1	Decile 1	Decile 10	D10/D1
(1) Their object: consumption net of rent	28.4	78.7	2.77		–	
(1a) chain view: informal food carries δ_g	38.6	81.8	2.12		–	
(1b) and informal non-food goods carry 0.85	44.3	83.9	1.89		–	
(2) Food, their store classes and deciles	3.0	26.8	9.04	3.8	20.5	5.40
(3) Purchased food only	4.9	28.5	5.88	5.4	22.3	4.10
(4) our store classes	1.5	26.1	17.85	2.2	19.6	8.86
(5) person-weighted deciles	1.1	26.6	24.50	1.5	20.3	13.36
(6) outside the Amazon regime	1.1	28.0	25.70	1.6	21.5	13.88
(7) each good's statutory rate	1.1	21.9	20.15	1.5	17.8	11.64
(8) ENAHO's imputed purchase values	1.3	25.8	20.23	1.7	22.7	13.11

Percent. Rows (1)–(1b): formal share of consumption net of rent, plus, in the chain view, δ_g times the informal purchases that carry it. Rows (2)–(8): share of food spending bought at formal stores, weighted in (7)–(8) by each good's statutory rate relative to 18 percent; row (8) is the sales-tax profile of Figure M3. Dashes: not computed for 2022 (see text).

Appendix R Store types

Table R1 classifies the store types reported in ENAHO as formal, informal or other, Figure R1 shows a retail street market in our sample, Figure R2 a corner store and a street vendor, both outside our sample, and Figure R3 two of the receipts we collected.

Table R1: Classification of vendor types reported in ENAHO as formal, informal, or other

Formal	**Informal**	**Others**
Bakery	Street vendor	Other places (counted as informal)
Supermarket	Corner store	
Restaurants and bars	Wholesale/retail street markets	
	Food trucks/vans	
	Fairs	

Figure R1: An example of a retail street market in our sample



Figure R2: Other examples of traditional markets (not in our sample)



(a) Corner store



(b) Street vendor

Figure R3: Examples of receipts



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F001-00000371

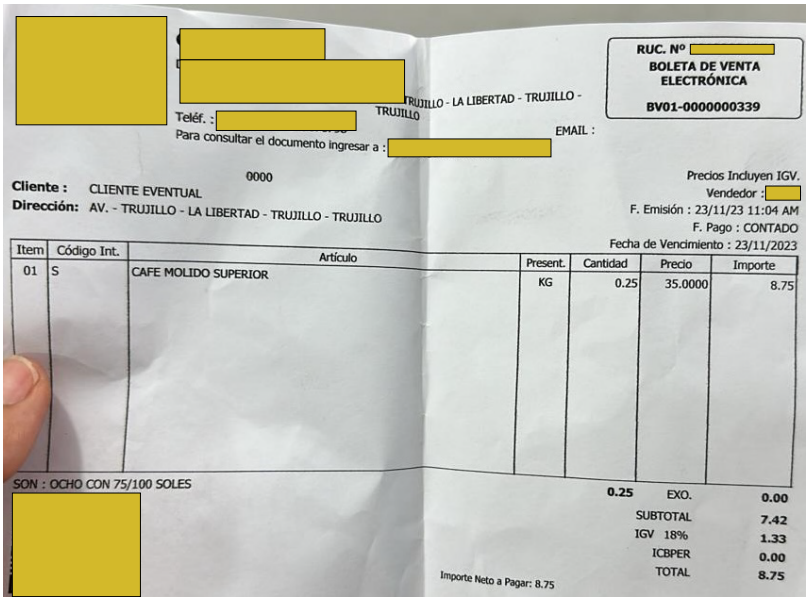
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ACEITUNA EXTRA	0.50	20.00	10.00
QUESO MANTECOSO	0.25	32.00	8.00
MAIZ PERLA	0.25	12.00	3.00
[illegible]LA SUELTA	0.05	100.00	5.00

OP. GRAVADAS:	S/	22.03
OP. EXONERADAS:	S/	0.00
OP. INAFECTAS:	S/	0.00
IGV 18%:	S/	3.97
ICBP (0):	S/	0.00
IMPORTE TOTAL:	S/	26.00

SON: VEINTISEIS SOLES CON 00/100
FORMA PAGO: Contado

(a) "Start date: October 10, 2022"



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Precios Incluyen IGV.
Vendedor : [redacted]
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F. Pago : CONTADO
Fecha de Vencimiento : 23/11/2023

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SON : OCHO CON 75/100 SOLES

0.25	EXO.	0.00
	SUBTOTAL	7.42
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Importe Neto a Pagar: 8.75

(b) "Start date: May 11, 2022"